Exemple d'utilisation du package DegradSim

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Préparation

Installation du package depuis GitHub :

```
remotes::install_github("rytracetito/levy_packages", subdir = "DegradSim")
```

Les parties I et II utilisent les sept **processus de Lévy** du package (`A = NULL` par défaut). La partie III montre les **processus de Lévy à échelle de temps** (*time-scaled Lévy processes*) $X_t = Y_{A(t)}$, obtenus avec une fonction d'échelle de temps A (*time-scaling function*) : gamma, Wiener et inverse gaussien. La partie IV reprend ces simulations avec la **méthode Bridge** (`method = "Bridge"`), pour les cinq processus qui ont un pont dans le package `levyBridge`.

Paramètres utilisés :

Processus	Fonction	Paramètres
Gamma	`sim_gamma`	`a = 2`, `b = 3` (taux)

Processus	Fonction	Paramètres
Wiener	`sim_wiener`	`mu = 1.5`, `sigma = 2`
Inverse gaussien	`sim_ig`	`mu = 0.5`, `eta = 2`
Variance-gamma	`sim_vg`	`theta = 0.5`, `sigma = 1.5`, `nu = 2`
Meixner	`sim_meixner`	`a = 1`, `b = 0.3`, `d = 1.5`, `m = 0.2`
Normal inverse gaussien	`sim_nig`	`alpha = 2`, `beta = 0.5`, `delta = 1`, `mu = 0.2`
Poisson composé	`sim_cpoisson`	`lambda = 0.5`, sauts exponentiels de moyenne 1 (`rjump` par défaut)

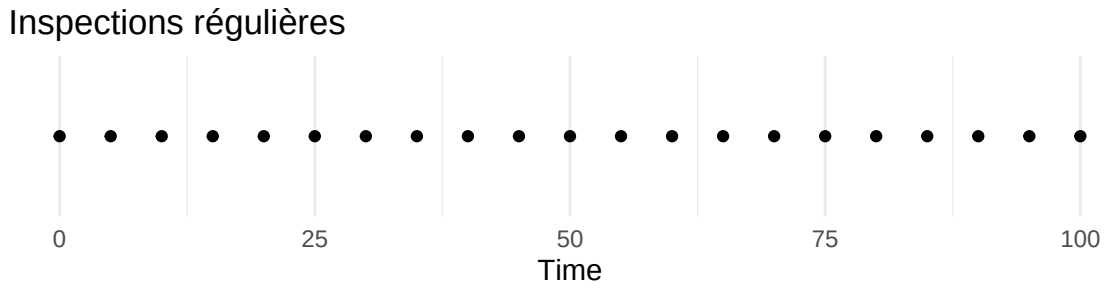
Dans chaque cas, les trajectoires simulées sont d'abord **stockées** dans un objet `x`, puis utilisées : premières valeurs, accroissements de la première trajectoire (arrondis à 3 décimales), puis graphique. Les accroissements montrent qu'un palier presque plat sur le graphique correspond à de petits accroissements, pas à des accroissements nuls (sauf pour le Poisson composé, constant entre deux chocs). Quand toutes les trajectoires ont les mêmes instants d'inspection, `x` est une matrice : une colonne `t`, puis une colonne par trajectoire (`X1`, `X2`, ...).

I - Une trajectoire

Une trajectoire inspectée 20 fois entre 0 et 100. `x` est une matrice à deux colonnes : `t` et `X1`.

Cas 1 : inspections régulières

```
t1 <- inspection_times("regular", n = 20, tmax = 100)$time
plot_plan(t1, "Inspections régulières")
```



Modèle 1 : gamma

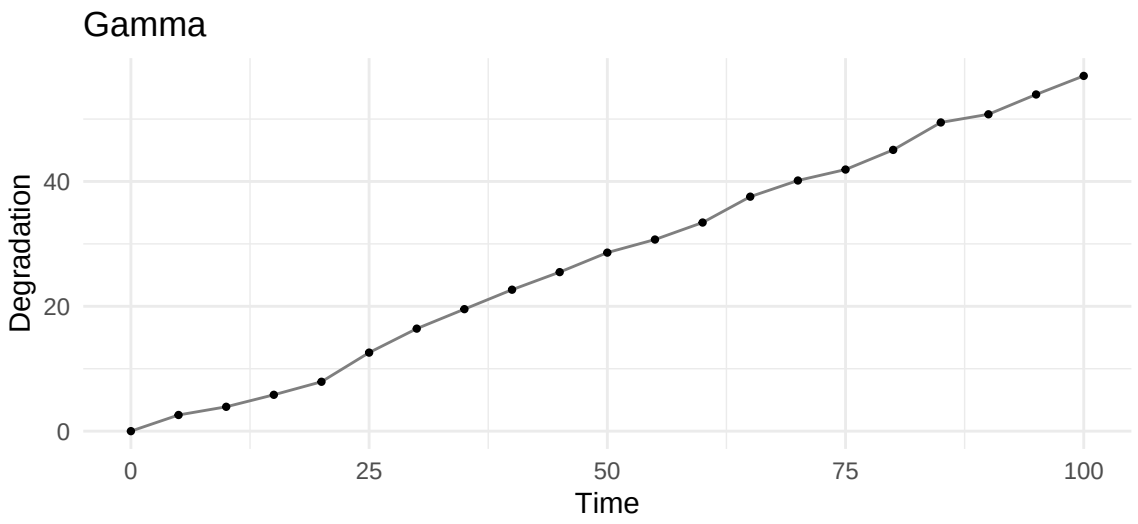
```
set.seed(11)
x <- sim_gamma(t1, a = 2, b = 3)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.588550
## [3,] 10  3.900945
## [4,] 15  5.822353
## [5,] 20  7.910152
## [6,] 25 12.582689
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.589 1.312 1.921 2.088 4.673 3.841 3.120 3.116 2.819 3.123 2.095 2.725
## [13] 4.144 2.588 1.757 3.150 4.395 1.307 3.174 2.977
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

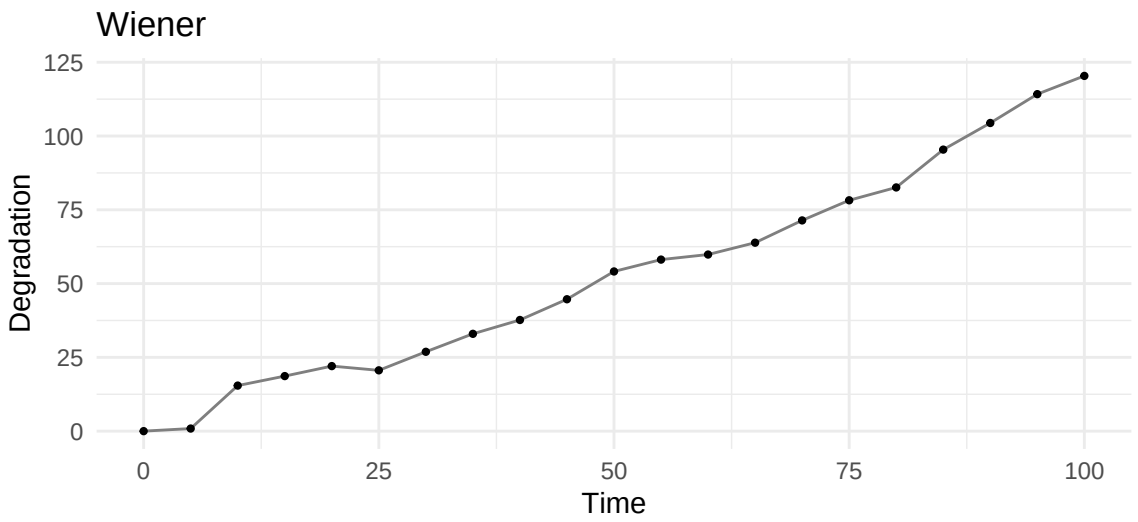
```
set.seed(12)
x <- sim_wiener(t1, mu = 1.5, sigma = 2)
head(x)
```

```
##       t         X1
## [1,]  0  0.0000000
## [2,]  5  0.8787004
## [3,] 10 15.4320167
## [4,] 15 18.6533253
## [5,] 20 22.0389368
## [6,] 25 20.6052097
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.879 14.553  3.221  3.386 -1.434  6.282  6.090  4.690  7.024  9.414
## [11]  4.022  1.714  4.014  7.553  6.818  4.354 12.817  9.023  9.767  6.188
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

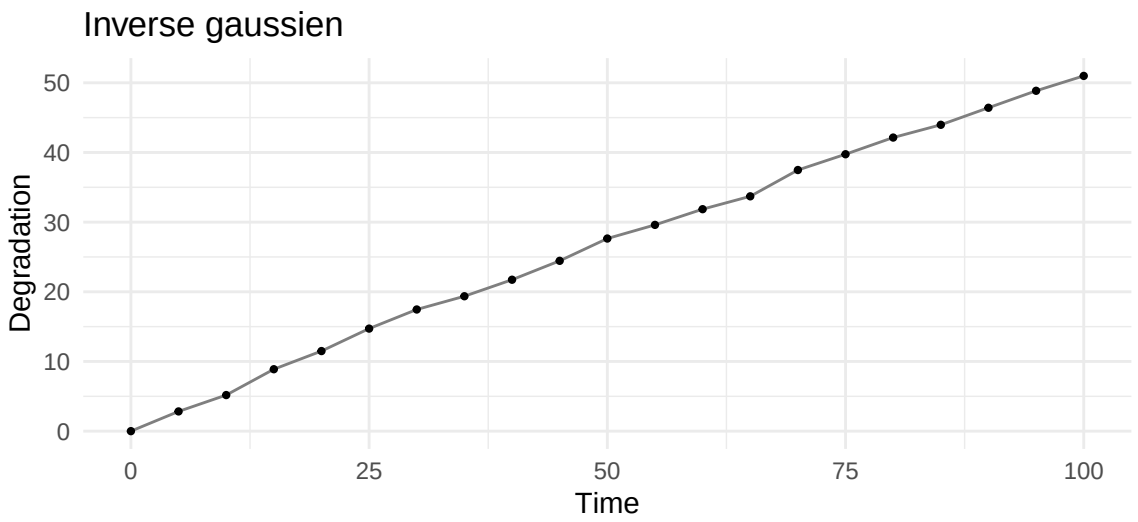
```
set.seed(13)
x <- sim_ig(t1, mu = 0.5, eta = 2)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.829678
## [3,] 10  5.177833
## [4,] 15  8.886486
## [5,] 20 11.493417
## [6,] 25 14.718884
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.830 2.348 3.709 2.607 3.225 2.743 1.901 2.371 2.713 3.199 1.959 2.255
## [13] 1.846 3.775 2.266 2.394 1.832 2.444 2.437 2.137
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

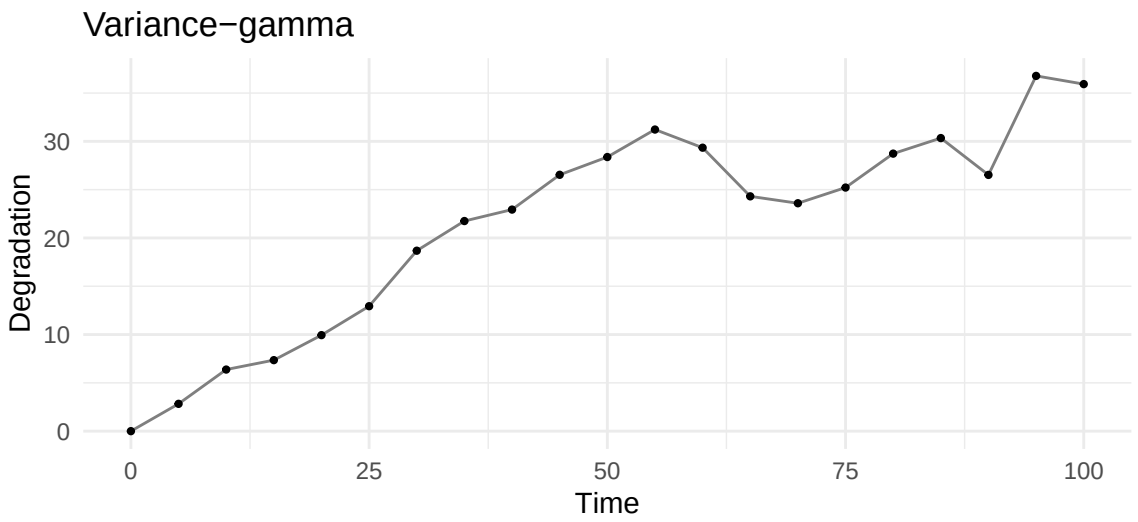
```
set.seed(14)
x <- sim_vg(t1, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.826550
## [3,] 10  6.375562
## [4,] 15  7.349937
## [5,] 20  9.939181
## [6,] 25 12.934933
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.827  3.549  0.974  2.589  2.996  5.745  3.070  1.187  3.601  1.838
## [11]  2.847 -1.879 -5.037 -0.715  1.625  3.514  1.609 -3.813 10.250 -0.850
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

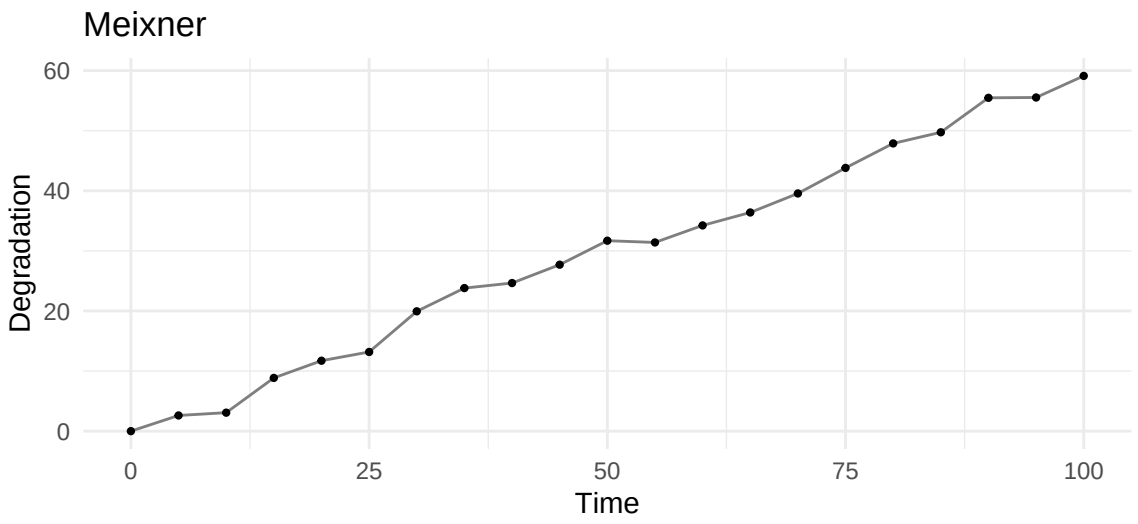
```
set.seed(15)
x <- sim_meixner(t1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.609114
## [3,] 10  3.073932
## [4,] 15  8.858509
## [5,] 20 11.718557
## [6,] 25 13.175516
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.609  0.465  5.785  2.860  1.457  6.765  3.865  0.841  3.056  3.990
## [11] -0.291  2.835  2.153  3.165  4.254  4.075  1.855  5.735  0.053  3.594
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

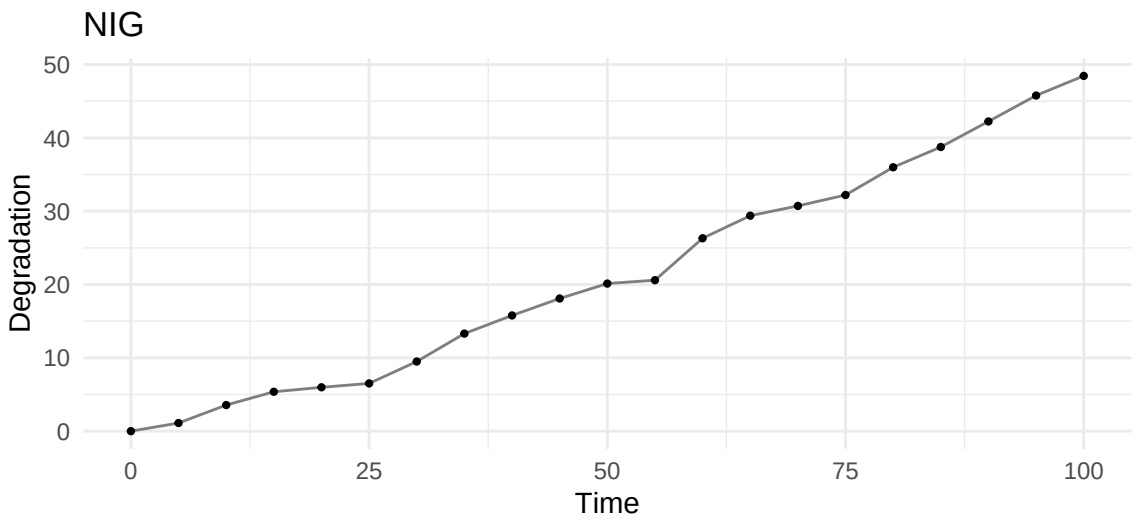
```
set.seed(16)
x <- sim_nig(t1, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##       t       X1
## [1,]  0 0.000000
## [2,]  5 1.106802
## [3,] 10 3.555632
## [4,] 15 5.369281
## [5,] 20 5.980709
## [6,] 25 6.508225
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.107 2.449 1.814 0.611 0.528 2.987 3.809 2.473 2.314 2.036 0.464 5.717
## [13] 3.078 1.331 1.489 3.784 2.767 3.484 3.534 2.677
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

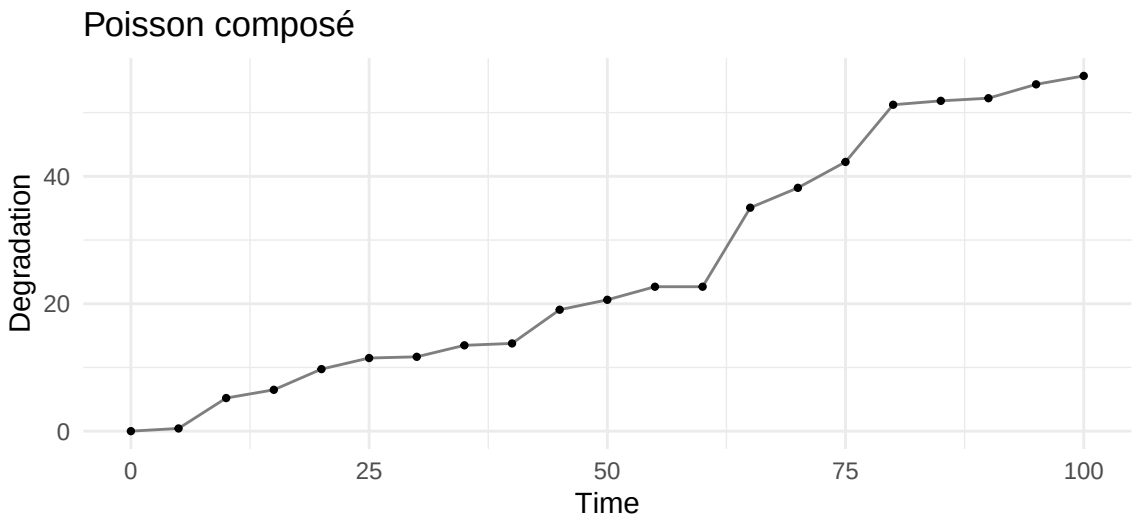
```
set.seed(17)
x <- sim_cpoisson(t1, lambda = 0.5)
head(x)
```

```
##       t         X1
## [1,]  0  0.0000000
## [2,]  5  0.4210759
## [3,] 10  5.1945031
## [4,] 15  6.4873965
## [5,] 20  9.7368579
## [6,] 25 11.4769556
```

```
round(diff(x[, "X1"]), 3)
```

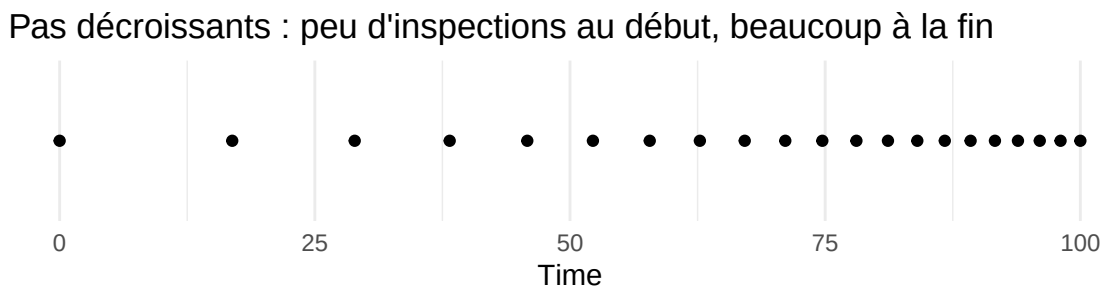
```
##  [1]  0.421  4.773  1.293  3.249  1.740  0.191  1.804  0.298  5.292  1.554
## [11]  2.055  0.000 12.406  3.120  4.068  8.975  0.614  0.413  2.177  1.325
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
20
40
0
25
50
75
100
Time

Cas 2 : pas décroissants

```
t2 <- inspection_times("decreasing", n = 20, tmax = 100)$time
plot_plan(t2, "Pas décroissants : peu d'inspections au début, beaucoup à la fin")
```



Modèle 1 : gamma

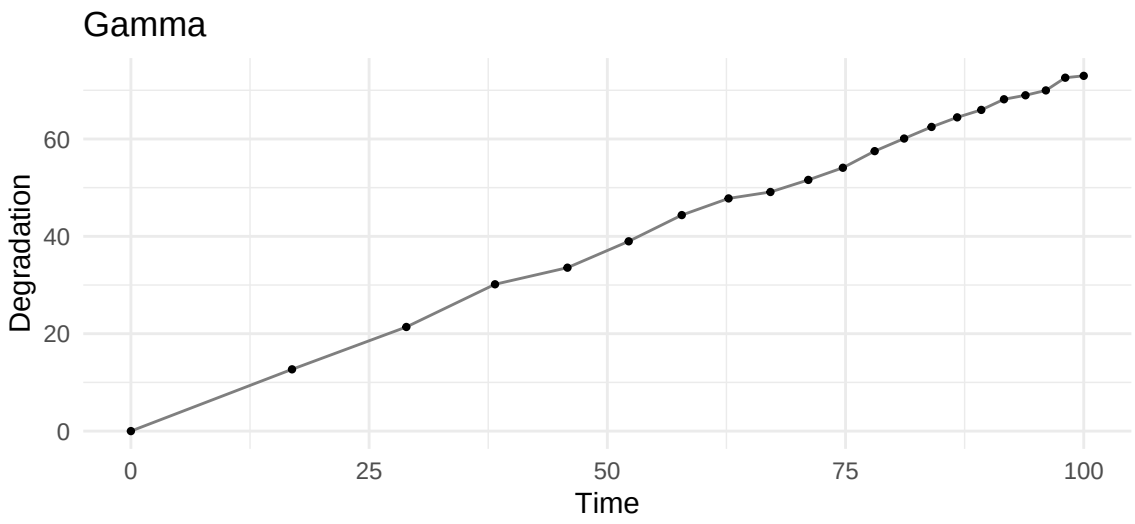
```
set.seed(21)
x <- sim_gamma(t2, a = 2, b = 3)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 12.68436
## [3,] 28.90648 21.38241
## [4,] 38.21229 30.15090
## [5,] 45.81569 33.56269
## [6,] 52.24427 38.98223
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 12.684  8.698  8.768  3.412  5.420  5.385  3.423  1.320  2.481  2.496
## [11]  3.416  2.586  2.386  1.966  1.530  2.184  0.821  0.999  2.611  0.380
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

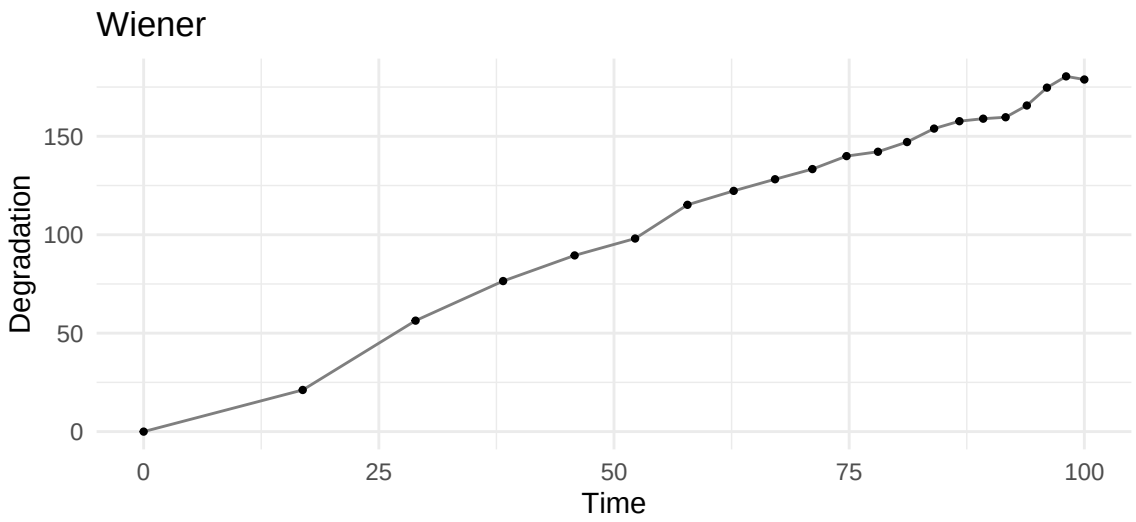
```
set.seed(22)
x <- sim_wiener(t2, mu = 1.5, sigma = 2)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 21.15190
## [3,] 28.90648 56.36371
## [4,] 38.21229 76.47126
## [5,] 45.81569 89.49119
## [6,] 52.24427 98.07444
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 21.152 35.212 20.108 13.020  8.583 17.123  7.075  5.908  5.165  6.588
## [11]  2.216  4.924  6.837  3.762  1.271  0.728  5.970  9.068  5.724 -1.591
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

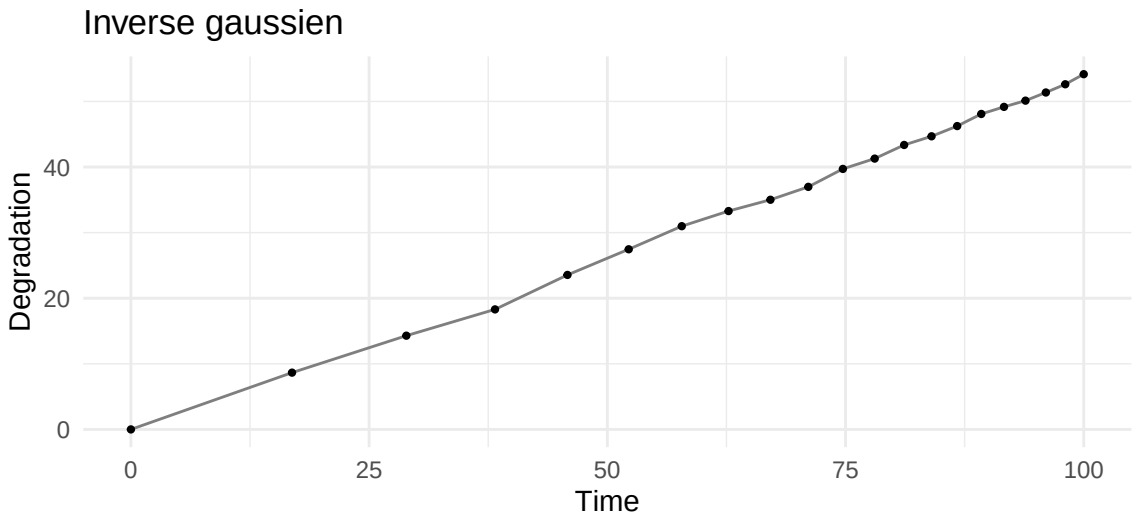
```
set.seed(23)
x <- sim_ig(t2, mu = 0.5, eta = 2)
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  8.655577
## [3,] 28.90648 14.289435
## [4,] 38.21229 18.296030
## [5,] 45.81569 23.551265
## [6,] 52.24427 27.462389
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 8.656 5.634 4.007 5.255 3.911 3.519 2.307 1.724 1.965 2.736 1.572 2.079
## [13] 1.321 1.558 1.849 1.079 0.948 1.244 1.255 1.538
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

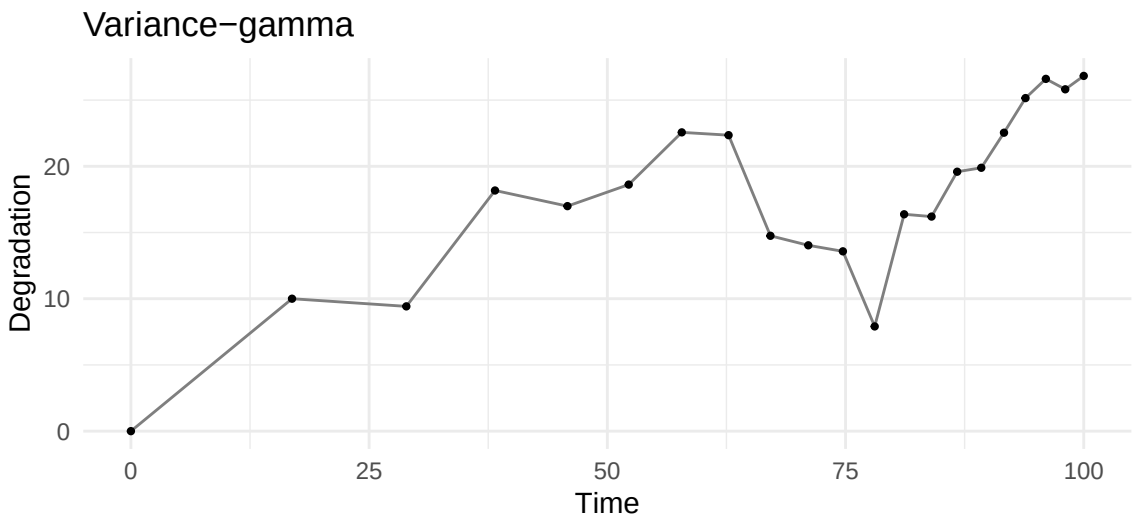
```
set.seed(24)
x <- sim_vg(t2, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 10.00188
## [3,] 28.90648  9.42331
## [4,] 38.21229 18.17461
## [5,] 45.81569 16.99351
## [6,] 52.24427 18.62047
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.002 -0.579  8.751 -1.181  1.627  3.944 -0.213 -7.600 -0.718 -0.455
## [11] -5.672  8.471 -0.174  3.388  0.298  2.649  2.613  1.452 -0.788  1.012
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

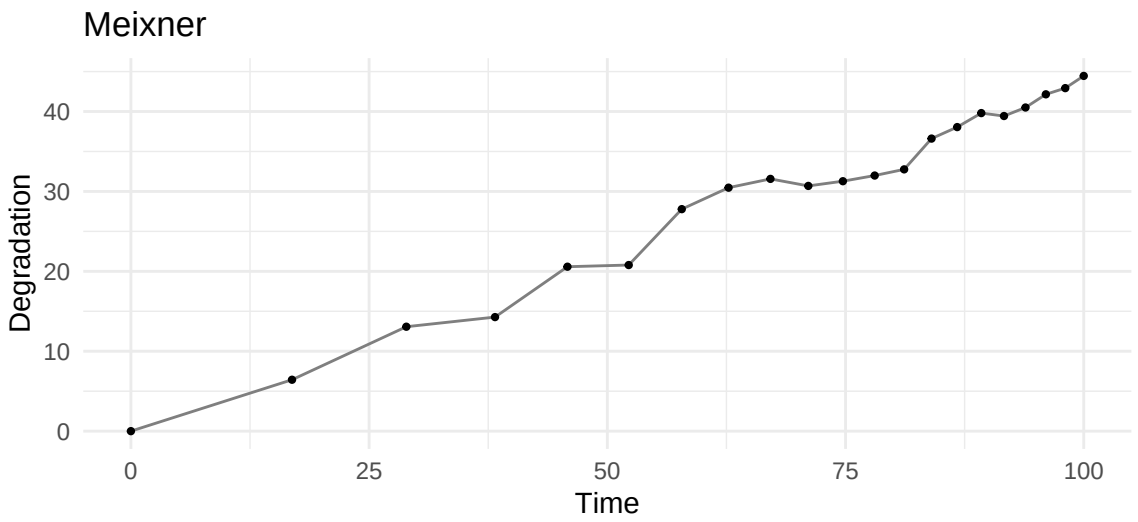
```
set.seed(25)
x <- sim_meixner(t2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  6.431984
## [3,] 28.90648 13.068405
## [4,] 38.21229 14.274293
## [5,] 45.81569 20.578044
## [6,] 52.24427 20.786662
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.432  6.636  1.206  6.304  0.209  6.996  2.688  1.098 -0.879  0.589
## [11]  0.708  0.773  3.853  1.438  1.752 -0.366  1.064  1.650  0.775  1.531
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

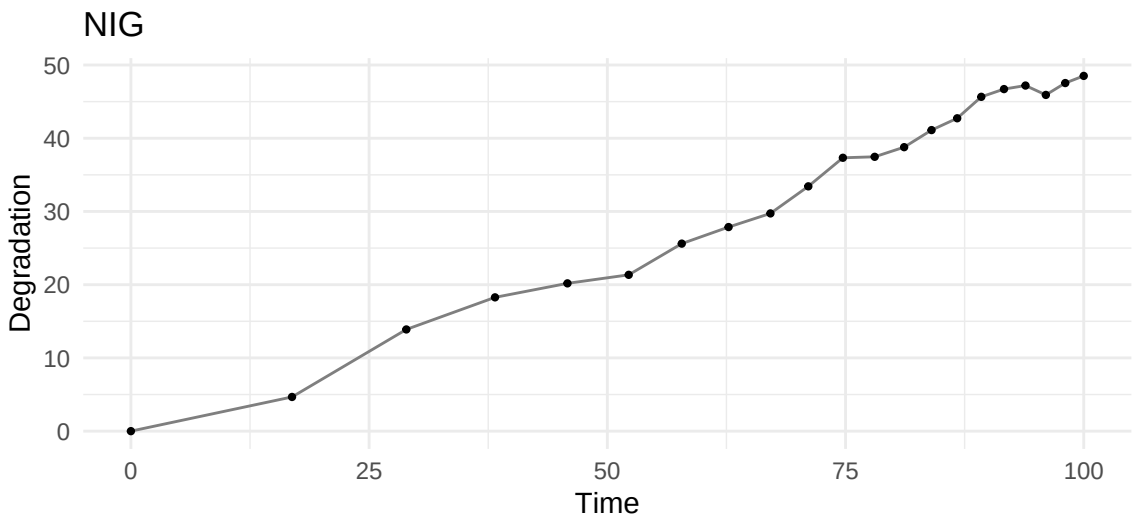
```
set.seed(26)
x <- sim_nig(t2, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  4.668869
## [3,] 28.90648 13.887229
## [4,] 38.21229 18.270337
## [5,] 45.81569 20.180364
## [6,] 52.24427 21.346607
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  4.669  9.218  4.383  1.910  1.166  4.261  2.262  1.864  3.696  3.902
## [11]  0.132  1.316  2.332  1.614  2.929  1.057  0.483 -1.279  1.621  0.976
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

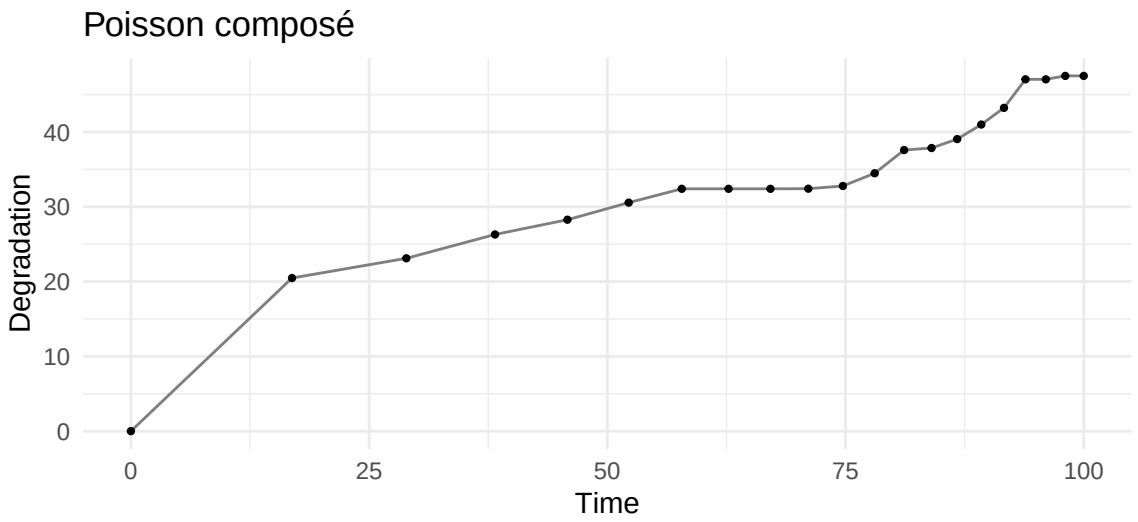
```
set.seed(27)
x <- sim_cpoisson(t2, lambda = 0.5)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 20.47580
## [3,] 28.90648 23.11248
## [4,] 38.21229 26.29892
## [5,] 45.81569 28.27167
## [6,] 52.24427 30.55641
```

```
round(diff(x[, "X1"]), 3)
```

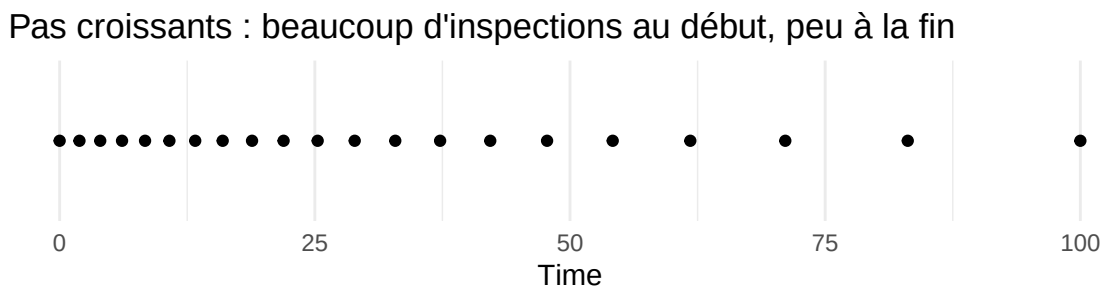
```
##  [1] 20.476  2.637  3.186  1.973  2.285  1.850  0.000  0.000  0.015  0.363
## [11]  1.700  3.106  0.269  1.196  1.943  2.232  3.814  0.000  0.459  0.000
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
10
20
30
40
0
25
50
75
100
Time

Cas 3 : pas croissants

```
t3 <- inspection_times("increasing", n = 20, tmax = 100)$time
plot_plan(t3, "Pas croissants : beaucoup d'inspections au début, peu à la fin")
```



Modèle 1 : gamma

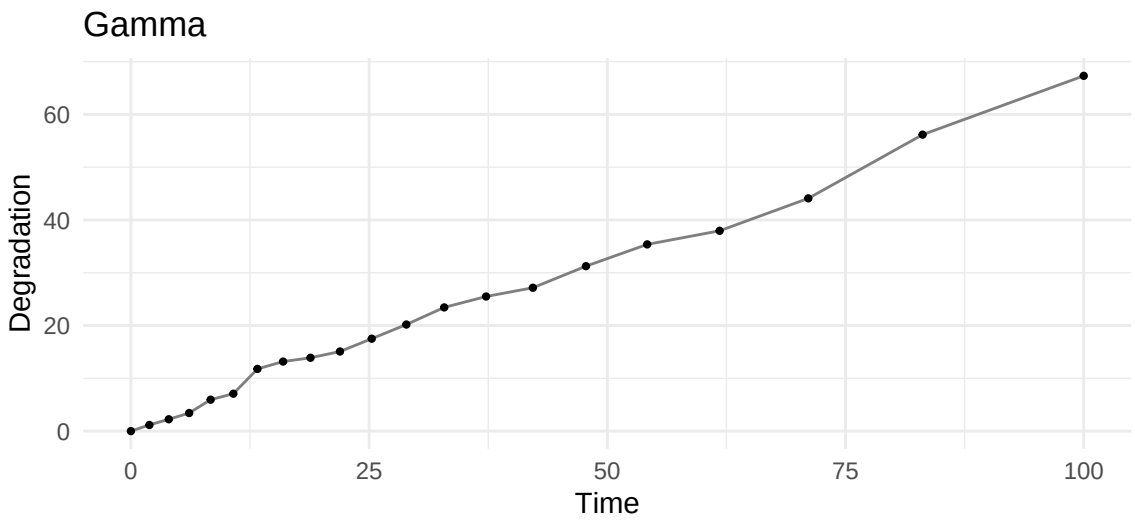
```
set.seed(31)
x <- sim_gamma(t3, a = 2, b = 3)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  1.940035 1.161003
## [3,]  3.974743 2.237610
## [4,]  6.113840 3.420305
## [5,]  8.368618 5.951271
## [6,] 10.752309 7.083982
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.161  1.077  1.183  2.531  1.133  4.699  1.389  0.714  1.189  2.432
## [11]  2.680  3.245  2.064  1.658  4.087  4.129  2.576  6.145 12.071 11.140
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

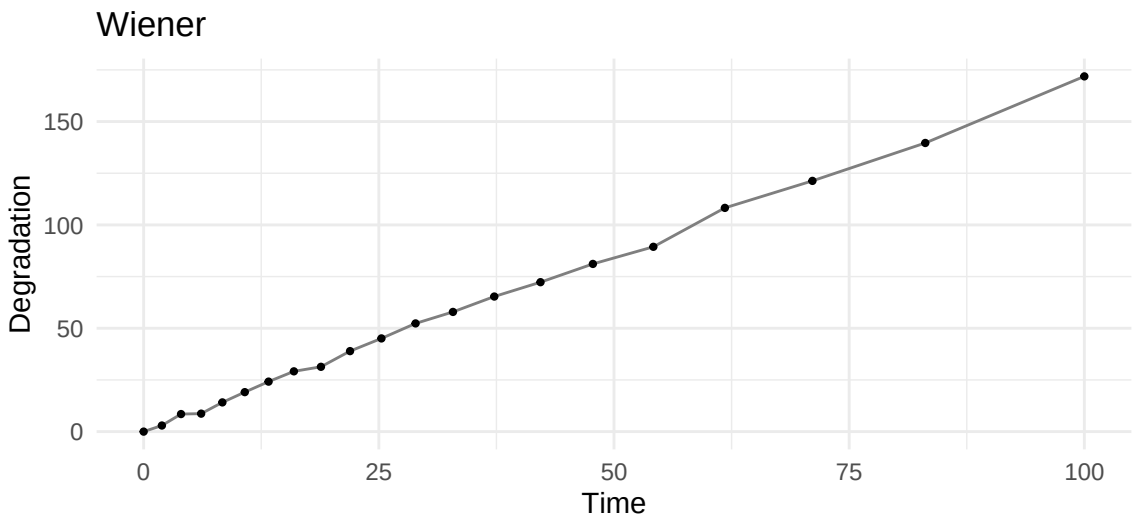
```
set.seed(32)
x <- sim_wiener(t3, mu = 1.5, sigma = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  1.940035  2.950837
## [3,]  3.974743  8.494273
## [4,]  6.113840  8.696041
## [5,]  8.368618 14.137385
## [6,] 10.752309 19.100712
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.951  5.543  0.202  5.441  4.963  5.087  4.971  2.198  7.588  6.144
## [11]  7.254  5.563  7.444  6.951  8.826  8.283 18.828 13.065 18.341 32.221
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

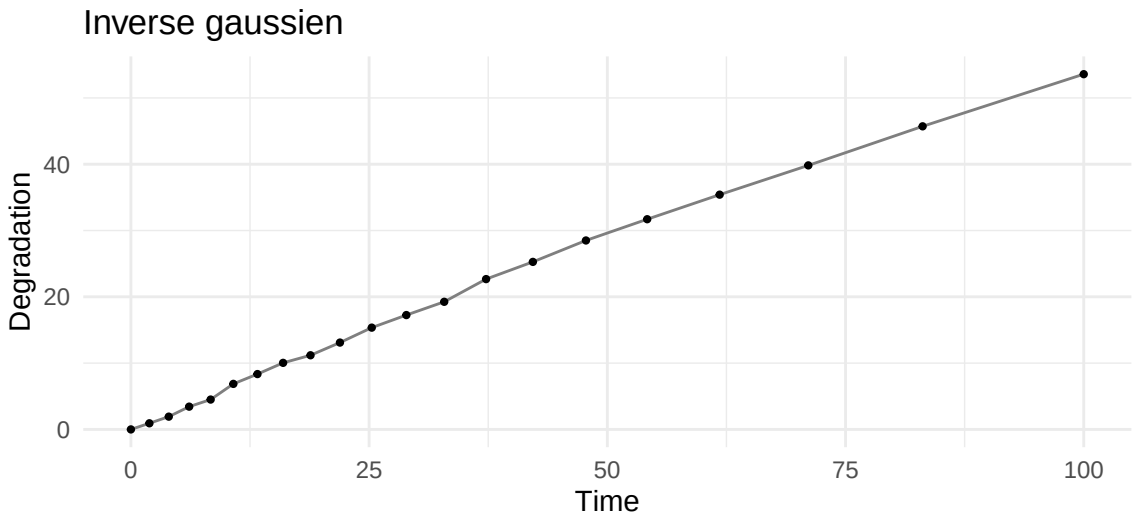
```
set.seed(33)
x <- sim_ig(t3, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  1.940035 0.9238276
## [3,]  3.974743 1.9267367
## [4,]  6.113840 3.4350758
## [5,]  8.368618 4.5045984
## [6,] 10.752309 6.8688533
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.924 1.003 1.508 1.070 2.364 1.478 1.693 1.144 1.914 2.252 1.896 2.002
## [13] 3.424 2.595 3.231 3.205 3.704 4.415 5.894 7.865
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

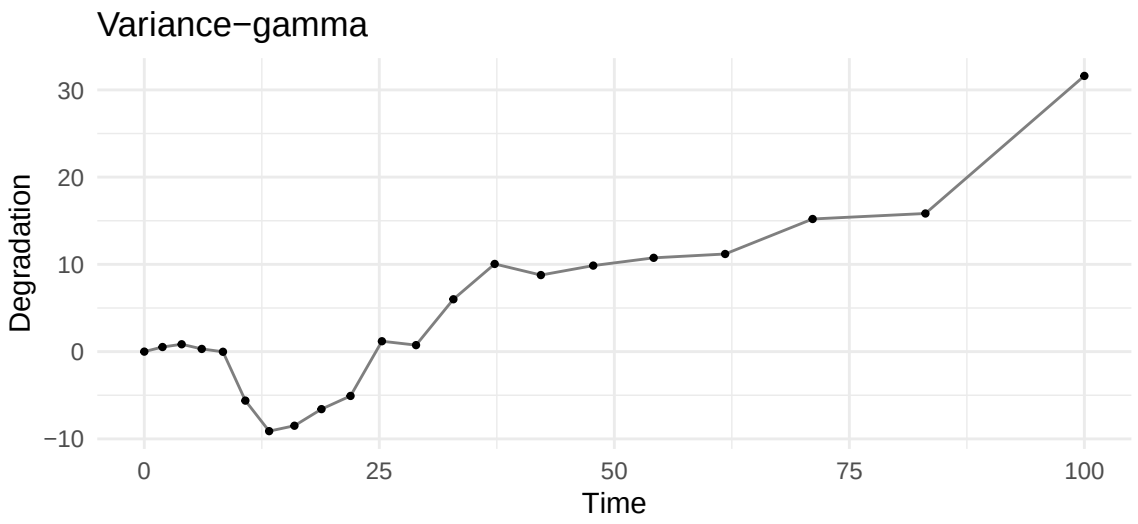
```
set.seed(34)
x <- sim_vg(t3, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t          X1
## [1,]  0.000000  0.00000000
## [2,]  1.940035  0.52692453
## [3,]  3.974743  0.84233589
## [4,]  6.113840  0.30912827
## [5,]  8.368618 -0.01932868
## [6,] 10.752309 -5.61595766
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.527  0.315 -0.533 -0.328 -5.597 -3.498  0.620  1.903  1.513  6.269
## [11] -0.446  5.257  4.048 -1.275  1.086  0.890  0.440  4.011  0.630 15.771
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

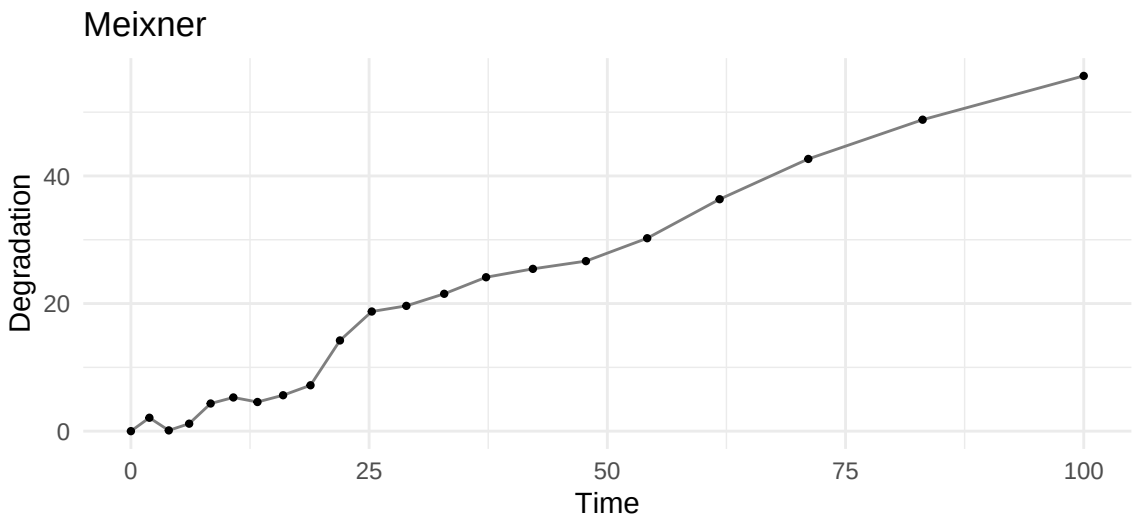
```
set.seed(35)
x <- sim_meixner(t3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  1.940035 2.0981428
## [3,]  3.974743 0.1163328
## [4,]  6.113840 1.1689796
## [5,]  8.368618 4.3318852
## [6,] 10.752309 5.2801270
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.098 -1.982  1.053  3.163  0.948 -0.713  1.061  1.563  7.025  4.535
## [11]  0.886  1.892  2.595  1.315  1.213  3.589  6.116  6.314  6.149  6.868
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

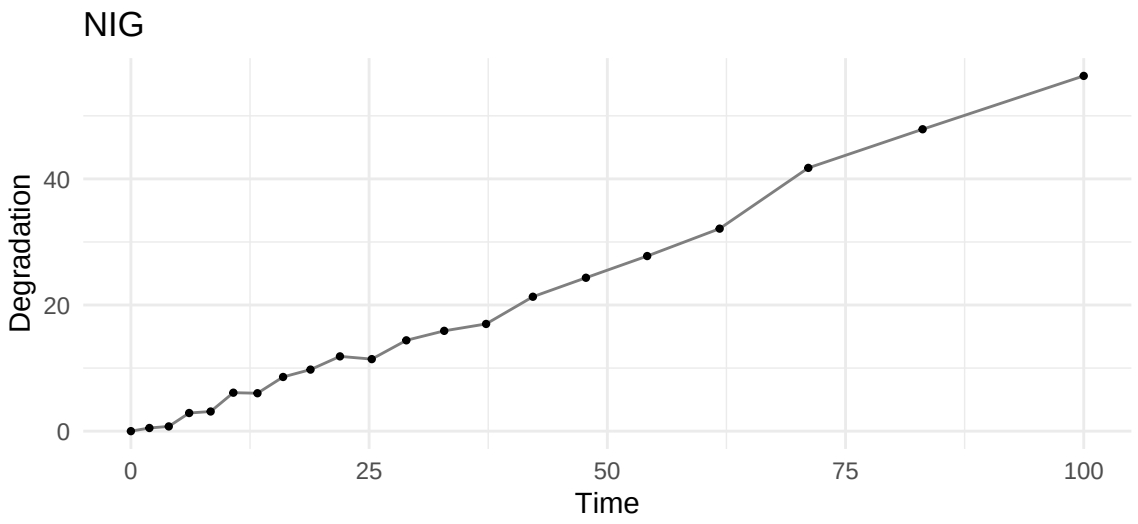
```
set.seed(36)
x <- sim_nig(t3, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  1.940035 0.4992997
## [3,]  3.974743 0.7488852
## [4,]  6.113840 2.8742631
## [5,]  8.368618 3.1134810
## [6,] 10.752309 6.0980221
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.499  0.250  2.125  0.239  2.985 -0.086  2.579  1.169  2.098 -0.438
## [11]  2.981  1.503  1.094  4.313  3.011  3.444  4.345  9.634  6.128  8.460
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

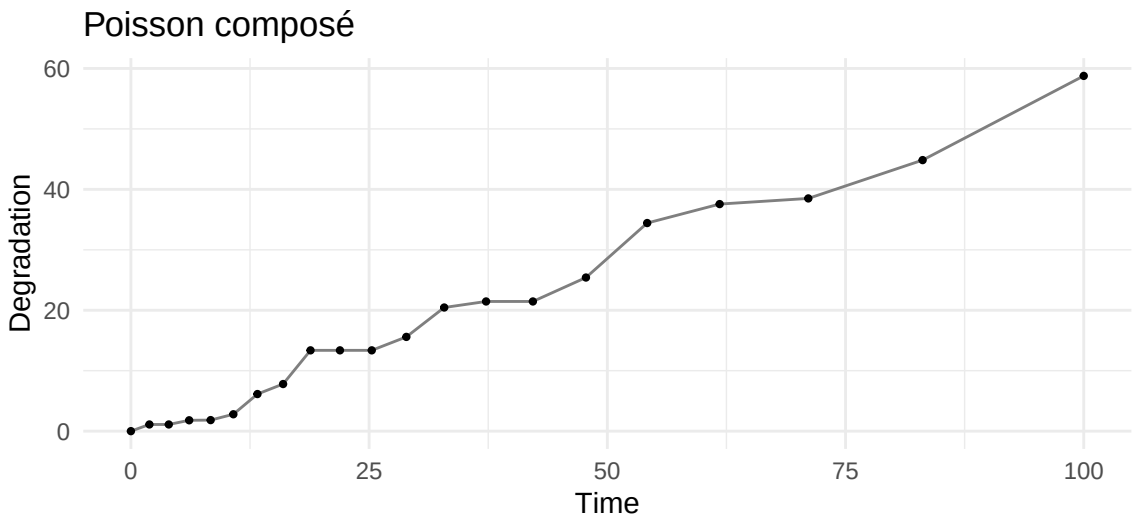
```
set.seed(37)
x <- sim_cpoisson(t3, lambda = 0.5)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  1.940035 1.104451
## [3,]  3.974743 1.104451
## [4,]  6.113840 1.796155
## [5,]  8.368618 1.827021
## [6,] 10.752309 2.792182
```

```
round(diff(x[, "X1"]), 3)
```

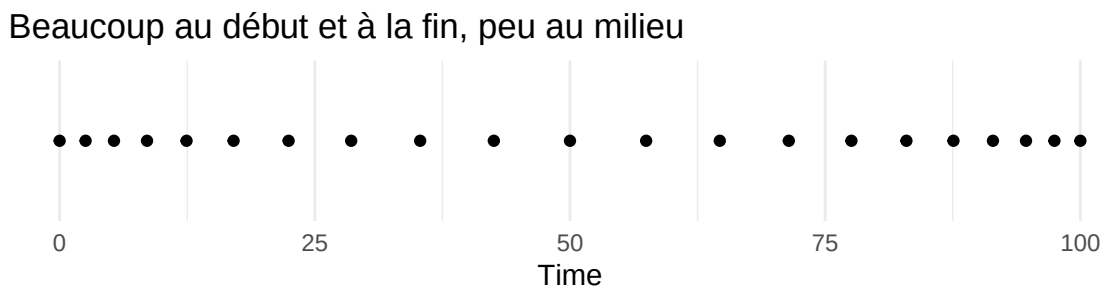
```
##  [1]  1.104  0.000  0.692  0.031  0.965  3.344  1.657  5.574  0.000  0.000
## [11]  2.224  4.862  0.999  0.000  3.960  9.014  3.136  0.937  6.341 13.930
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 4 : mixtes, beaucoup au début et à la fin

```
t4 <- inspection_times("inc-dec", n = 20, tmax = 100)$time
plot_plan(t4, "Beaucoup au début et à la fin, peu au milieu")
```



Modèle 1 : gamma

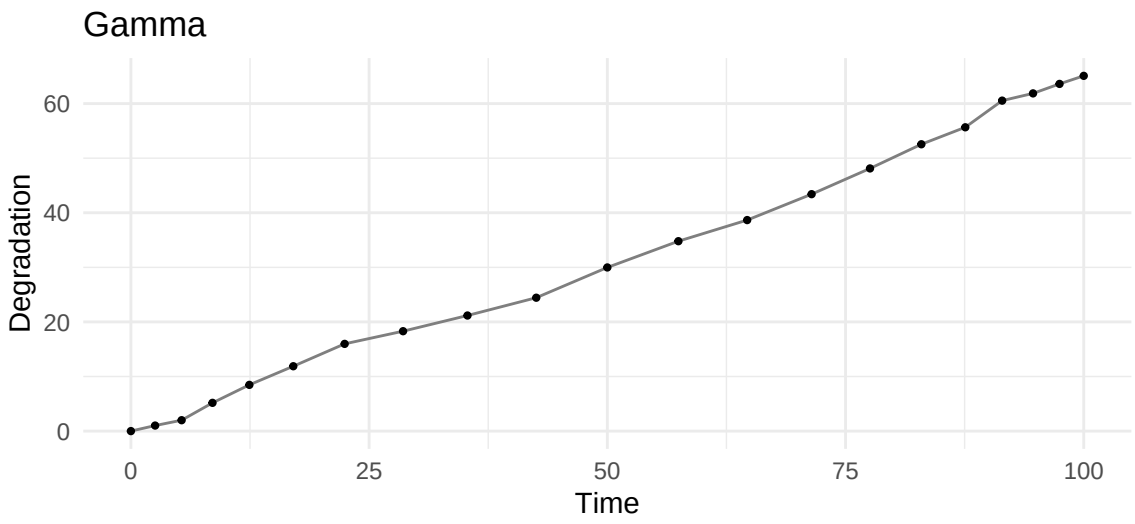
```
set.seed(41)
x <- sim_gamma(t4, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.540921  1.013078
## [3,]  5.322554  1.993097
## [4,]  8.562047  5.174992
## [5,] 12.431733  8.468824
## [6,] 17.042253 11.882103
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.013 0.980 3.182 3.294 3.413 4.097 2.322 2.872 3.263 5.544 4.811 3.869
## [13] 4.746 4.715 4.429 3.110 4.884 1.327 1.738 1.476
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

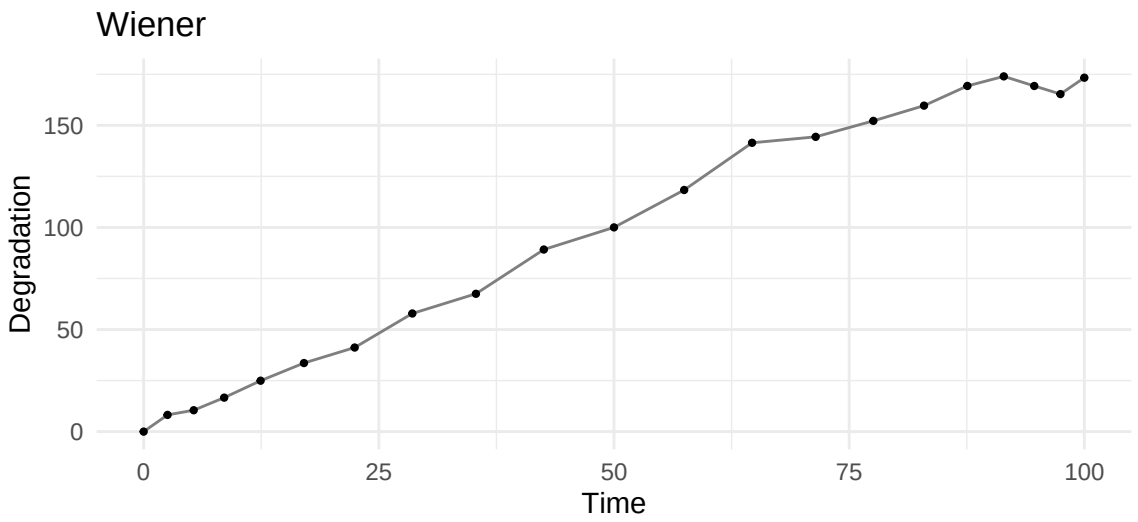
```
set.seed(42)
x <- sim_wiener(t4, mu = 1.5, sigma = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000  0.00000
## [2,]  2.540921  8.18207
## [3,]  5.322554 10.47089
## [4,]  8.562047 16.63729
## [5,] 12.431733 24.93169
## [6,] 17.042253 33.58357
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  8.182  2.289  6.166  8.294  8.652  7.591 16.680  9.649 21.673 10.846
## [11] 18.316 23.115  2.918  7.815  7.465  9.647  4.686 -4.703 -3.968  8.020
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

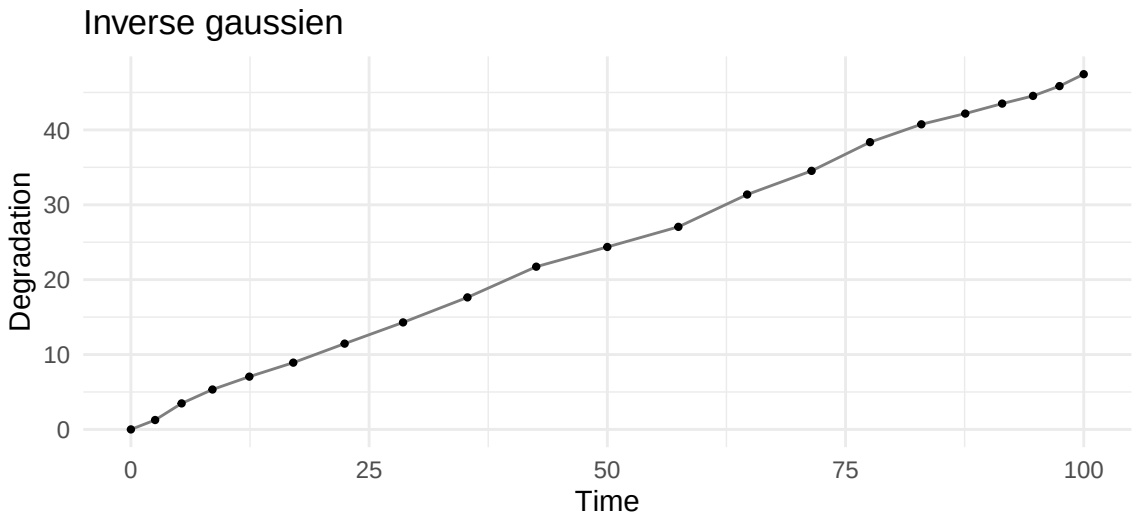
```
set.seed(43)
x <- sim_ig(t4, mu = 0.5, eta = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.540921 1.255599
## [3,]  5.322554 3.475956
## [4,]  8.562047 5.329629
## [5,] 12.431733 7.048824
## [6,] 17.042253 8.917167
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.256 2.220 1.854 1.719 1.868 2.538 2.835 3.341 4.100 2.635 2.685 4.320
## [13] 3.158 3.831 2.385 1.432 1.335 1.028 1.309 1.592
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

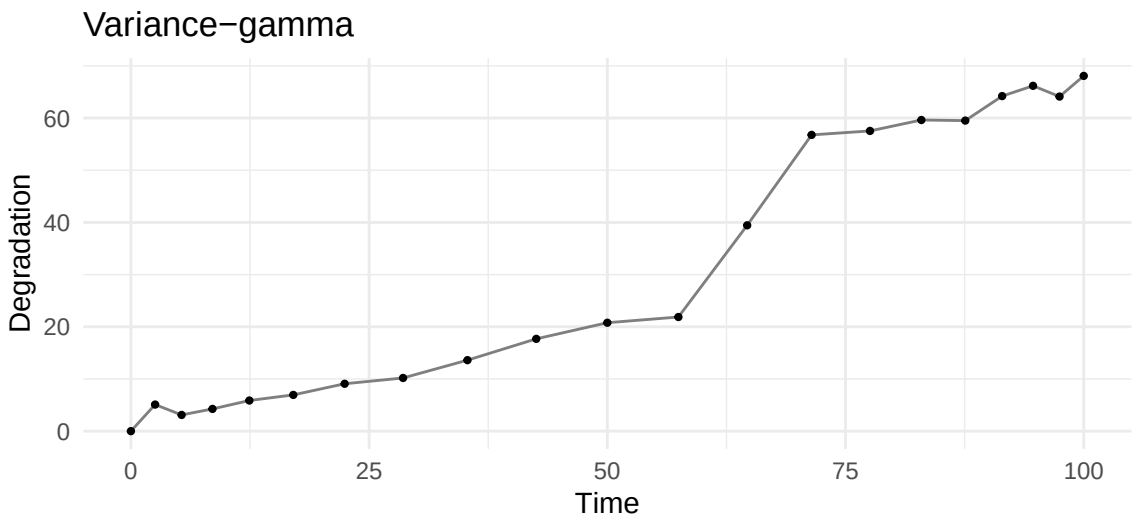
```
set.seed(44)
x <- sim_vg(t4, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.540921 5.095509
## [3,]  5.322554 3.104549
## [4,]  8.562047 4.258139
## [5,] 12.431733 5.869560
## [6,] 17.042253 6.941949
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.096 -1.991  1.154  1.611  1.072  2.137  1.112  3.414  4.080  3.092
## [11]  1.093 17.575 17.326  0.764  2.093 -0.115  4.696  1.969 -2.059  3.960
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

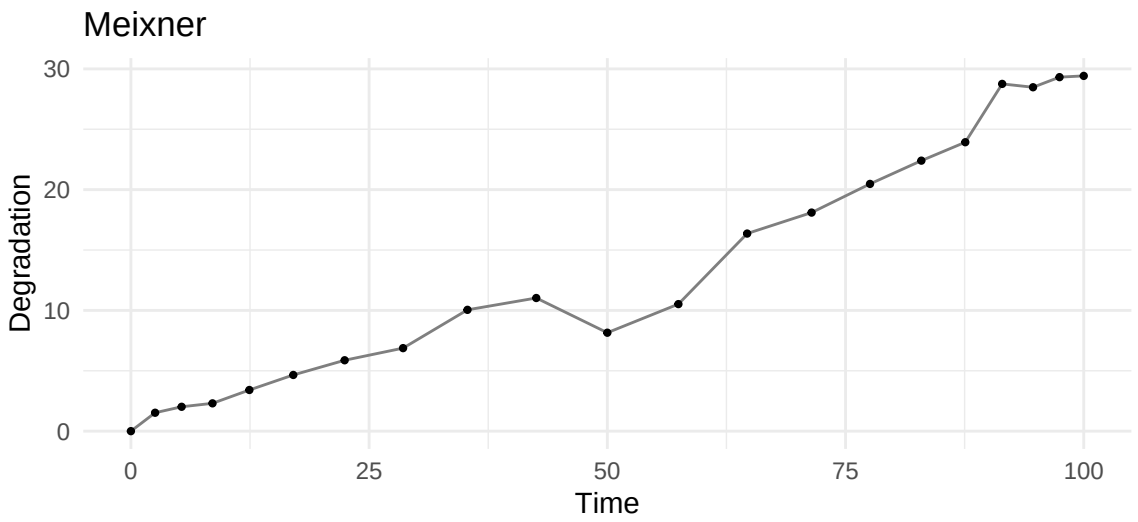
```
set.seed(45)
x <- sim_meixner(t4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.540921 1.523821
## [3,]  5.322554 2.017361
## [4,]  8.562047 2.301981
## [5,] 12.431733 3.408298
## [6,] 17.042253 4.652765
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.524  0.494  0.285  1.106  1.244  1.217  1.005  3.171  0.980 -2.874
## [11]  2.368  5.843  1.737  2.374  1.926  1.526  4.830 -0.278  0.837  0.103
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

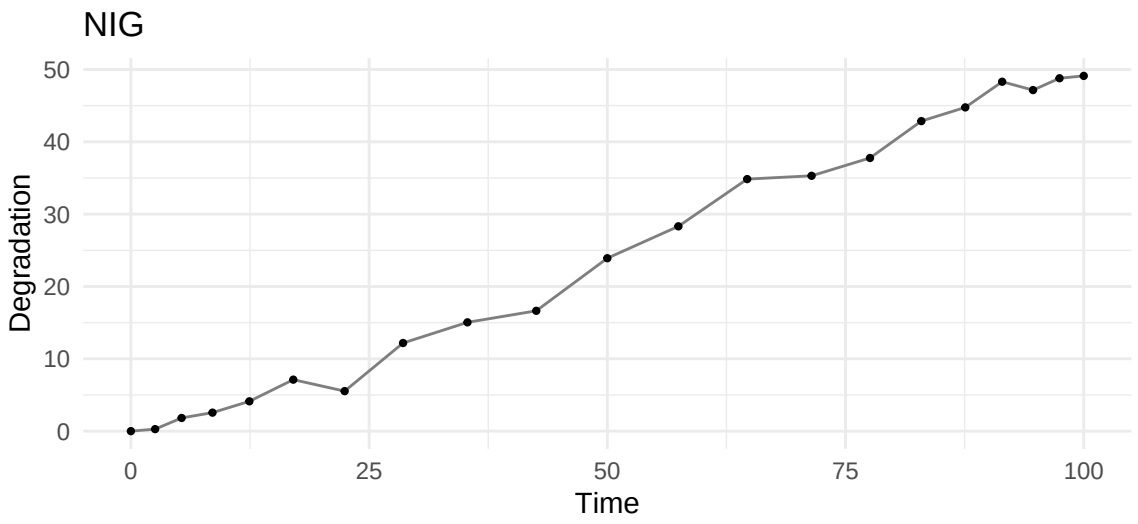
```
set.seed(46)
x <- sim_nig(t4, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  2.540921 0.2773168
## [3,]  5.322554 1.8228331
## [4,]  8.562047 2.5582928
## [5,] 12.431733 4.1226725
## [6,] 17.042253 7.1190832
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.277  1.546  0.735  1.564  2.996 -1.589  6.654  2.858  1.593  7.274
## [11]  4.408  6.528  0.456  2.463  5.101  1.895  3.545 -1.151  1.634  0.325
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

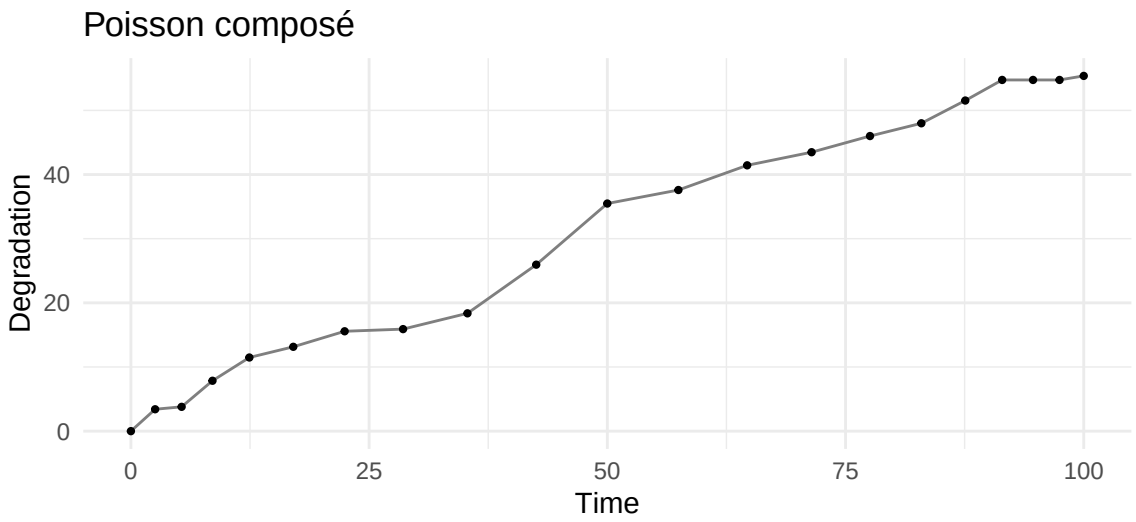
```
set.seed(47)
x <- sim_cpoisson(t4, lambda = 0.5)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.540921  3.410377
## [3,]  5.322554  3.788944
## [4,]  8.562047  7.855023
## [5,] 12.431733 11.480214
## [6,] 17.042253 13.146896
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.410 0.379 4.066 3.625 1.667 2.421 0.337 2.461 7.588 9.529 2.107 3.847
## [13] 2.046 2.519 1.991 3.545 3.210 0.000 0.000 0.631
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
20
40
0
25
50
75
100
Time

Cas 5 : mixtes, beaucoup au milieu

```
t5 <- inspection_times("dec-inc", n = 20, tmax = 100)$time
plot_plan(t5, "Peu au début et à la fin, beaucoup au milieu")
```



Modèle 1 : gamma

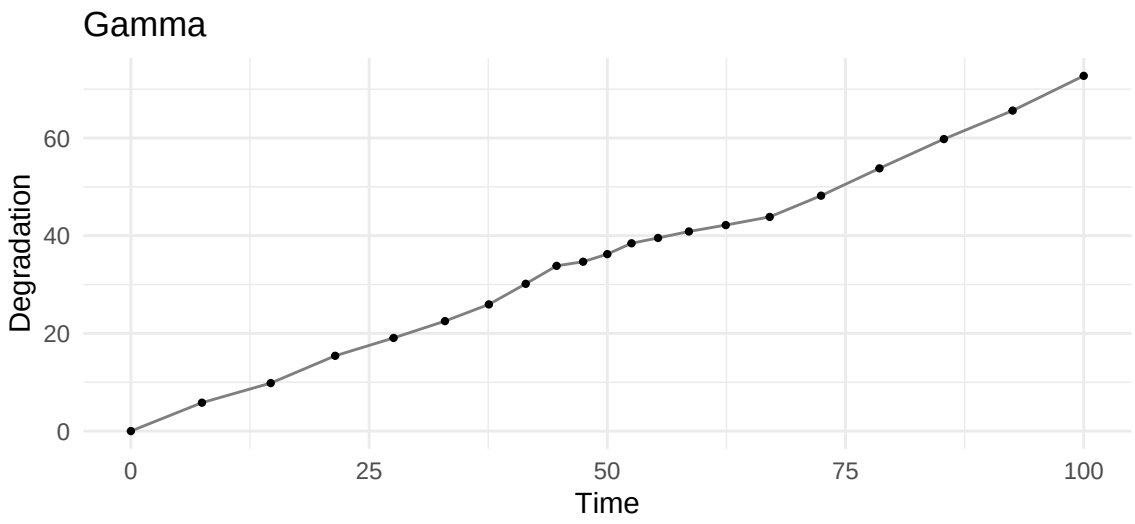
```
set.seed(51)
x <- sim_gamma(t5, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  5.814275
## [3,] 14.677446  9.824488
## [4,] 21.437953 15.412420
## [5,] 27.568267 19.065485
## [6,] 32.957747 22.511943
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 5.814 4.010 5.588 3.653 3.446 3.420 4.216 3.677 0.845 1.554 2.221 1.091
## [13] 1.322 1.314 1.673 4.344 5.600 5.990 5.831 7.110
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

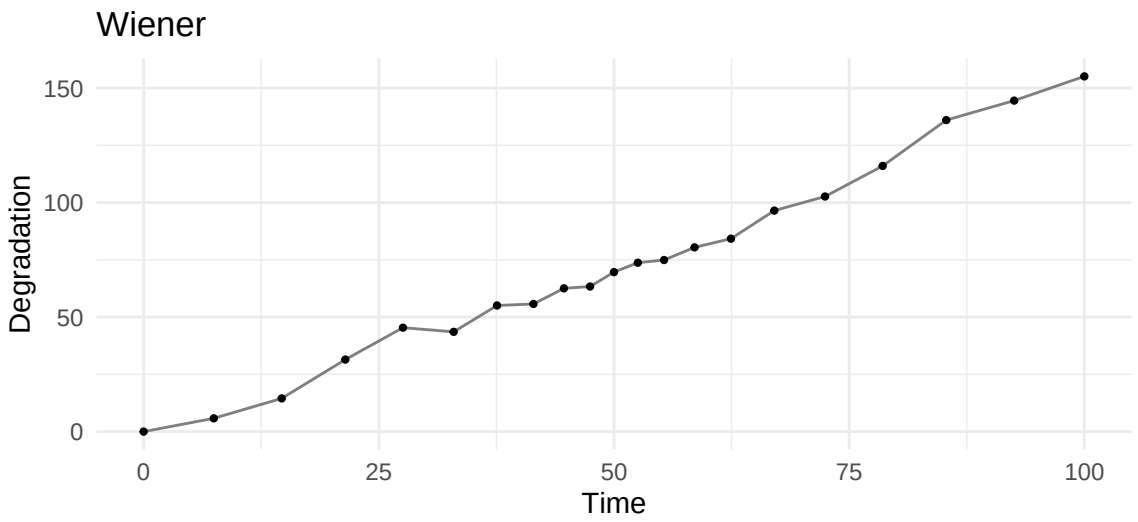
```
set.seed(52)
x <- sim_wiener(t5, mu = 1.5, sigma = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  5.801785
## [3,] 14.677446 14.507619
## [4,] 21.437953 31.452913
## [5,] 27.568267 45.376256
## [6,] 32.957747 43.556482
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.802  8.706 16.945 13.923 -1.820 11.506  0.661  6.829  0.794  6.298
## [11]  4.099  1.176  5.517  3.794 12.261  6.155 13.353 20.002  8.512 10.582
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

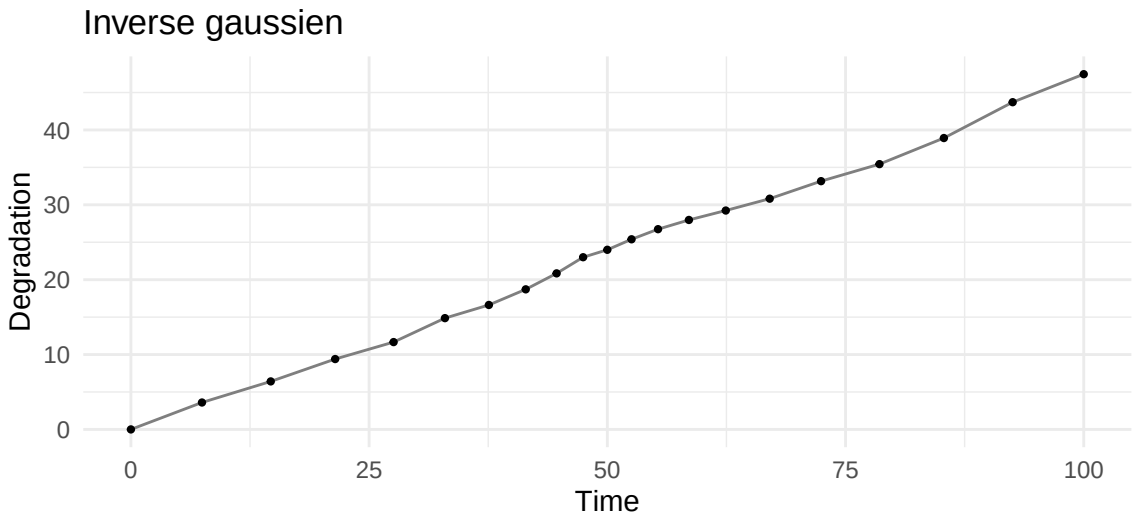
```
set.seed(53)
x <- sim_ig(t5, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  3.595579
## [3,] 14.677446  6.412297
## [4,] 21.437953  9.399752
## [5,] 27.568267 11.663964
## [6,] 32.957747 14.863754
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.596 2.817 2.987 2.264 3.200 1.759 2.088 2.132 2.158 0.992 1.403 1.350
## [13] 1.232 1.260 1.581 2.342 2.275 3.485 4.787 3.742
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

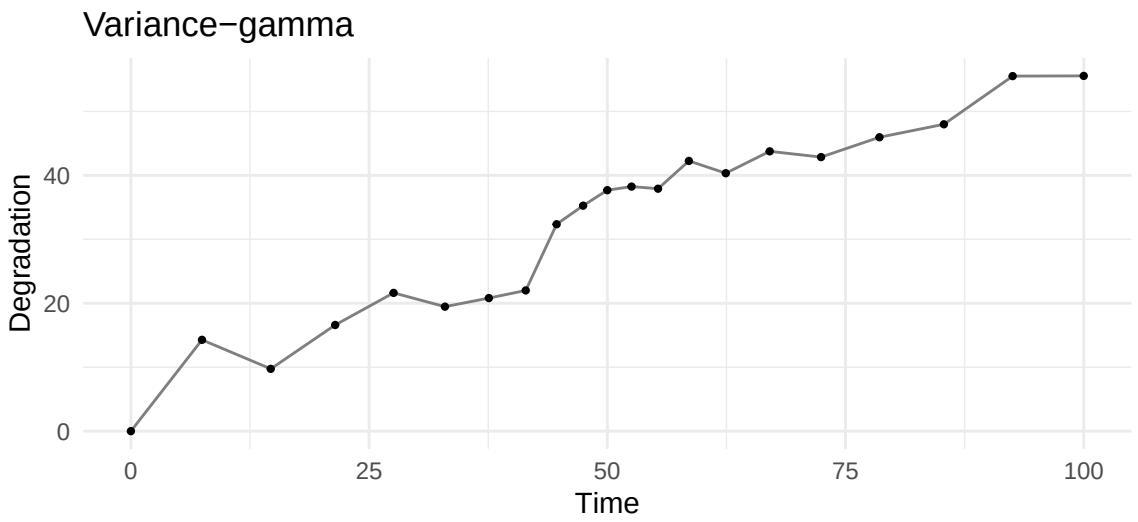
```
set.seed(54)
x <- sim_vg(t5, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079 14.277776
## [3,] 14.677446  9.742506
## [4,] 21.437953 16.599526
## [5,] 27.568267 21.619767
## [6,] 32.957747 19.463350
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 14.278 -4.535  6.857  5.020 -2.156  1.346  1.193 10.348  2.910  2.420
## [11]  0.563 -0.331  4.342 -1.921  3.421 -0.900  3.103  2.027  7.536  0.031
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

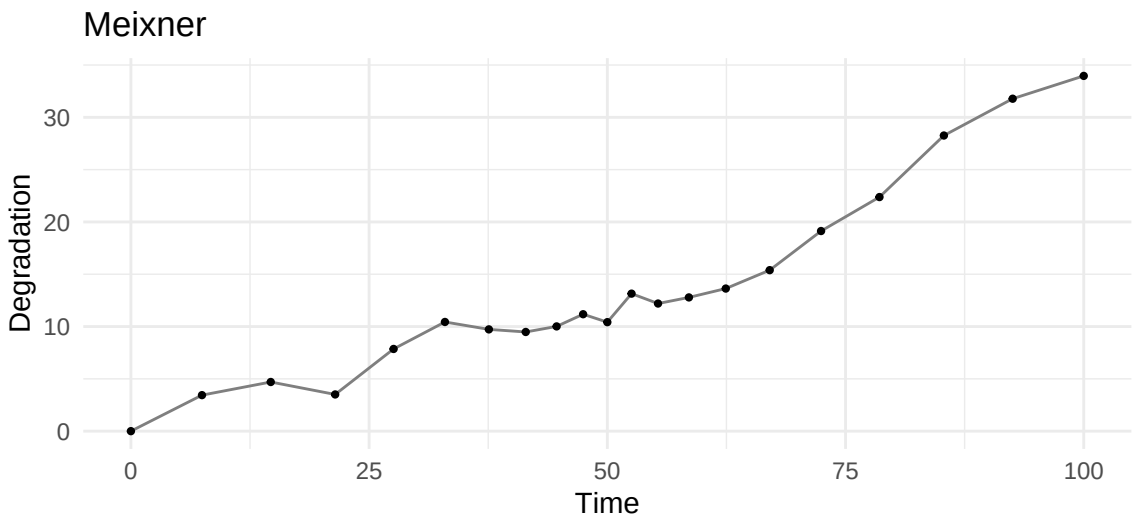
```
set.seed(55)
x <- sim_meixner(t5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  3.442434
## [3,] 14.677446  4.697757
## [4,] 21.437953  3.507490
## [5,] 27.568267  7.857074
## [6,] 32.957747 10.436038
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.442  1.255 -1.190  4.350  2.579 -0.708 -0.247  0.531  1.165 -0.758
## [11]  2.727 -0.946  0.586  0.846  1.759  3.740  3.247  5.884  3.518  2.182
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

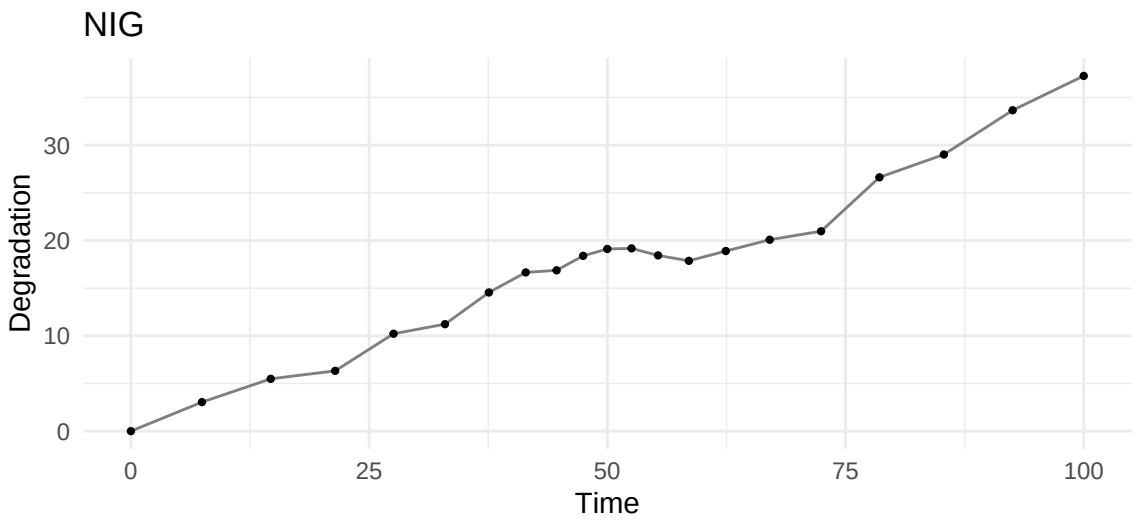
```
set.seed(56)
x <- sim_nig(t5, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  3.047164
## [3,] 14.677446  5.487323
## [4,] 21.437953  6.322425
## [5,] 27.568267 10.219941
## [6,] 32.957747 11.218525
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.047  2.440  0.835  3.898  0.999  3.328  2.105  0.220  1.519  0.720
## [11]  0.061 -0.735 -0.571  1.027  1.187  0.890  5.661  2.391  4.643  3.604
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

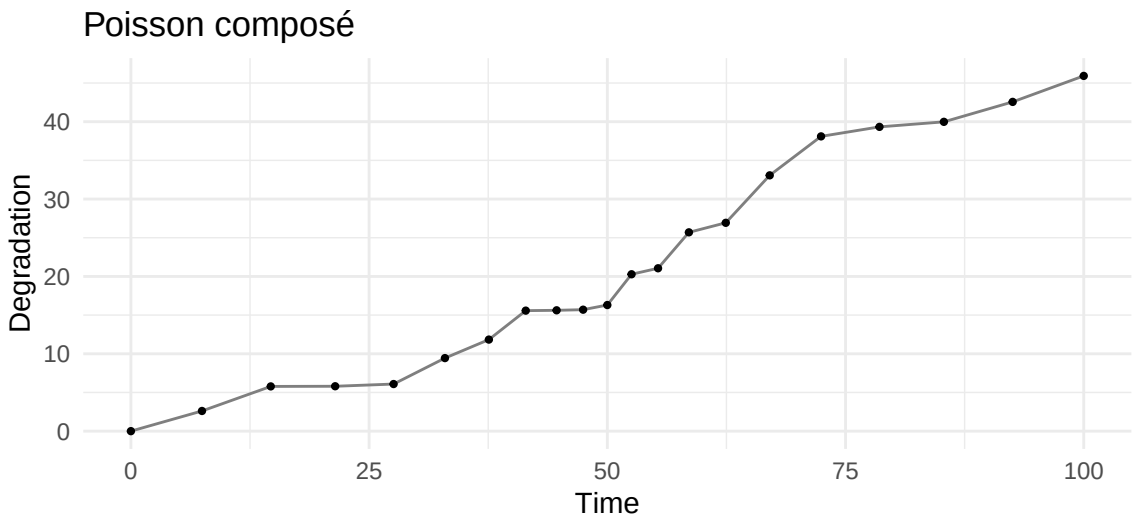
```
set.seed(57)
x <- sim_cpoisson(t5, lambda = 0.5)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  7.459079 2.606789
## [3,] 14.677446 5.781682
## [4,] 21.437953 5.802607
## [5,] 27.568267 6.088267
## [6,] 32.957747 9.442441
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.607 3.175 0.021 0.286 3.354 2.392 3.743 0.043 0.080 0.602 3.977 0.774
## [13] 4.646 1.231 6.136 5.036 1.230 0.651 2.577 3.361
```

```
plot_traj(x, "Poisson composé")
```

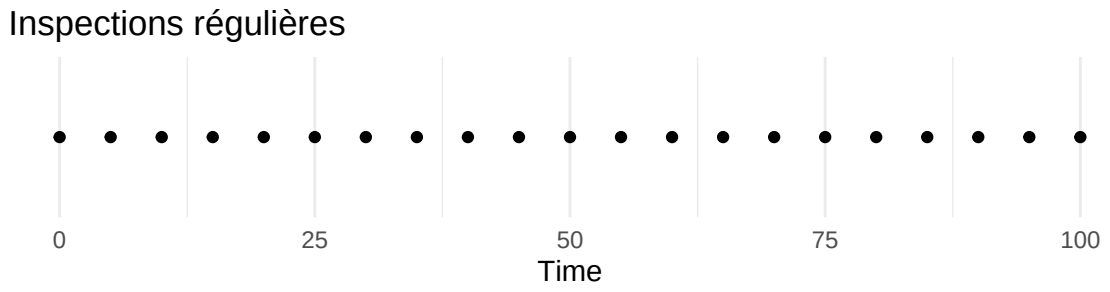

Poisson composé
Degradation
0
10
20
30
40
0
25
50
75
100
Time

II - Plusieurs trajectoires

5 trajectoires, chacune inspectée 20 fois entre 0 et 100, aux mêmes instants. `x` est une matrice : une colonne `t`, puis `X1`, ..., `X5`. On affiche les premières lignes des trois premières trajectoires.

Cas 1 : inspections régulières

```
p1 <- inspection_times("regular", n = 20, n_traj = 5, tmax = 100)
plot_plan(p1$time[p1$id == 1], "Inspections régulières")
```



Modèle 1 : gamma

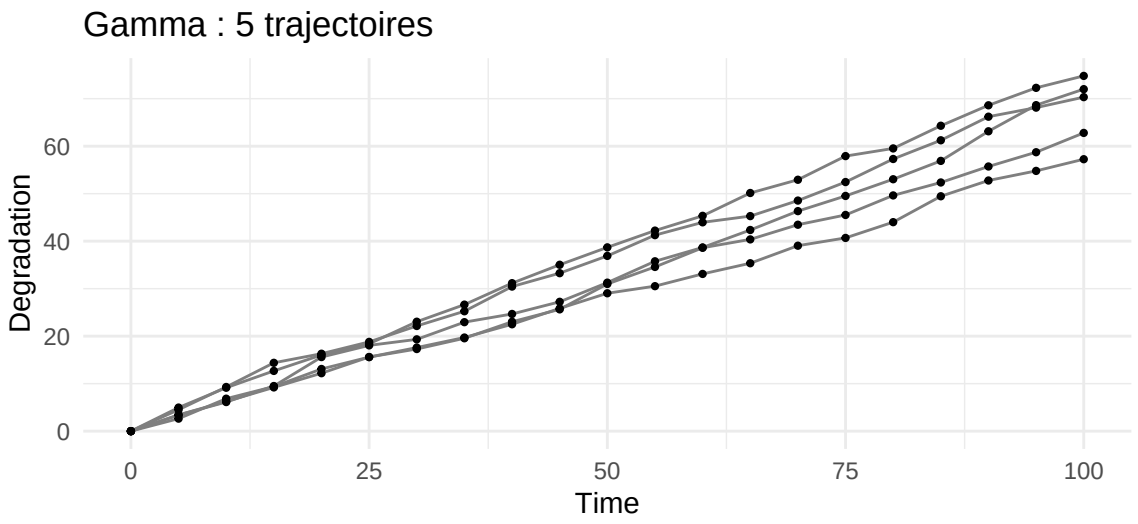
```
set.seed(111)
x <- sim_gamma(p1, a = 2, b = 3)
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  3.412944  3.391215  4.962620
## [3,] 10  6.248927  6.129922  9.138235
## [4,] 15  9.207337  9.496121 12.684072
## [5,] 20 12.200879 15.598724 16.033187
## [6,] 25 15.606267 18.057525 18.389191
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.413 2.836 2.958 2.994 3.405 1.684 2.267 3.473 2.611 5.347 3.584 4.039
## [13] 1.762 3.088 2.059 4.112 2.727 3.365 3.005 4.065
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

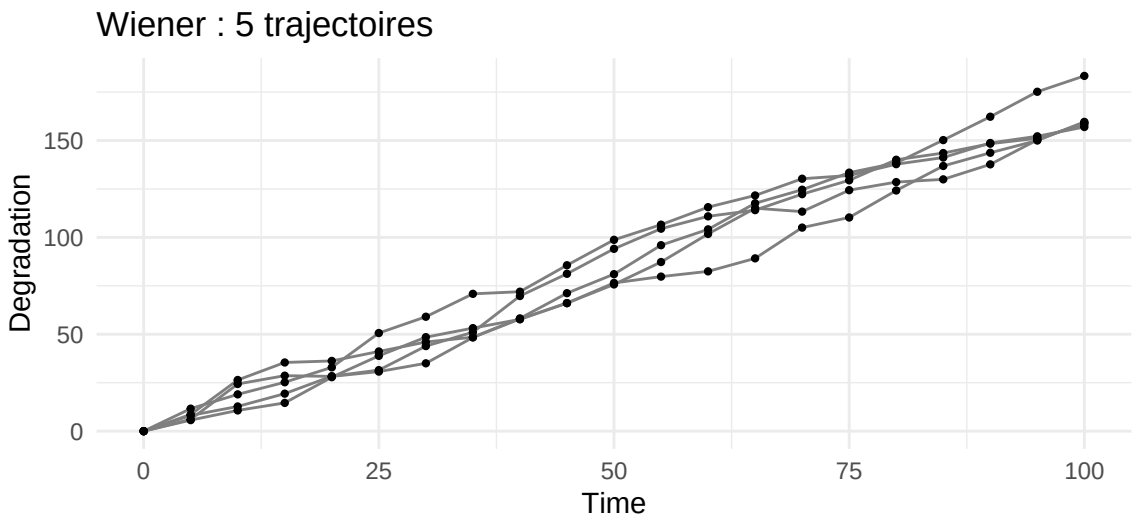
```
set.seed(112)
x <- sim_wiener(p1, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##       t        X1        X2       X3
## [1,]  0  0.000000  0.000000  0.00000
## [2,]  5  6.094713  8.613025 11.60806
## [3,] 10 24.342933 26.400095 18.97625
## [4,] 15 28.630759 35.438839 25.23648
## [5,] 20 28.257068 36.240624 32.90835
## [6,] 25 30.724657 41.112867 50.63640
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.095 18.248  4.288 -0.374  2.468  4.282 13.359  9.521  8.181 10.422
## [11]  3.263  2.710  6.713 15.872  5.218 13.935 12.657  6.808  6.338  9.608
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

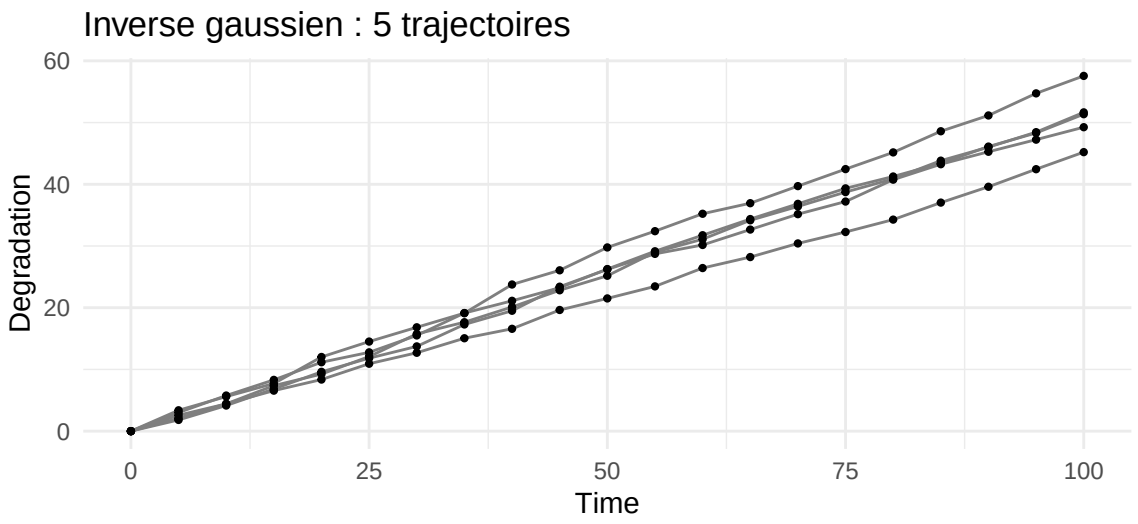
```
set.seed(113)
x <- sim_ig(p1, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  2.575667  1.816995  3.380367
## [3,] 10  4.416063  4.148258  5.600797
## [4,] 15  7.371107  6.834152  7.721204
## [5,] 20  9.244923  9.608309 12.018158
## [6,] 25 12.077408 11.820806 14.513946
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.576 1.840 2.955 1.874 2.832 3.674 1.901 2.465 2.708 2.349 3.822 2.106
## [13] 3.062 2.242 2.329 2.141 2.951 2.186 2.432 3.209
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

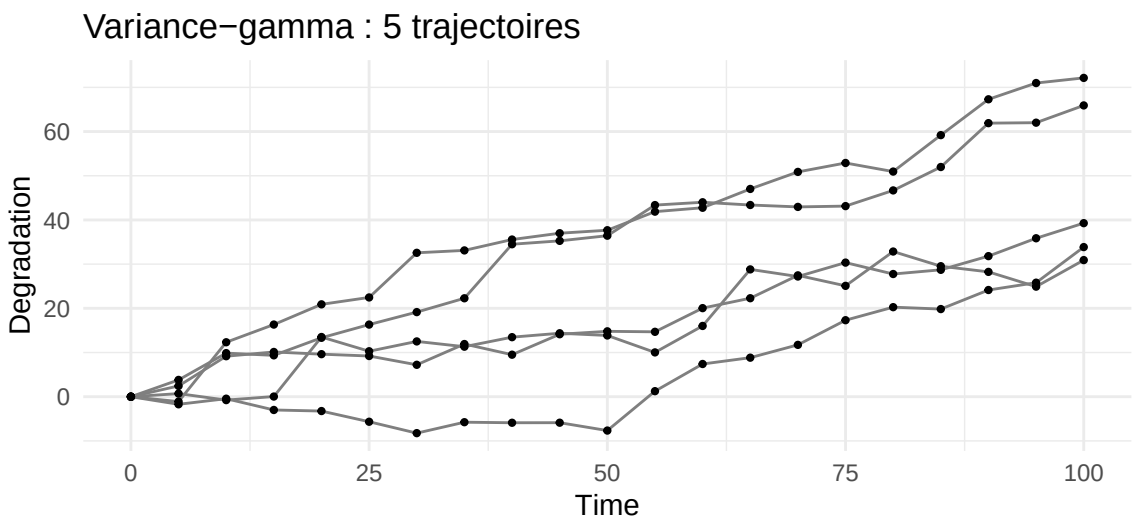
```
set.seed(114)
x <- sim_vg(p1, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##       t        X1          X2        X3
## [1,]  0  0.000000  0.00000000  0.000000
## [2,]  5  2.436627  0.70296417 -1.107265
## [3,] 10  9.162189 -0.77011502 12.323899
## [4,] 15 10.122663  0.03664116 16.329080
## [5,] 20  9.624831 13.50917123 20.901106
## [6,] 25  9.211656 10.28751843 22.462301
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.437  6.726  0.960 -0.498 -0.413 -1.990  4.699 -2.413  4.626  0.661
## [11] -0.109  5.361  2.242  5.178 -2.377  7.766 -3.320 -1.284 -3.340  6.006
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

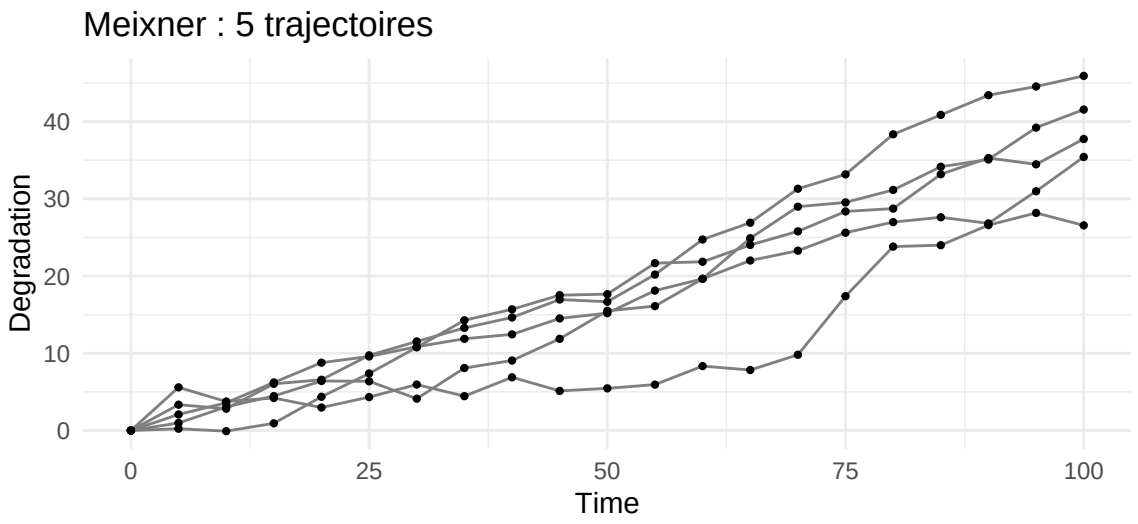
```
set.seed(115)
x <- sim_meixner(p1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##       t       X1       X2          X3
## [1,]  0 0.000000 0.000000  0.00000000
## [2,]  5 3.339643 2.078708  0.22800433
## [3,] 10 2.823348 3.582343 -0.08350747
## [4,] 15 6.044890 6.217424  0.93099359
## [5,] 20 6.571270 8.782078  4.36570414
## [6,] 25 9.730140 9.586900  7.38028101
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.340 -0.516  3.222  0.526  3.159  1.800  1.755  1.354  2.323 -0.290
## [11]  3.520  4.541  2.174  4.399  1.858  5.186  2.518  2.551  1.127  1.372
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

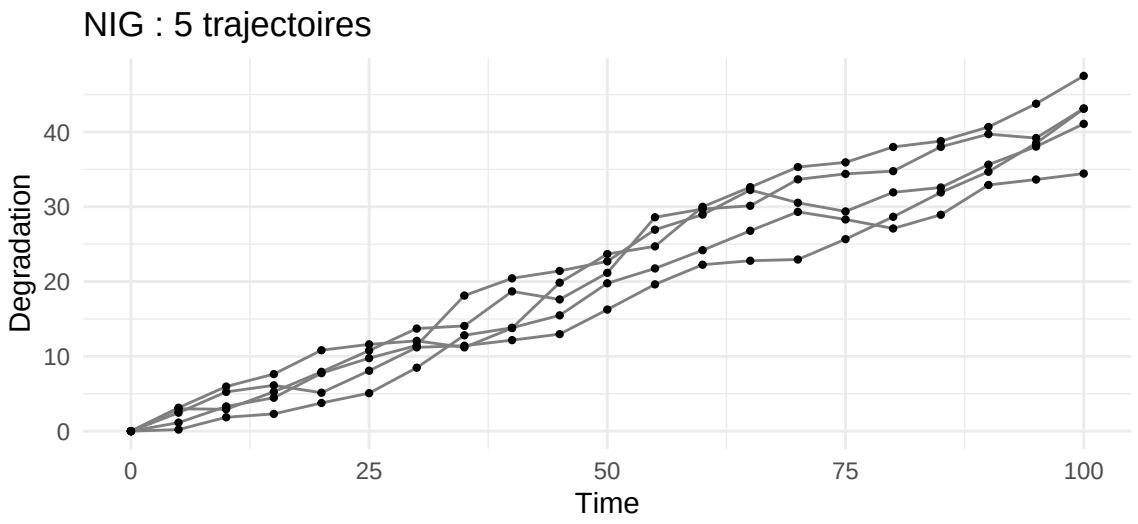
```
set.seed(116)
x <- sim_nig(p1, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##       t       X1        X2        X3
## [1,]  0 0.000000  0.000000 0.0000000
## [2,]  5 2.468533  2.990295 0.2074206
## [3,] 10 5.246575  2.923195 1.8555426
## [4,] 15 6.122326  5.263634 2.3019116
## [5,] 20 5.136418  7.929152 3.7644948
## [6,] 25 8.077892 10.777178 5.0749732
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.469  2.778  0.876 -0.986  2.941  3.123  0.221  0.743  0.804  3.282
## [11]  3.372  2.628  0.529  0.166  2.722  2.988  3.257  2.767  3.812  4.617
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

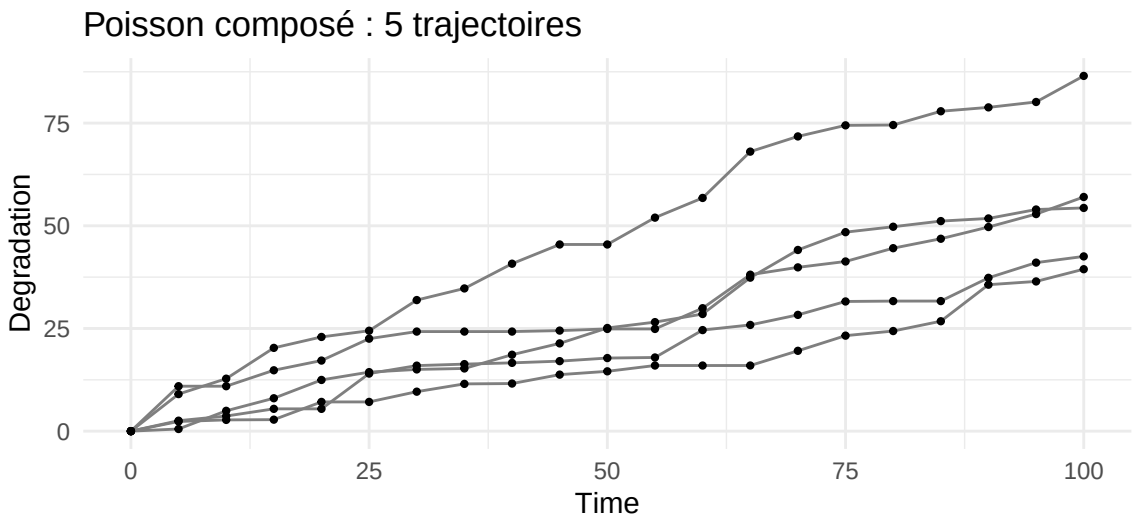
```
set.seed(117)
x <- sim_cpoisson(p1, lambda = 0.5)
head(x[, 1:4])
```

```
##       t       X1       X2        X3
## [1,]  0  0.00000  0.00000  0.000000
## [2,]  5 10.93951  9.02175  2.547582
## [3,] 10 10.93951 12.77248  3.677322
## [4,] 15 14.81978 20.27313  5.428941
## [5,] 20 17.18333 22.93611  5.428941
## [6,] 25 22.51752 24.44392 14.017366
```

```
round(diff(x[, "X1"]), 3)
```

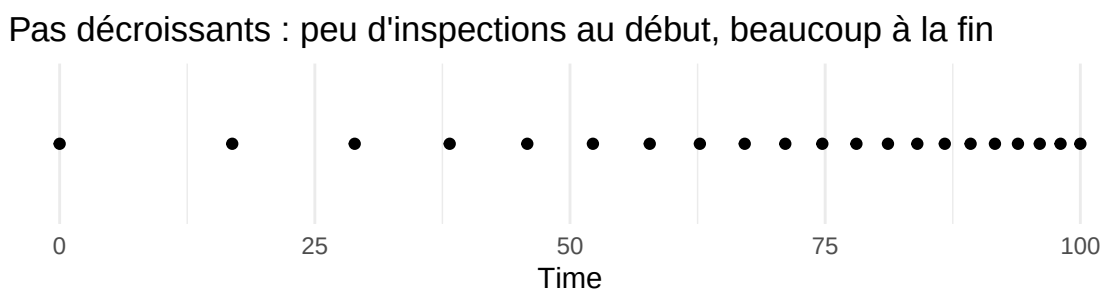
```
##  [1] 10.940  0.000  3.880  2.364  5.334  1.725  0.000  0.000  0.238  0.407
## [11]  0.000  5.049  8.156  1.785  1.432  3.239  2.295  2.852  3.165  4.153
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
0
25
50
75
0
25
50
75
100
Time

Cas 2 : pas décroissants

```
p2 <- inspection_times("decreasing", n = 20, n_traj = 5, tmax = 100)
plot_plan(p2$time[p2$id == 1], "Pas décroissants : peu d'inspections au début, beaucoup à la fin")
```



Modèle 1 : gamma

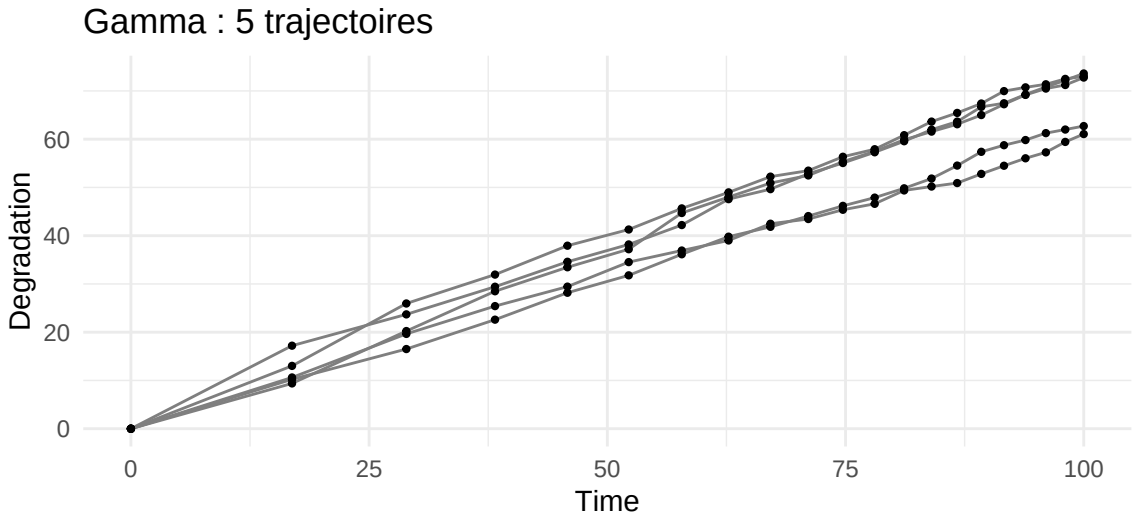
```
set.seed(121)
x <- sim_gamma(p2, a = 2, b = 3)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.000000
## [2,] 16.90921 10.62024 17.20485  9.386244
## [3,] 28.90648 19.65015 23.69326 20.236937
## [4,] 38.21229 25.40798 29.41898 28.486875
## [5,] 45.81569 29.44038 34.59453 33.456933
## [6,] 52.24427 34.52621 38.18268 37.192447
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.620  9.030  5.758  4.032  5.086  2.398  2.089  3.466  0.972  1.918
## [11]  1.235  2.772  0.809  0.720  1.897  1.689  1.545  1.233  2.161  1.626
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

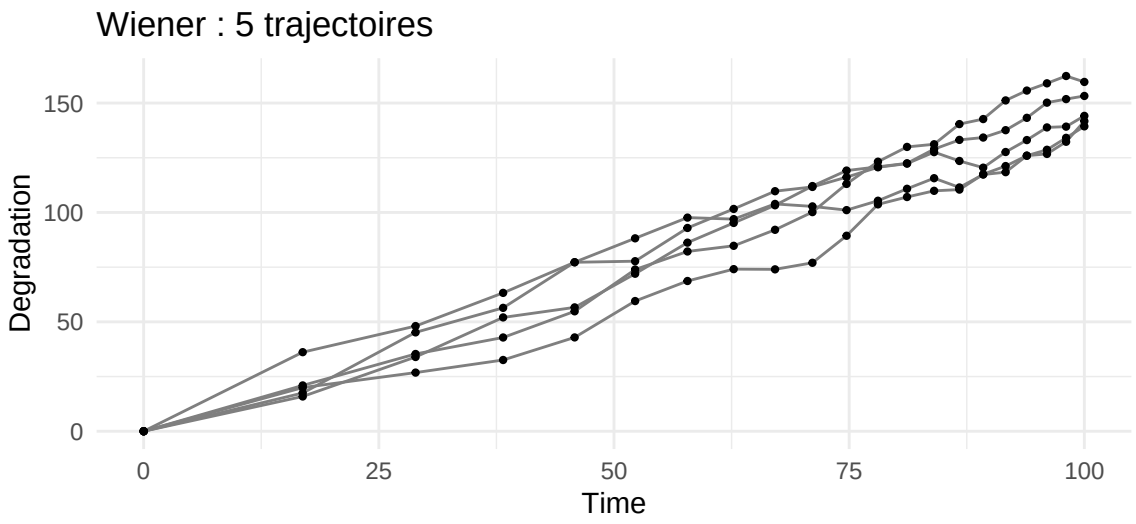
```
set.seed(122)
x <- sim_wiener(p2, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 16.90921 36.14323 20.94065 19.95223
## [3,] 28.90648 48.07174 35.31596 26.80538
## [4,] 38.21229 63.24777 42.85070 32.56368
## [5,] 45.81569 77.22253 54.81566 42.88415
## [6,] 52.24427 77.72730 73.90821 59.50291
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 36.143 11.929 15.176 13.975  0.505 15.191  8.693  8.108  1.947  4.464
## [11]  4.524  1.677  5.246 -4.043 -3.056  7.155  5.454  5.783  0.361  4.913
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

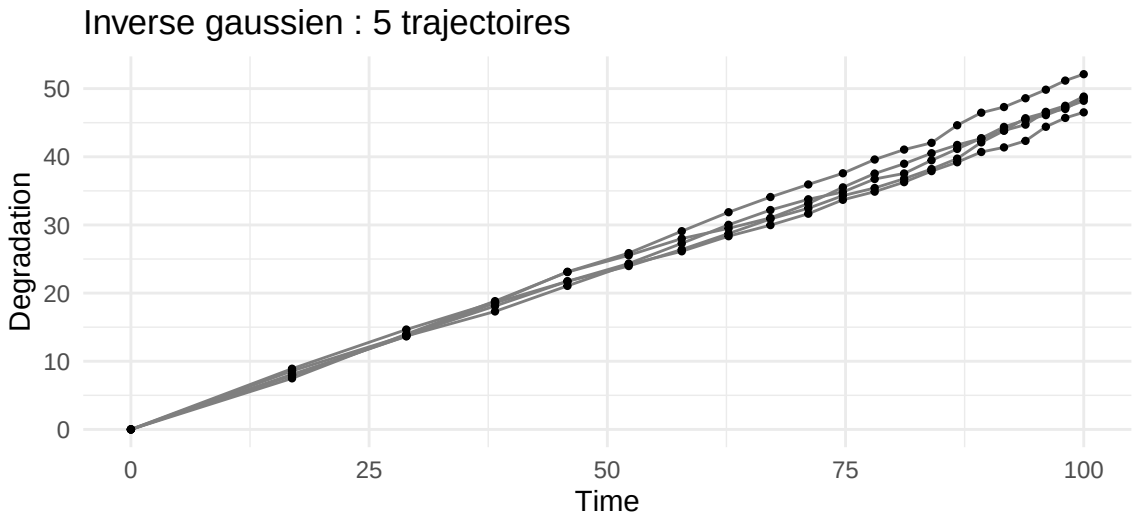
```
set.seed(123)
x <- sim_ig(p2, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 16.90921  7.897723  8.904532  8.073231
## [3,] 28.90648 13.700327 14.653043 13.652377
## [4,] 38.21229 17.306694 18.671536 18.566432
## [5,] 45.81569 21.060098 23.128694 21.727826
## [6,] 52.24427 24.193476 25.862694 24.330757
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 7.898 5.803 3.606 3.753 3.133 1.940 2.214 1.627 1.673 2.039 1.196 1.395
## [13] 1.619 1.301 1.505 0.673 0.955 2.070 1.300 0.819
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

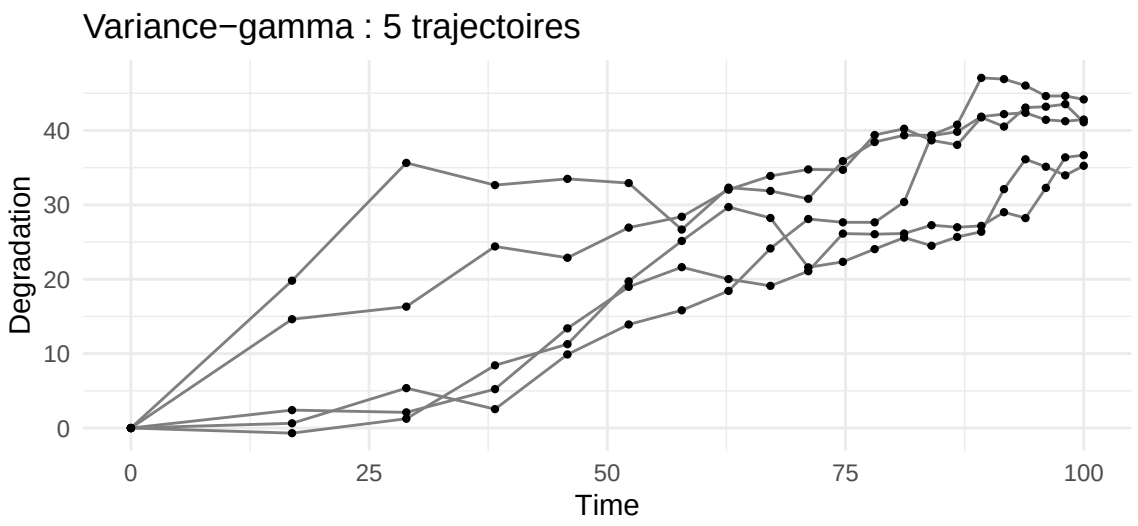
```
set.seed(124)
x <- sim_vg(p2, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##             t        X1        X2       X3
## [1,]  0.00000  0.000000  0.000000  0.00000
## [2,] 16.90921  0.637251  2.409797 19.80395
## [3,] 28.90648  5.363706  2.105576 35.63543
## [4,] 38.21229  2.540566  5.230256 32.65205
## [5,] 45.81569  9.891368 13.399796 33.49435
## [6,] 52.24427 13.917200 18.976941 32.92566
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.637  4.726 -2.823  7.351  4.026  1.908  2.606  5.707  3.958 -0.445
## [11] -0.007  2.737  8.907  0.533  2.031  0.344  0.170 -0.934 -0.195  0.226
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

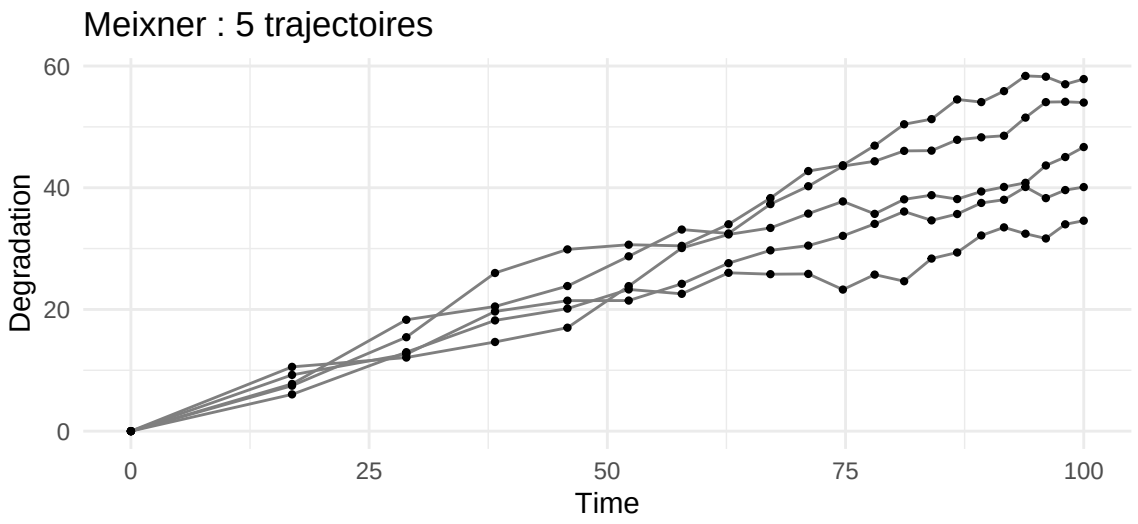
```
set.seed(125)
x <- sim_meixner(p2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##             t       X1        X2       X3
## [1,]  0.00000  0.00000  0.000000  0.00000
## [2,] 16.90921 10.56185  6.039943  9.24308
## [3,] 28.90648 12.09210 12.976338 12.64901
## [4,] 38.21229 14.65336 18.195877 19.65971
## [5,] 45.81569 16.99743 20.136227 21.44034
## [6,] 52.24427 23.83157 23.285497 21.45917
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.562  1.530  2.561  2.344  6.834  6.259  2.230  1.072  2.353  2.001
## [11] -2.044  2.387  0.686 -0.641  1.247  0.744  0.680  2.850  1.385  1.643
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

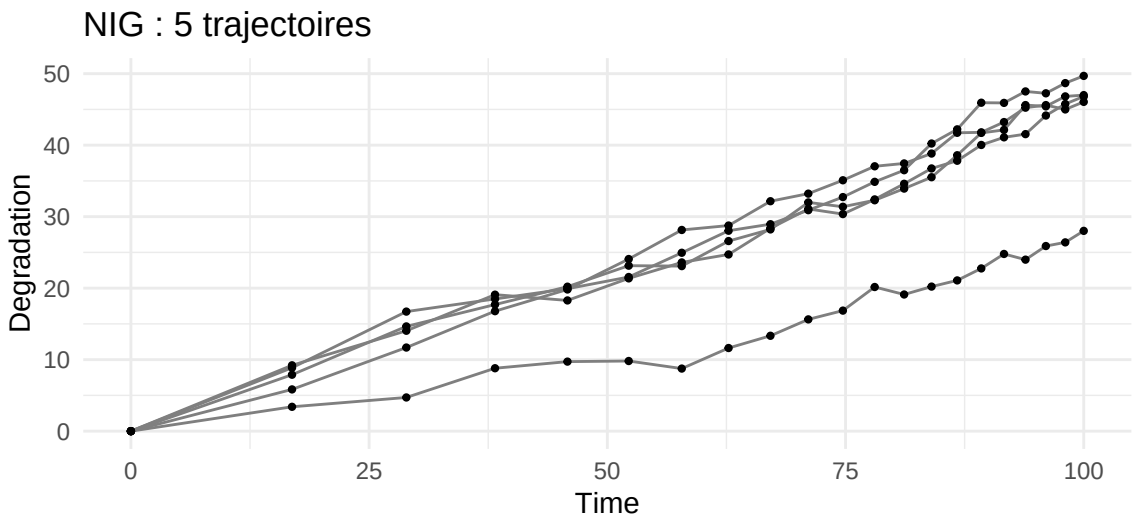
```
set.seed(126)
x <- sim_nig(p2, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##             t       X1        X2        X3
## [1,]  0.00000 0.000000  0.000000  0.000000
## [2,] 16.90921 3.403669  9.210465  8.863915
## [3,] 28.90648 4.711353 14.040403 16.720638
## [4,] 38.21229 8.797905 19.090496 18.494623
## [5,] 45.81569 9.725299 18.282901 19.921991
## [6,] 52.24427 9.809557 21.367375 21.559535
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.404  1.308  4.087  0.927  0.084 -1.058  2.875  1.712  2.294  1.226
## [11]  3.301 -1.046  1.104  0.866  1.675  2.024 -0.795  1.903  0.514  1.608
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

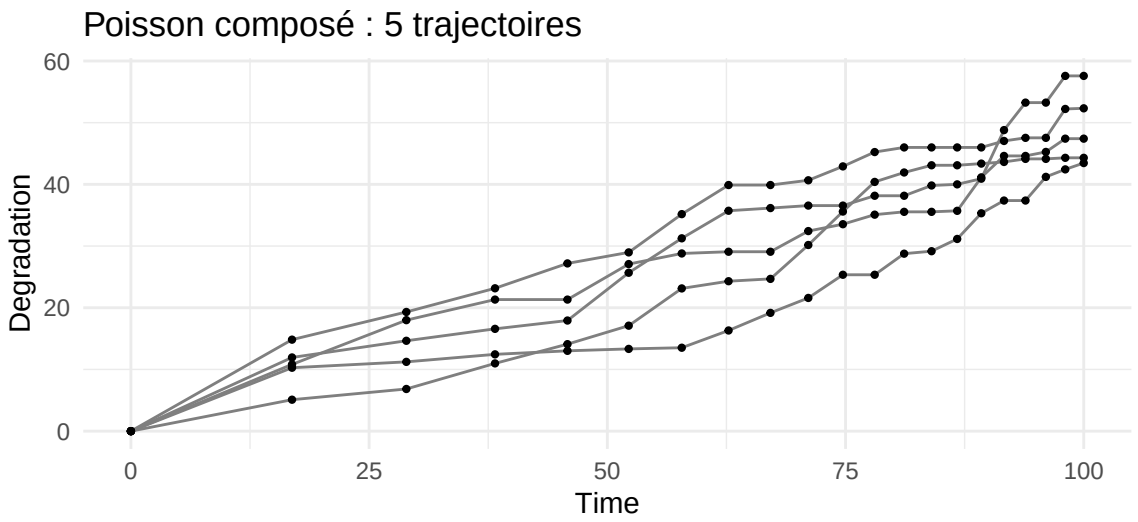
```
set.seed(127)
x <- sim_cpoisson(p2, lambda = 0.5)
head(x[, 1:4])
```

```
##             t        X1       X2       X3
## [1,]  0.00000  0.000000  0.00000  0.00000
## [2,] 16.90921  5.094637 10.27714 14.82955
## [3,] 28.90648  6.832984 11.22789 19.31514
## [4,] 38.21229 10.979469 12.44772 23.15368
## [5,] 45.81569 14.098460 13.01757 27.18597
## [6,] 52.24427 17.094252 13.32985 28.97341
```

```
round(diff(x[, "X1"]), 3)
```

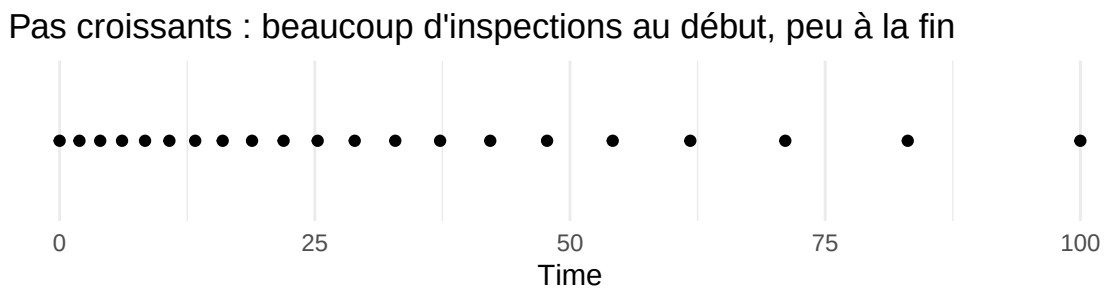
```
##  [1] 5.095 1.738 4.146 3.119 2.996 6.038 1.170 0.377 5.492 5.420 4.804 1.528
## [13] 1.169 0.000 0.274 0.266 0.499 0.000 0.170 0.000
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 3 : pas croissants

```
p3 <- inspection_times("increasing", n = 20, n_traj = 5, tmax = 100)
plot_plan(p3$time[p3$id == 1], "Pas croissants : beaucoup d'inspections au début, peu à la fin")
```



Modèle 1 : gamma

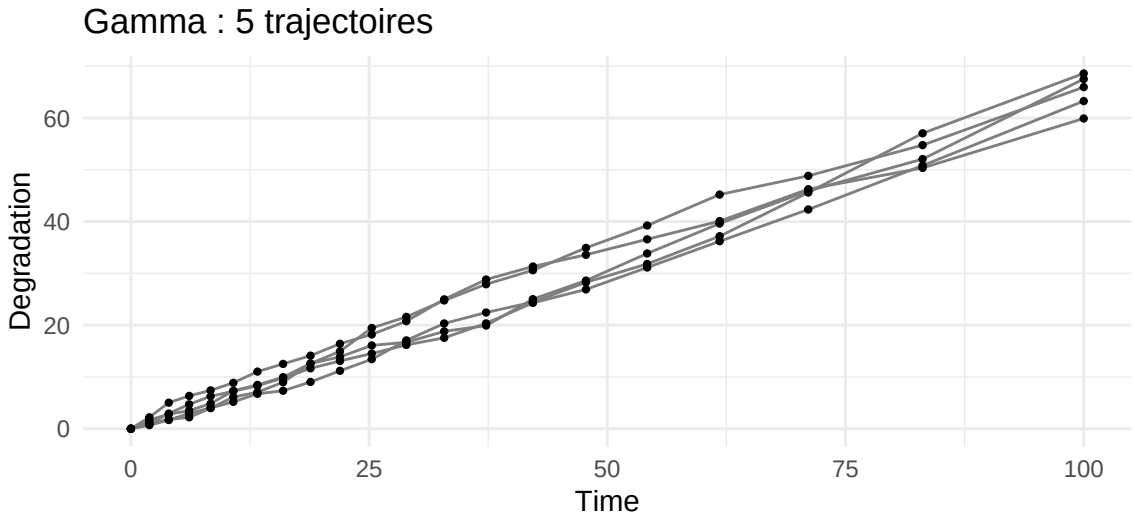
```
set.seed(131)
x <- sim_gamma(p3, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1       X2       X3
## [1,]  0.000000 0.0000000 0.000000 0.000000
## [2,]  1.940035 0.6807504 1.060241 2.206014
## [3,]  3.974743 1.6795349 1.791835 5.027377
## [4,]  6.113840 2.9693747 2.202258 6.336759
## [5,]  8.368618 4.0223494 3.975035 7.405013
## [6,] 10.752309 5.1828148 6.103820 8.862926
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.681  0.999  1.290  1.053  1.160  1.560  0.585  1.699  2.156  2.235
## [11]  3.633  3.264  2.122  1.999  3.815  3.560  5.353  8.435 11.436 11.550
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

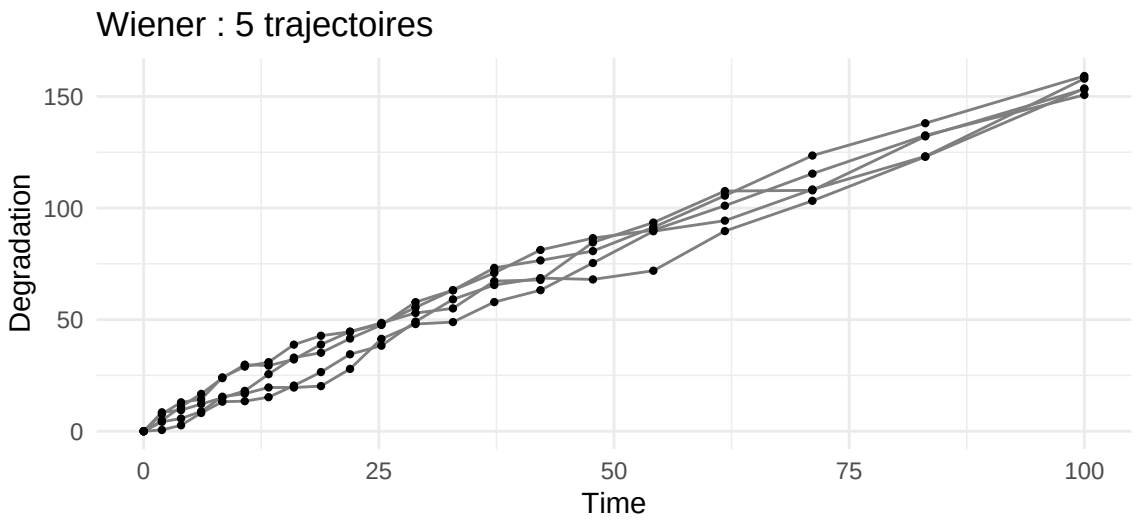
```
set.seed(132)
x <- sim_wiener(p3, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##              t        X1         X2        X3
## [1,]  0.000000  0.000000  0.0000000  0.000000
## [2,]  1.940035  4.230787  0.5691491  8.489047
## [3,]  3.974743  5.699667  2.6729907  9.454022
## [4,]  6.113840  8.879185  8.1968896 12.194740
## [5,]  8.368618 15.465729 13.2503463 14.987435
## [6,] 10.752309 16.842018 13.4664873 18.080960
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  4.231  1.469  3.180  6.587  1.376  2.773 -0.025  0.635  7.691 13.470
## [11]  6.643  0.893  8.950  5.330 12.177 14.230  4.739 13.966 14.832 34.817
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

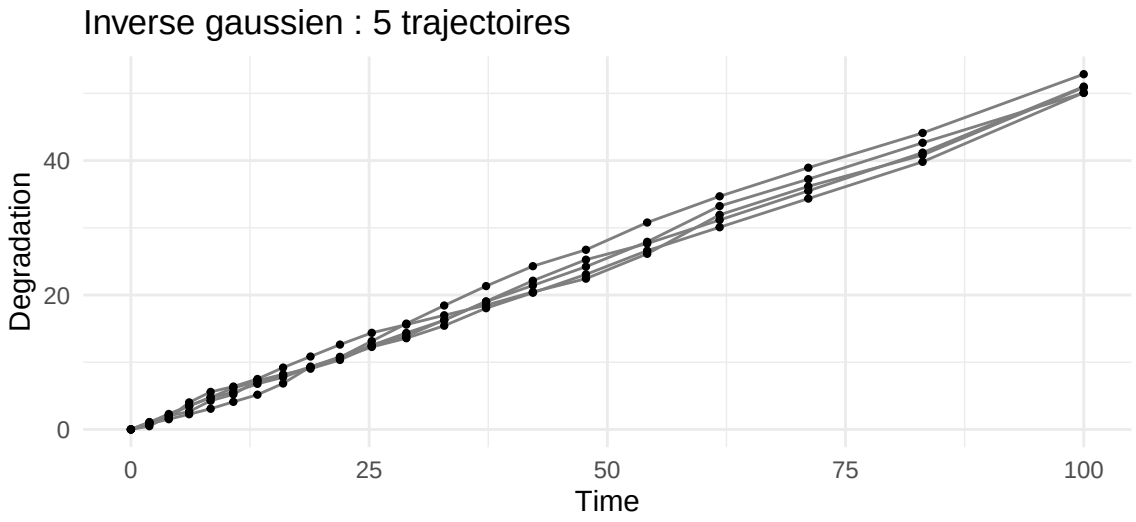
```
set.seed(133)
x <- sim_ig(p3, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000 0.0000000 0.0000000 0.000000
## [2,]  1.940035 0.9390493 0.4884332 1.088656
## [3,]  3.974743 2.0880034 2.0127334 1.694220
## [4,]  6.113840 3.4899909 2.6631953 4.018758
## [5,]  8.368618 4.7335340 4.2683565 5.584057
## [6,] 10.752309 5.5338600 5.2210800 6.378180
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.939 1.149 1.402 1.244 0.800 1.245 0.987 1.482 1.473 2.417 2.603 2.697
## [13] 2.891 2.963 2.438 4.059 3.901 4.255 5.170 8.749
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

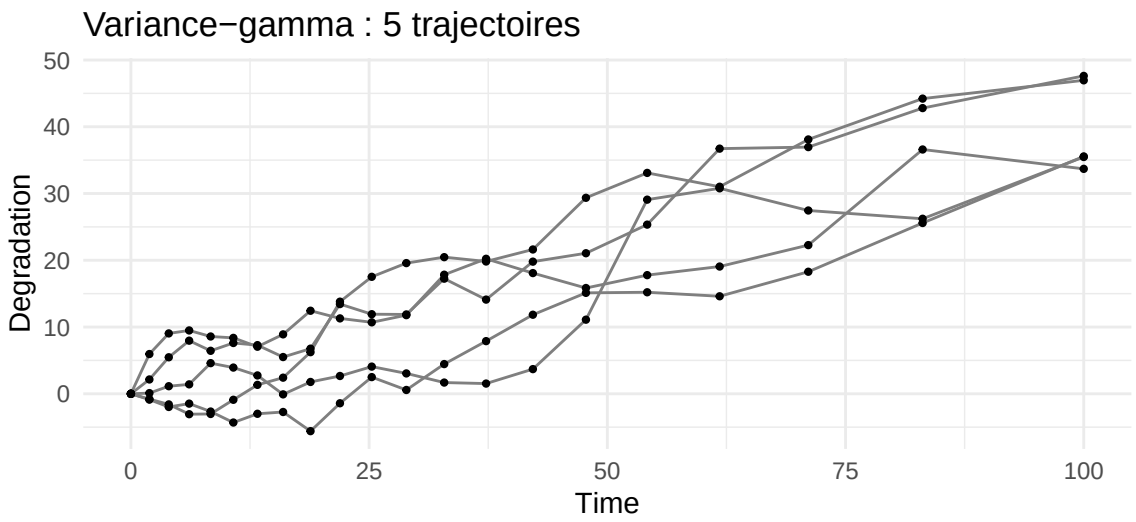
```
set.seed(134)
x <- sim_vg(p3, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t       X1        X2         X3
## [1,]  0.000000 0.000000 0.0000000  0.0000000
## [2,]  1.940035 5.953242 0.0991426 -0.8006998
## [3,]  3.974743 9.043546 1.1334917 -1.5900090
## [4,]  6.113840 9.487525 1.4108693 -3.0584259
## [5,]  8.368618 8.582190 4.5890451 -3.0247012
## [6,] 10.752309 8.374895 3.9255079 -0.8998503
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.953  3.090  0.444 -0.905 -0.207 -1.325  1.852  3.537 -1.146 -0.587
## [11]  1.073  6.059  2.383 -2.140 -2.239  1.929  1.291  3.208 14.324 -2.901
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

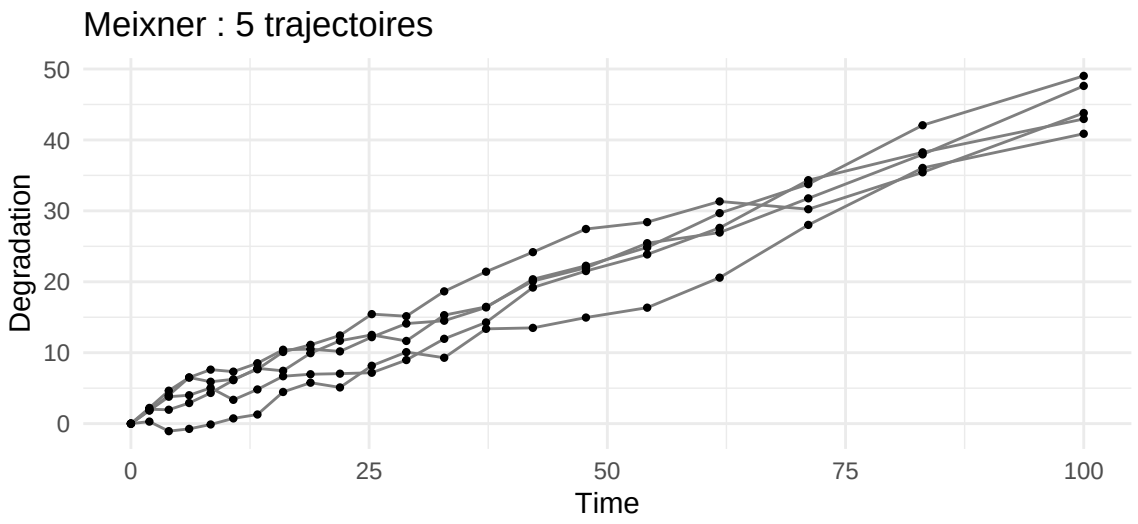
```
set.seed(135)
x <- sim_meixner(p3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t         X1       X2       X3
## [1,]  0.000000  0.0000000 0.000000 0.000000
## [2,]  1.940035  0.2866694 2.024976 1.842790
## [3,]  3.974743 -1.0583616 1.958050 3.793688
## [4,]  6.113840 -0.7404516 2.908604 4.004078
## [5,]  8.368618 -0.1171312 4.326521 5.041161
## [6,] 10.752309  0.7393661 6.128973 3.356289
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.287 -1.345  0.318  0.623  0.856  0.548  3.176  1.307 -0.667  3.054
## [11]  1.925 -0.796  4.077  0.133  1.458  1.393  4.235  7.435  8.031  4.820
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

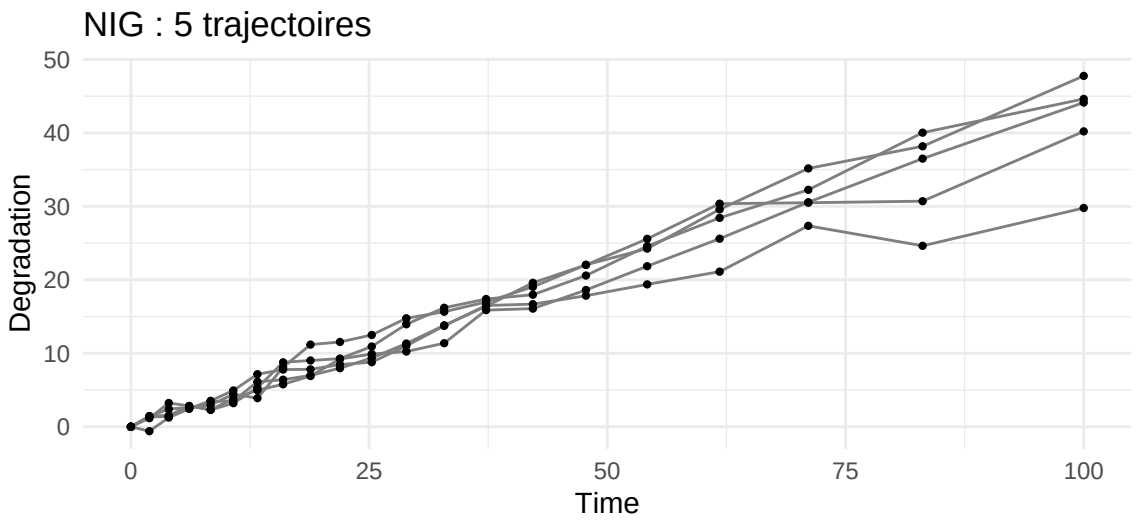
```
set.seed(136)
x <- sim_nig(p3, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t       X1       X2         X3
## [1,]  0.000000 0.000000 0.000000  0.0000000
## [2,]  1.940035 1.416675 1.163461 -0.5974616
## [3,]  3.974743 2.398329 3.233627  1.2486142
## [4,]  6.113840 2.639810 2.835734  2.4510788
## [5,]  8.368618 2.962988 2.302184  3.4433145
## [6,] 10.752309 4.418144 3.207621  3.5021879
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.417  0.982  0.241  0.323  1.455 -0.528  4.205  3.088  0.357  0.954
## [11]  2.269  0.891  1.326  2.062  2.999  2.201  5.348  5.570  3.014  9.580
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

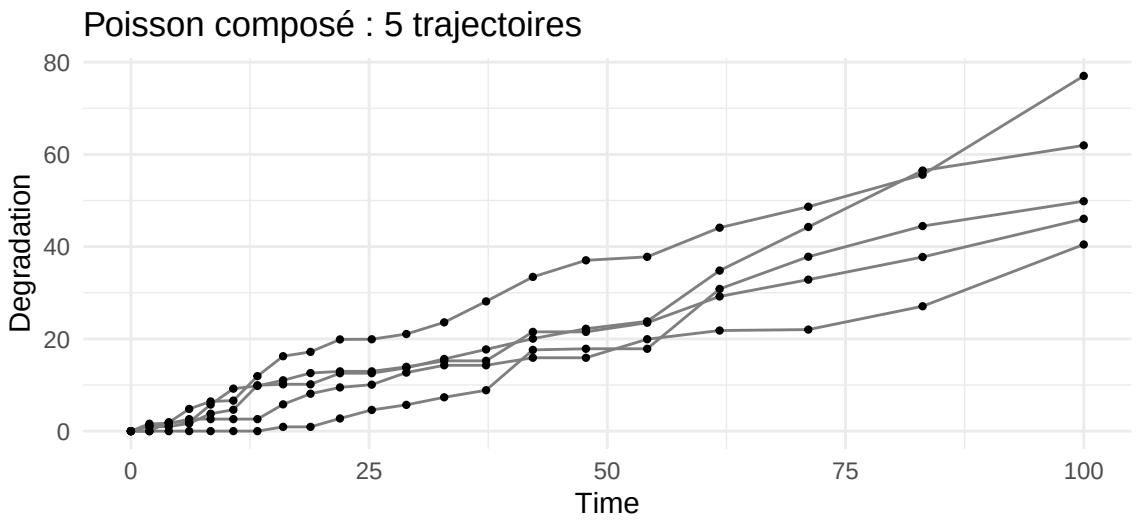
```
set.seed(137)
x <- sim_cpoisson(p3, lambda = 0.5)
head(x[, 1:4])
```

```
##              t       X1        X2       X3
## [1,]  0.000000 0.000000 0.0000000 0.000000
## [2,]  1.940035 1.126505 0.9923666 0.000000
## [3,]  3.974743 1.856240 0.9923666 1.922071
## [4,]  6.113840 4.813572 1.6737306 1.922071
## [5,]  8.368618 6.442241 3.7726864 5.759813
## [6,] 10.752309 6.619919 4.6405605 9.211759
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.127  0.730  2.957  1.629  0.178  5.304  4.321  0.932  2.706  0.037
## [11]  1.134  2.535  4.539  5.310  3.599  0.752  6.321  4.543  6.941 21.428
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
Time
0
20
40
60
80
0
25
50
75
100

Cas 4 : mixtes, beaucoup au début et à la fin

```
p4 <- inspection_times("inc-dec", n = 20, n_traj = 5, tmax = 100)
plot_plan(p4$time[p4$id == 1], "Beaucoup au début et à la fin, peu au milieu")
```



Modèle 1 : gamma

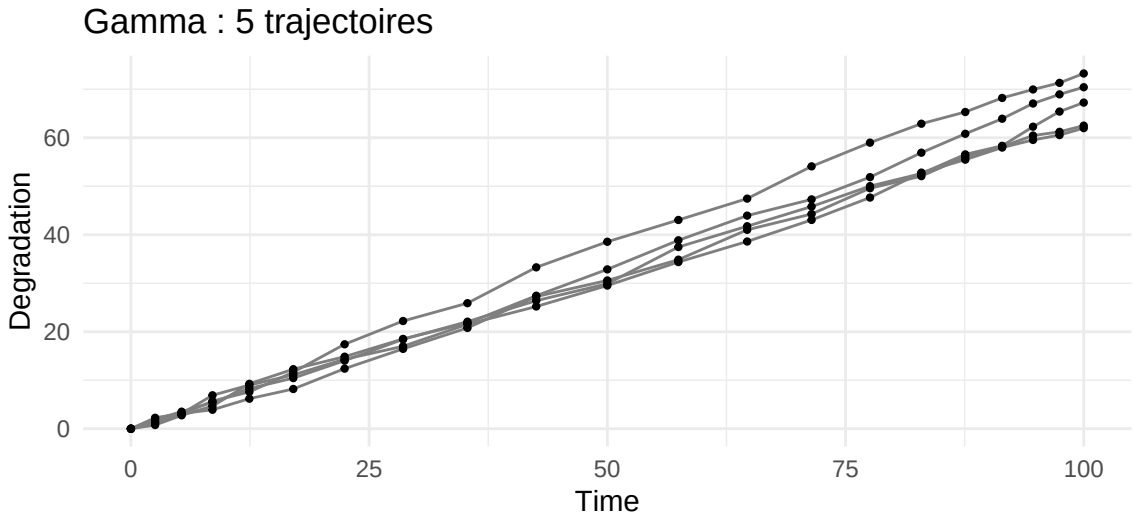
```
set.seed(141)
x <- sim_gamma(p4, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000  0.000000 0.000000  0.000000
## [2,]  2.540921  1.916327 1.174257  2.256682
## [3,]  5.322554  3.520553 3.137588  3.099718
## [4,]  8.562047  5.407939 3.909497  6.897871
## [5,] 12.431733  8.292020 6.188806  8.993691
## [6,] 17.042253 10.434968 8.191329 11.098034
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.916 1.604 1.887 2.884 2.143 3.602 4.399 3.640 4.331 3.499 7.567 4.281
## [13] 4.062 4.209 2.591 3.954 1.772 3.936 3.120 1.856
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

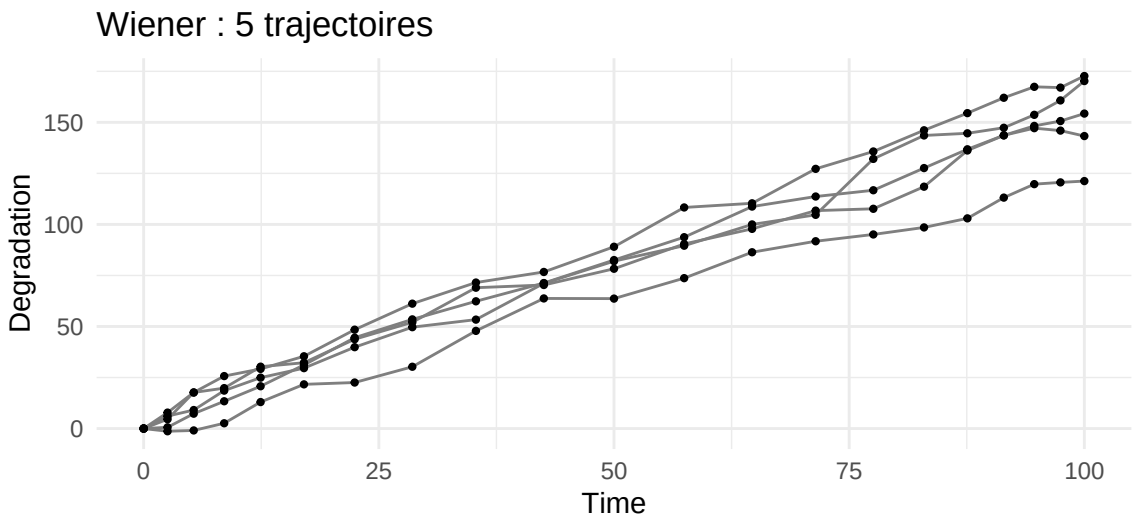
```
set.seed(142)
x <- sim_wiener(p4, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##              t        X1         X2        X3
## [1,]  0.000000  0.000000  0.0000000  0.000000
## [2,]  2.540921  7.811498 -1.3223131  4.565071
## [3,]  5.322554 17.668172 -0.9279949 17.670392
## [4,]  8.562047 25.696328  2.5898009 19.801467
## [5,] 12.431733 29.212882 12.9757735 30.310208
## [6,] 17.042253 35.402095 21.6501585 32.251385
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  7.811  9.857  8.028  3.517  6.189 13.028 12.753 10.340  5.187 12.348
## [11] 19.250  2.036 16.878  8.513 10.390  8.398  7.528  5.360 -0.391  5.692
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

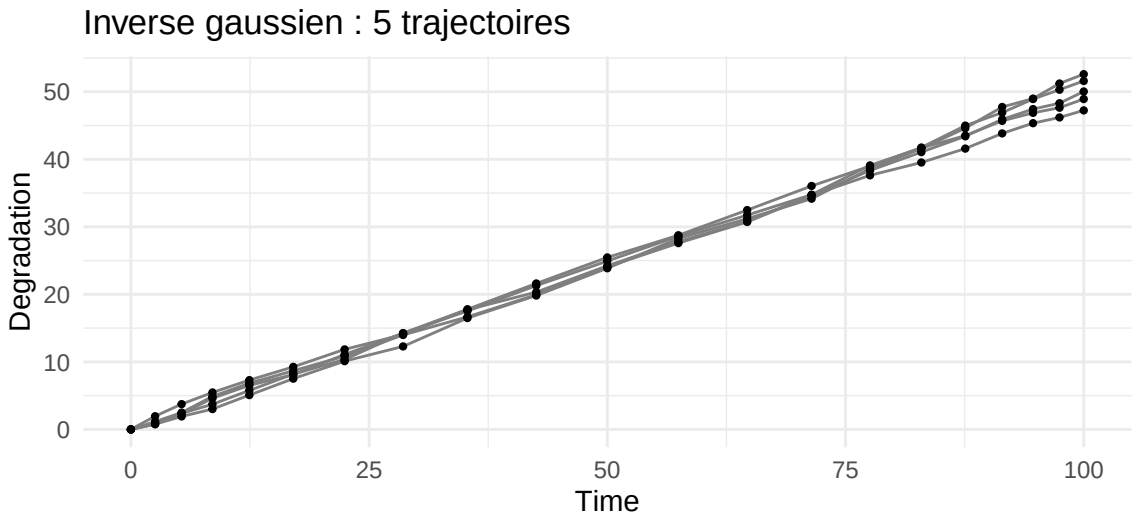
```
set.seed(143)
x <- sim_ig(p4, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000 0.0000000 0.000000 0.0000000
## [2,]  2.540921 0.7772242 1.956862 0.9506147
## [3,]  5.322554 1.9233357 3.745237 2.3784685
## [4,]  8.562047 3.0144983 5.472639 3.7276869
## [5,] 12.431733 5.0832297 7.300213 5.7749032
## [6,] 17.042253 7.5167033 9.259178 8.1877298
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.777 1.146 1.091 2.069 2.433 2.597 2.175 4.202 3.330 4.050 4.402 2.917
## [13] 2.977 4.438 2.984 3.016 3.141 1.239 2.242 1.378
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

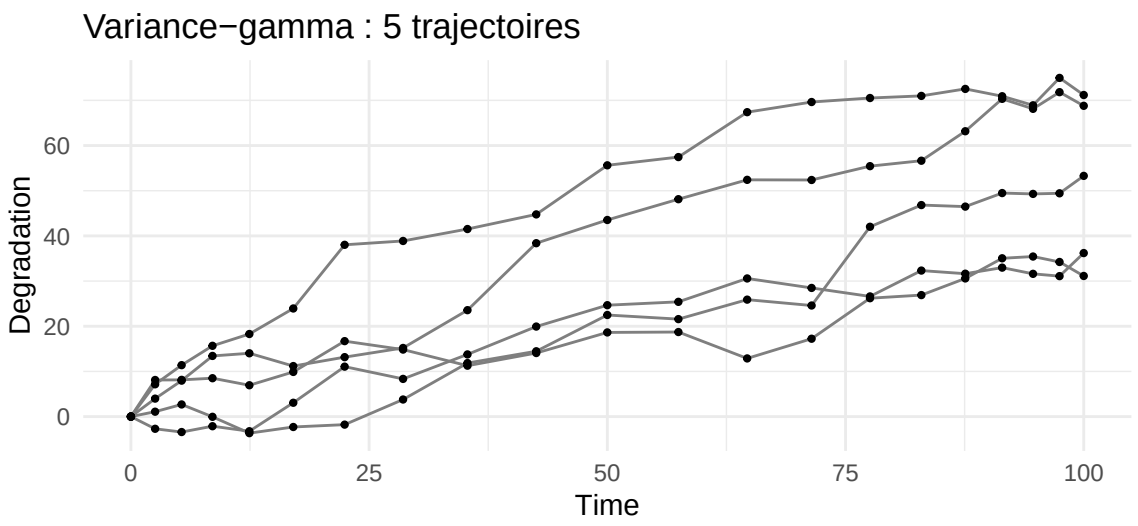
```
set.seed(144)
x <- sim_vg(p4, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t       X1        X2        X3
## [1,]  0.000000 0.000000  0.000000  0.000000
## [2,]  2.540921 8.101337  7.146385 -2.701680
## [3,]  5.322554 8.131089 11.386395 -3.421002
## [4,]  8.562047 8.493392 15.665319 -2.130697
## [5,] 12.431733 6.936565 18.279288 -3.241100
## [6,] 17.042253 9.887127 23.929366  3.061084
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  8.101  0.030  0.362 -1.557  2.951  6.814 -1.871 -3.565  2.809  4.555
## [11]  0.087 -5.843  4.361  8.967  0.692  3.686  4.459  0.392 -1.206 -3.099
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

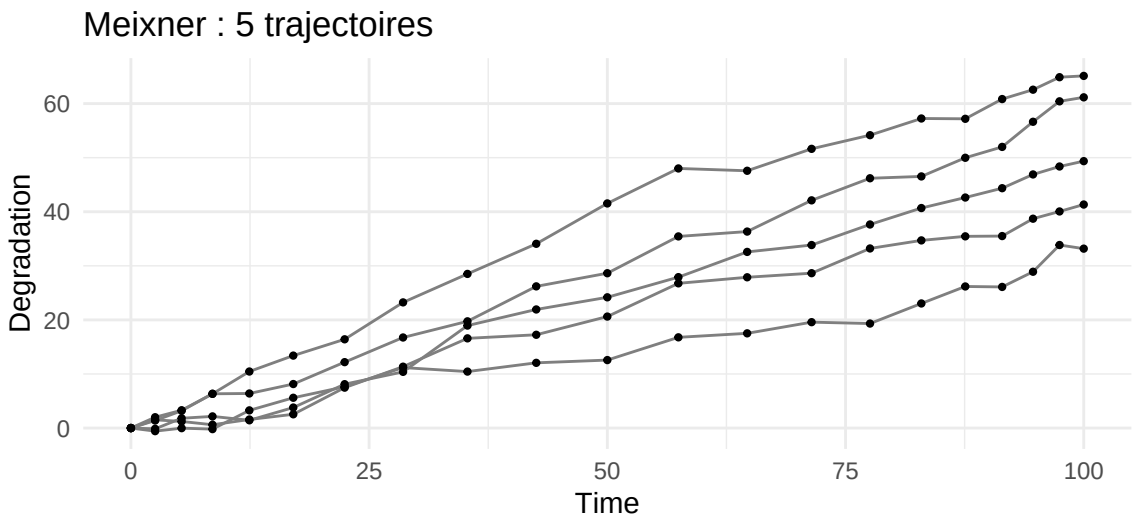
```
set.seed(145)
x <- sim_meixner(p4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1          X2        X3
## [1,]  0.000000  0.000000  0.00000000 0.0000000
## [2,]  2.540921  2.004889 -0.57081430 1.5262858
## [3,]  5.322554  3.288517 -0.01945328 1.2183414
## [4,]  8.562047  6.329637 -0.19546133 0.6123311
## [5,] 12.431733 10.456133  3.27044396 1.5840534
## [6,] 17.042253 13.399033  5.59940925 2.5455475
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.005  1.284  3.041  4.126  2.943  3.014  6.826  5.267  5.568  7.465
## [11]  6.469 -0.429  4.055  2.525  3.087 -0.058  3.650  1.735  2.309  0.241
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

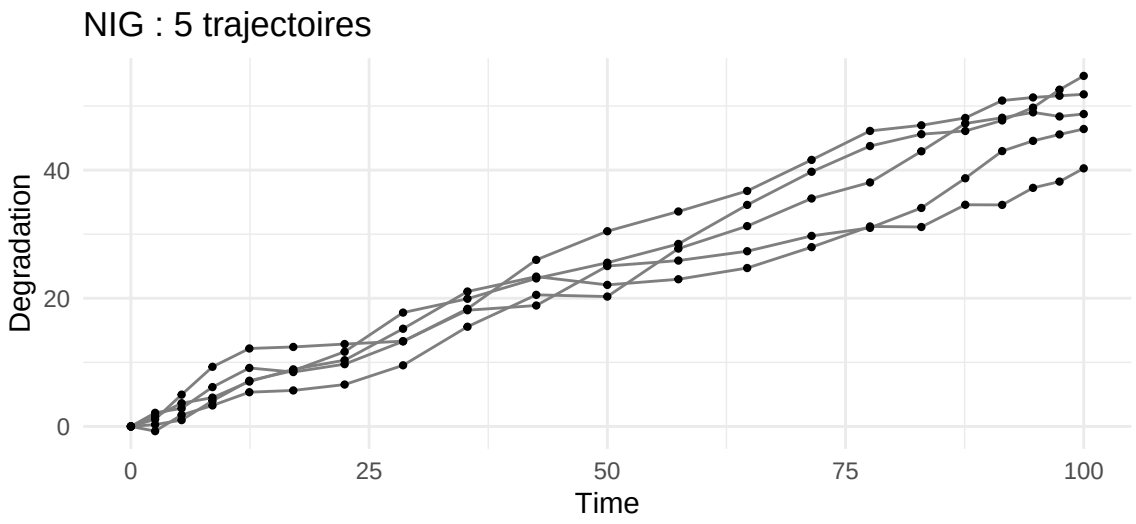
```
set.seed(146)
x <- sim_nig(p4, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t       X1         X2        X3
## [1,]  0.000000 0.000000  0.0000000 0.0000000
## [2,]  2.540921 1.561046 -0.7367223 0.2876020
## [3,]  5.322554 3.624618  1.7851754 0.9848451
## [4,]  8.562047 4.495267  3.2748986 3.9989290
## [5,] 12.431733 7.001454  5.3411814 7.1301402
## [6,] 17.042253 8.906618  5.6123716 8.7004438
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.561  2.064  0.871  2.506  1.905  1.425  4.912  5.809  2.327 -1.294
## [11]  0.883  1.751  3.257  3.229 -0.084  3.469 -0.022  2.659  0.981  2.066
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

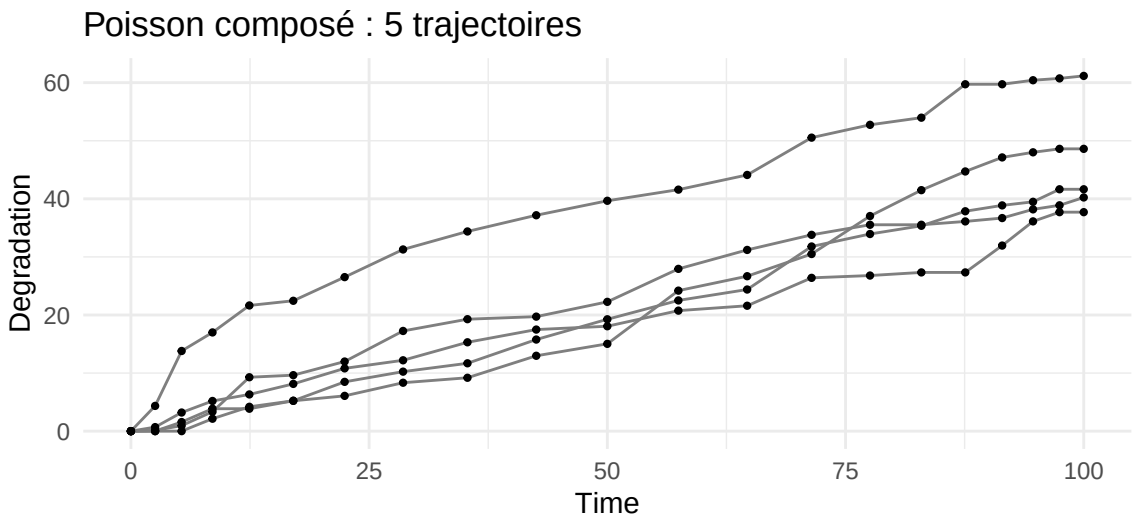
```
set.seed(147)
x <- sim_cpoisson(p4, lambda = 0.5)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000 0.0000000  0.000000 0.0000000
## [2,]  2.540921 0.1210714  4.345207 0.6949144
## [3,]  5.322554 0.9519051 13.802905 3.2070645
## [4,]  8.562047 3.3447454 16.996240 5.1935449
## [5,] 12.431733 9.2809750 21.638766 6.3193773
## [6,] 17.042253 9.6390935 22.445798 8.1355463
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.121 0.831 2.393 5.936 0.358 2.346 5.264 2.027 0.440 2.544 5.680 3.261
## [13] 2.598 1.733 0.000 0.576 0.580 1.499 0.698 1.351
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 5 : mixtes, beaucoup au milieu

```
p5 <- inspection_times("dec-inc", n = 20, n_traj = 5, tmax = 100)
plot_plan(p5$time[p5$id == 1], "Peu au début et à la fin, beaucoup au milieu")
```



Modèle 1 : gamma

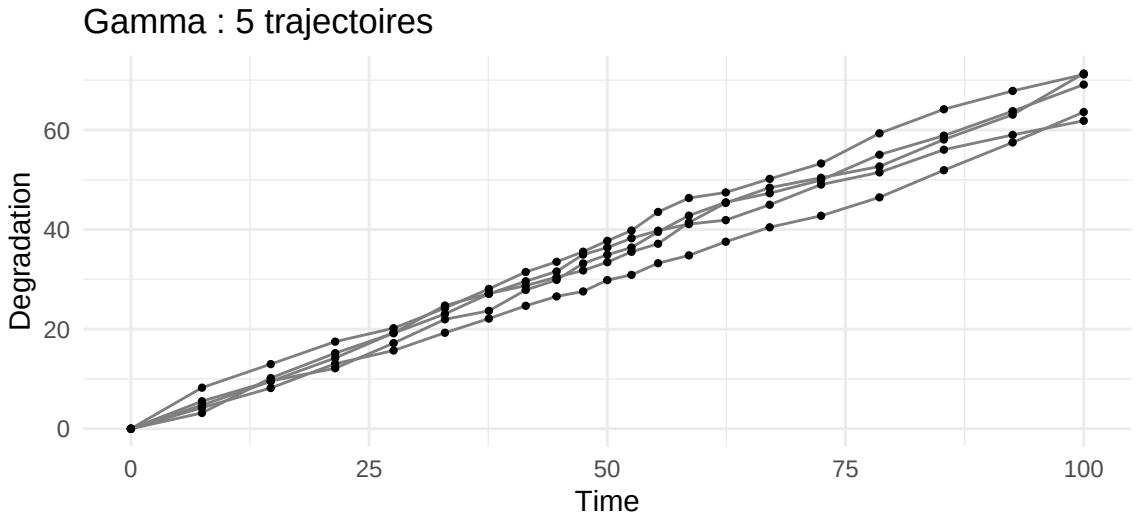
```
set.seed(151)
x <- sim_gamma(p5, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.459079  4.741041  5.521434  4.228339
## [3,] 14.677446  9.508096  9.575761  8.175986
## [4,] 21.437953 14.203284 12.146490 13.038912
## [5,] 27.568267 19.244251 17.192410 15.703649
## [6,] 32.957747 24.736728 21.962976 19.268742
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 4.741 4.767 4.695 5.041 5.492 2.407 1.586 1.710 1.344 1.661 2.085 1.614
## [13] 4.256 3.951 3.052 1.993 2.272 5.417 4.993 8.263
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

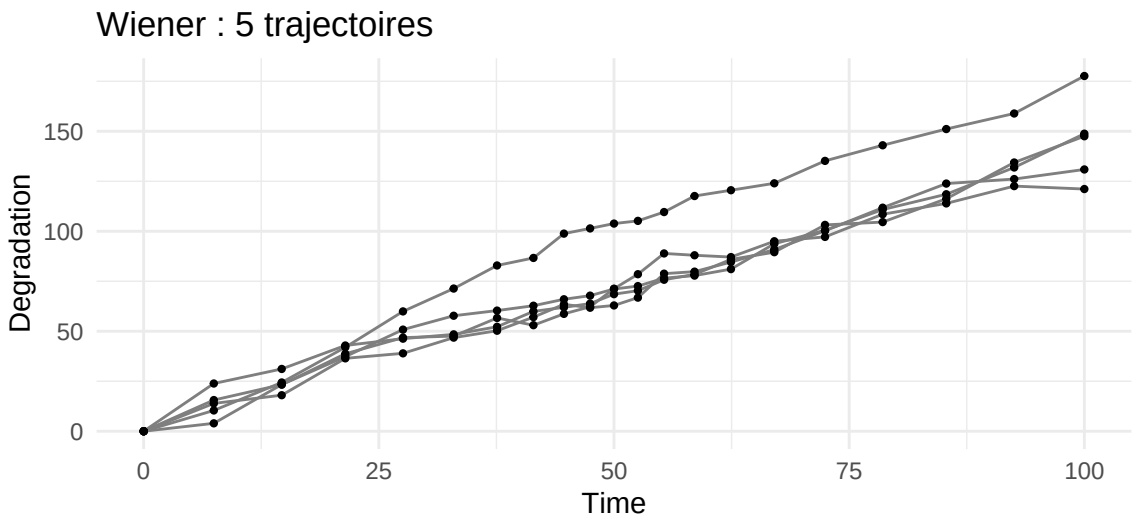
```
set.seed(152)
x <- sim_wiener(p5, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##              t       X1       X2       X3
## [1,]  0.000000  0.00000  0.00000  0.000000
## [2,]  7.459079 23.86975 15.59424  3.958588
## [3,] 14.677446 31.14916 23.32680 23.360176
## [4,] 21.437953 42.87639 38.71552 37.302685
## [5,] 27.568267 46.29445 46.82211 50.815925
## [6,] 32.957747 48.44725 47.57995 57.756556
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 23.870  7.279 11.727  3.418  2.153  3.777  7.718  2.015  1.883  4.704
## [11]  1.753  5.451  2.693  7.475  3.619 13.646  1.412 11.651 18.147 13.120
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

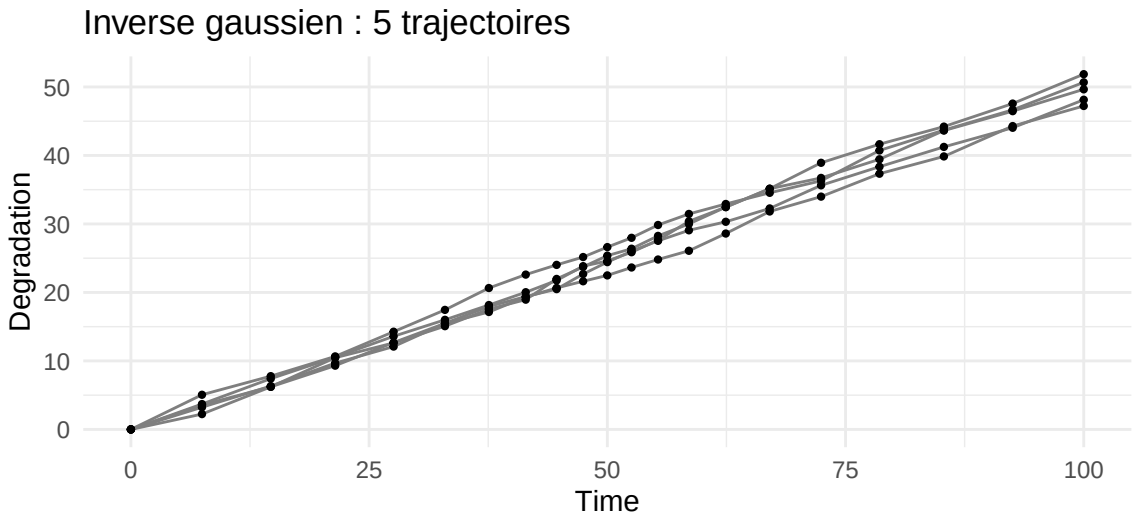
```
set.seed(153)
x <- sim_ig(p5, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.459079  2.238359  3.240561  3.557905
## [3,] 14.677446  6.217203  6.325747  6.219868
## [4,] 21.437953 10.468404  9.677937  9.307220
## [5,] 27.568267 12.628815 12.094830 12.680646
## [6,] 32.957747 15.074764 15.487383 15.478020
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.238 3.979 4.251 2.160 2.446 2.483 1.386 3.026 1.761 1.638 1.012 1.874
## [13] 1.749 2.454 2.684 3.788 2.710 2.572 3.362 4.294
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

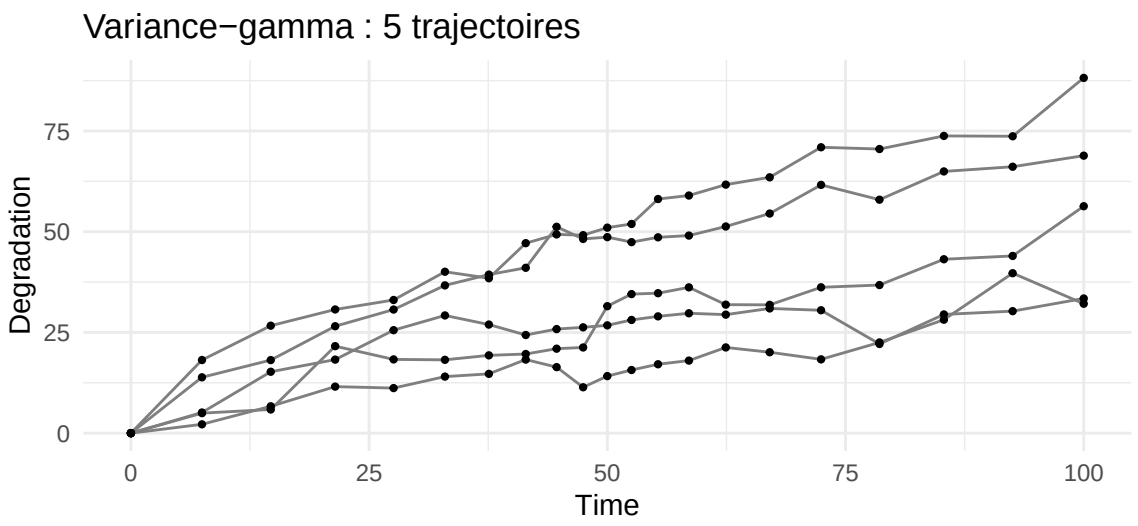
```
set.seed(154)
x <- sim_vg(p5, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000  0.00000
## [2,]  7.459079  2.190125  4.960023 13.85976
## [3,] 14.677446  6.667534  5.870367 18.13231
## [4,] 21.437953 11.550031 21.591089 26.53618
## [5,] 27.568267 11.182291 18.299595 30.67430
## [6,] 32.957747 14.030616 18.197226 36.67401
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.190  4.477  4.882 -0.368  2.848  0.685  3.542 -1.873 -4.993  2.766
## [11]  1.512  1.428  0.910  3.264 -1.194 -1.765  4.196  5.645 11.551 -7.583
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

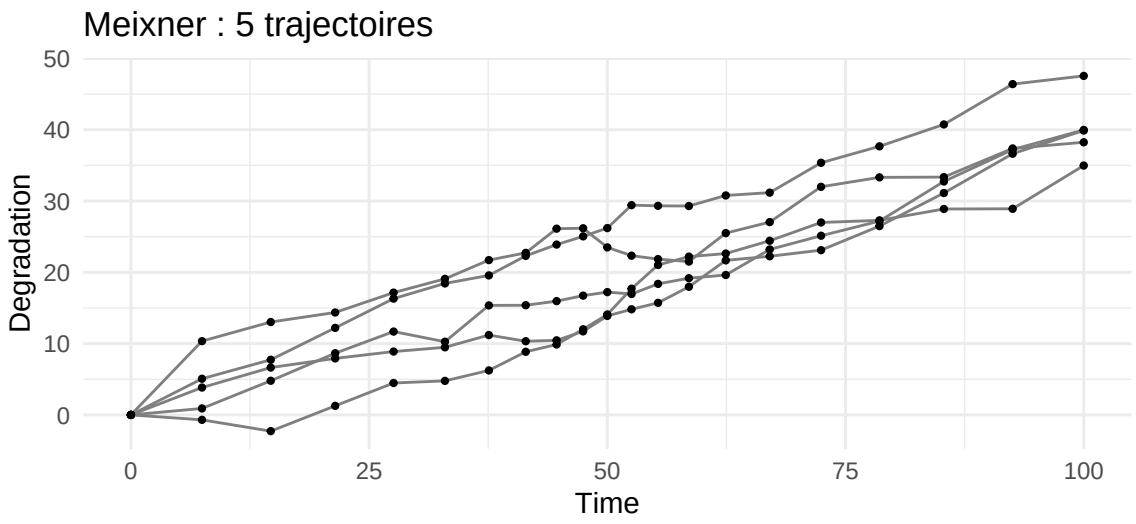
```
set.seed(155)
x <- sim_meixner(p5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1       X2       X3
## [1,]  0.000000  0.000000 0.000000  0.00000
## [2,]  7.459079  5.071196 3.839912 10.35035
## [3,] 14.677446  7.739227 6.643030 13.02948
## [4,] 21.437953 12.206749 7.927603 14.36807
## [5,] 27.568267 16.315425 8.886013 17.15169
## [6,] 32.957747 18.441474 9.482853 19.08732
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.071  2.668  4.468  4.109  2.126  1.128  2.741  1.584  1.152  1.161
## [11]  3.216 -0.093 -0.024  1.478  0.398  4.188  2.306  3.073  5.657  1.151
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

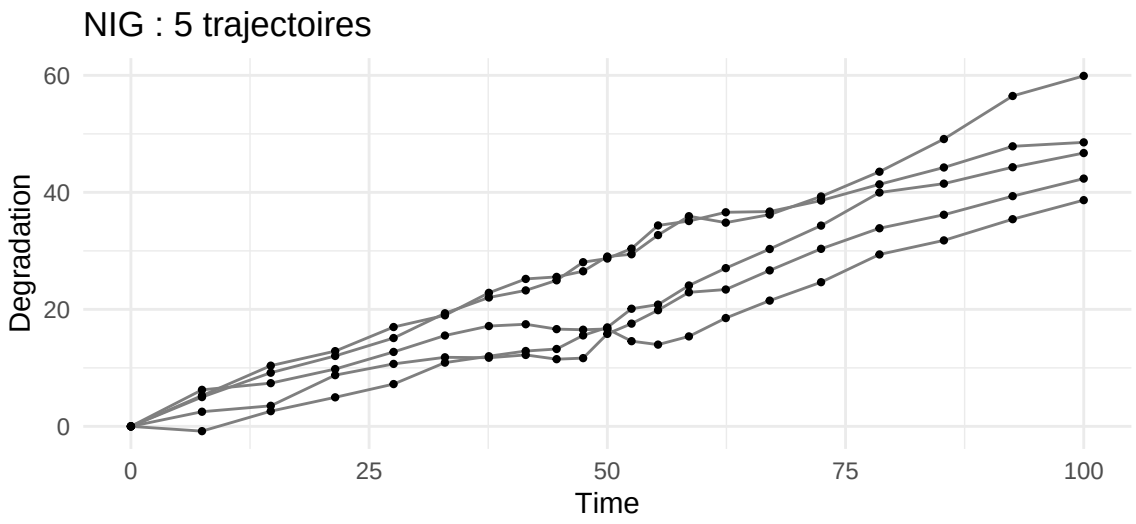
```
set.seed(156)
x <- sim_nig(p5, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t         X1        X2        X3
## [1,]  0.000000  0.0000000  0.000000  0.000000
## [2,]  7.459079 -0.8154205  6.241988  5.009794
## [3,] 14.677446  2.6002278  7.387626  9.169580
## [4,] 21.437953  4.9651138  9.796440 12.038473
## [5,] 27.568267  7.2308250 12.724568 15.099153
## [6,] 32.957747 10.8952491 15.527961 19.331089
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -0.815  3.416  2.365  2.266  3.664  1.096  0.888  0.346  2.322  1.370
## [11]  3.189  0.709  3.267  2.964  3.271  4.003  5.652  1.520  2.809  2.423
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

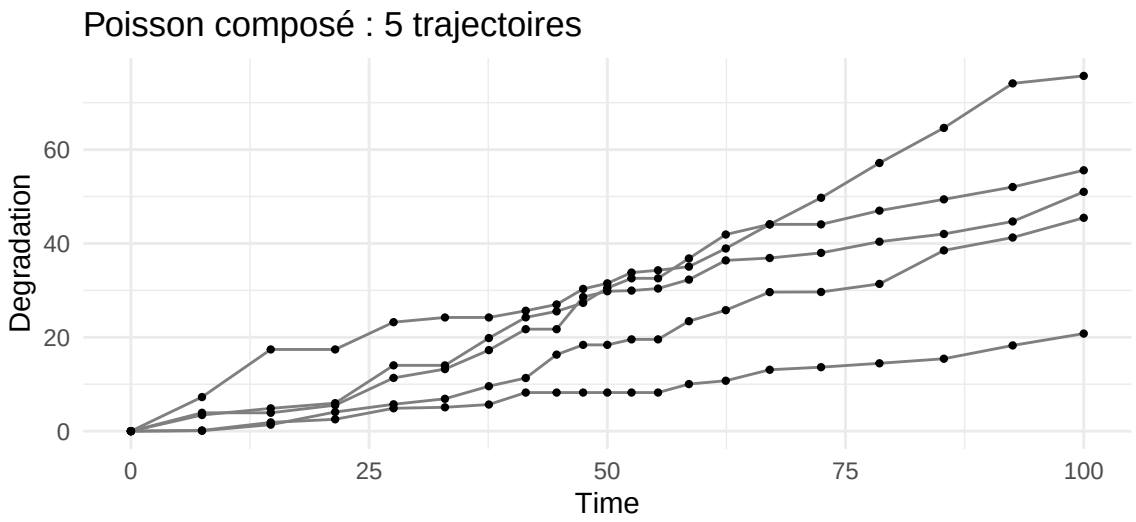
```
set.seed(157)
x <- sim_cpoisson(p5, lambda = 0.5)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000 0.0000000
## [2,]  7.459079  3.911296  3.438826 0.1384595
## [3,] 14.677446  3.911296  4.852867 1.8628205
## [4,] 21.437953  5.528340  5.961461 2.5230237
## [5,] 27.568267 11.344923 14.017337 4.8769255
## [6,] 32.957747 13.224447 14.017337 5.0783270
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.911 0.000 1.617 5.817 1.880 4.040 4.465 0.000 6.858 1.222 0.144 0.439
## [13] 1.896 4.099 0.505 1.094 2.385 1.647 2.663 6.315
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
0
20
40
60
0
25
50
75
100
Time

III - Processus de Lévy à échelle de temps

Une fonction d'échelle de temps A (fonction croissante, continue à droite avec des limites à gauche, $A(0) = 0$ et $A(t) \to +\infty$) donne le processus de Lévy à échelle de temps $X_t = Y_{A(t)}$, où Y est un processus de Lévy. Le paramètre que A remplace (identifiabilité) n'est alors pas donné :

Processus	Fonction	$X_t - X_s$, $\Delta A = A(t) - A(s)$	Avec $A(t) = at$ ou μt
Gamma	`sim_gamma(t, b, A)`	$\Gamma(\Delta A, b)$	gamma de forme a
Wiener	`sim_wiener(t, sigma, A)`	$\mathcal{N}(\Delta A, \sigma^2 \Delta A)$	Wiener $(\mu, \sigma\sqrt{\mu})$
Inverse gaussien	`sim_ig(t, eta, A)`	$IG(\Delta A, \eta \Delta A^2)$	IG $(\mu, \eta\mu^2)$

Pour le Wiener, $X_t = \mu t + \sigma B_{\mu t}$ a la loi de $\mu t + \sigma\sqrt{\mu}\, B_t$, car $(B_{\mu t})_{t\geq 0}$ a la loi de $(\sqrt{\mu}\, B_t)_{t\geq 0}$.

On prend $A(t) = 0.02\, t^{1.5}$, de sorte que $A(100) = 20$, et les 5 trajectoires du cas 2 de la partie II (pas décroissants).

Modèle 1 : gamma à échelle de temps (Cha et Mercier, 2019, exemple 2)

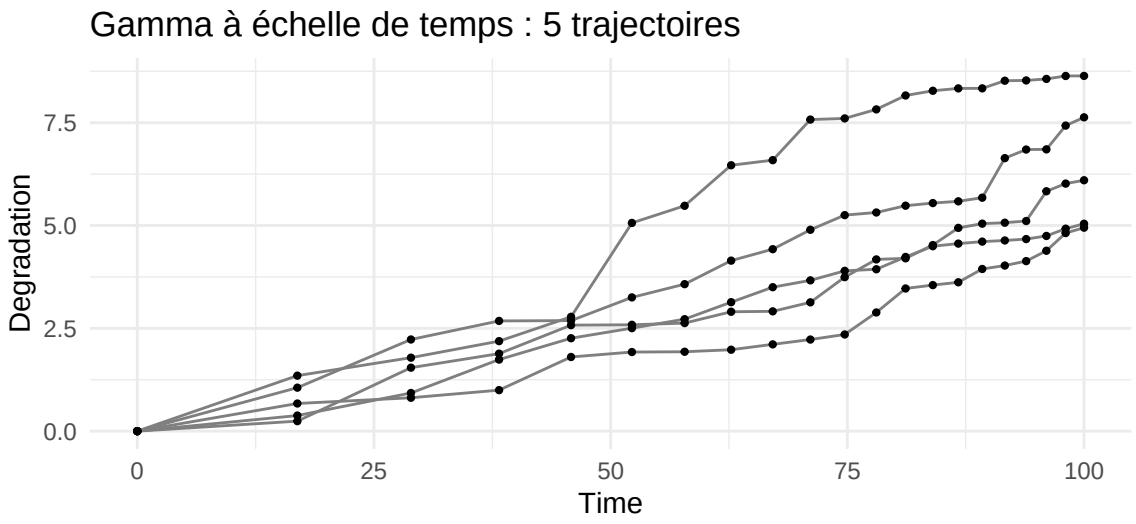
```
A <- function(t) 0.02 * t^1.5
set.seed(301)
x <- sim_gamma(p2, b = 3, A = A)
x
```

```
## 5 simulated trajectory(ies) of the time-scaled gamma process, same inspection times (matrix)
## parameters: b = 3
## simulation method: RWA
## time-scaling function: A = function (t)  0.02 * t^1.5
##
##             t        X1       X2        X3        X4       X5
## [1,]  0.00000 0.0000000 0.000000 0.0000000 0.0000000 0.000000
## [2,] 16.90921 0.3792614 1.349112 0.2429382 0.6721473 1.057136
## [3,] 28.90648 0.9275601 1.786393 1.5432021 0.8117168 2.228763
## [4,] 38.21229 1.7404953 2.188722 1.8841561 0.9973612 2.680303
## [5,] 45.81569 2.2589926 2.780103 2.5772813 1.8032195 2.687744
## [6,] 52.24427 2.5044998 5.063078 2.5860069 1.9234865 3.252174
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.379 0.548 0.813 0.518 0.246 0.218 0.413 0.364 0.166 0.231 0.039 0.299
## [13] 0.261 0.064 0.048 0.028 0.034 0.076 0.176 0.120
```

```
plot_traj(x, "Gamma à échelle de temps : 5 trajectoires")
```



Modèle 2 : Wiener à échelle de temps (Cha et Mercier, 2019, exemple 1)

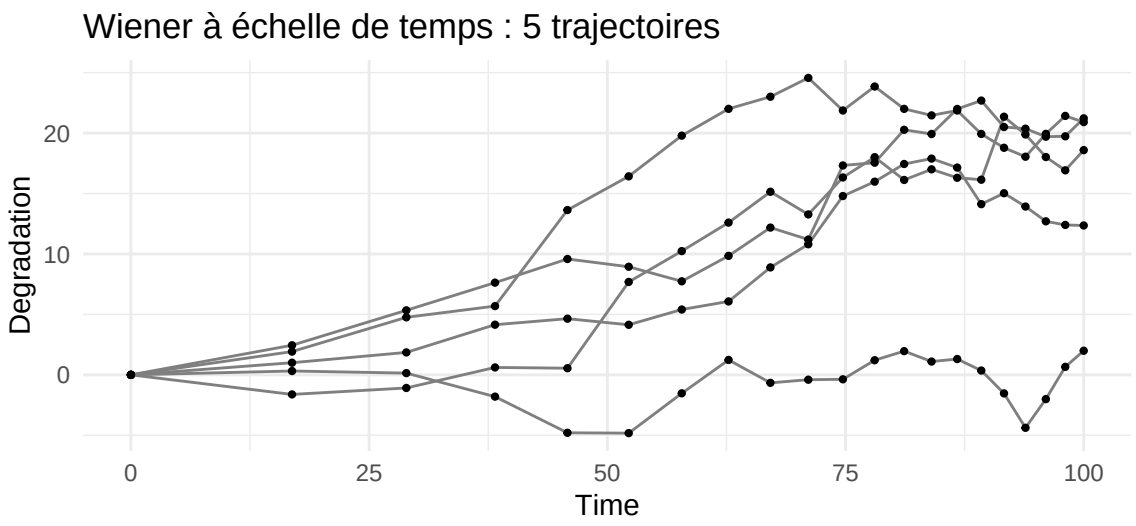
```
set.seed(302)
x <- sim_wiener(p2, sigma = 2, A = A)
head(x[, 1:4])
```

```
##             t        X1       X2         X3
## [1,]  0.00000  0.000000 0.000000  0.0000000
## [2,] 16.90921  1.923276 2.448910  0.3174737
## [3,] 28.90648  4.757541 5.337586  0.1435598
## [4,] 38.21229  5.687673 7.626493 -1.8019695
## [5,] 45.81569 13.631897 9.583937 -4.7940957
## [6,] 52.24427 16.422778 8.941656 -4.8223706
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.923  2.834  0.930  7.944  2.791  3.365  2.218  1.002  1.557 -2.697
## [11]  1.984 -1.849 -0.540  0.405 -1.941 -1.144 -0.739  1.888  1.488 -0.512
```

```
plot_traj(x, "Wiener à échelle de temps : 5 trajectoires")
```



Modèle 3 : inverse gaussien à échelle de temps (Cha et Mercier, 2019, exemple 3)

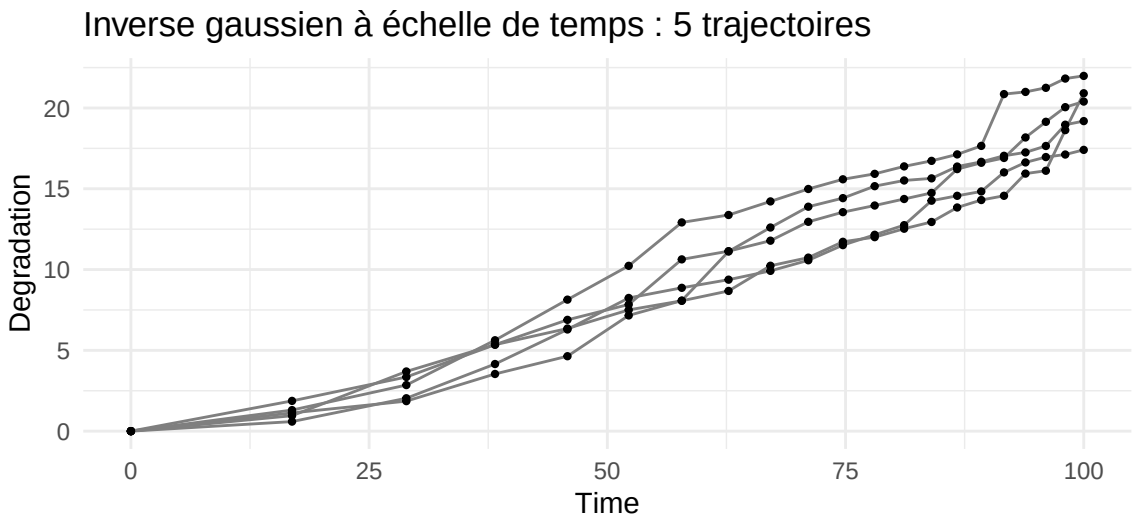
```
set.seed(303)
x <- sim_ig(p2, eta = 2, A = A)
head(x[, 1:4])
```

```
##             t       X1        X2       X3
## [1,]  0.00000 0.000000 0.0000000 0.000000
## [2,] 16.90921 1.871329 0.9560403 1.128459
## [3,] 28.90648 3.349287 3.6921305 1.856956
## [4,] 38.21229 5.349771 5.3504349 3.538414
## [5,] 45.81569 6.887926 6.3527608 4.637236
## [6,] 52.24427 7.839307 7.5044765 7.160367
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.871 1.478 2.000 1.538 0.951 2.791 0.508 0.648 1.168 0.597 0.412 0.403
## [13] 0.375 1.475 0.390 0.299 1.268 0.969 0.905 0.348
```

```
plot_traj(x, "Inverse gaussien à échelle de temps : 5 trajectoires")
```



Vérification : avec $A(t) = at$, on retrouve le processus gamma de forme a

Avec la même graine, le gamma à échelle de temps avec $A(t) = 2t$ et le gamma $(a, b) = (2, 3)$ donnent les mêmes trajectoires :

```
set.seed(304)
y1 <- sim_gamma(p2, b = 3, A = function(t) 2 * t)
set.seed(304)
y2 <- sim_gamma(p2, a = 2, b = 3)
all.equal(unclass(y1)[, -1], unclass(y2)[, -1], check.attributes = FALSE)
```

```
## [1] TRUE
```

IV - Méthode Bridge

Les parties I à III utilisent la méthode RWA, qui cumule des accroissements indépendants. Avec `method = "Bridge"`, la valeur à la dernière inspection est tirée d'abord. Les autres valeurs sont ensuite tirées une à une, en prenant chaque fois la date du milieu de la liste, par les fonctions `gamma_bridge`, `wiener_bridge`, `ig_bridge`, `vg_bridge` et `meixner_bridge` du package `levyBridge`, installé avec `DegradSim`.

Les trajectoires ont la même loi qu'avec la méthode RWA. Avec une même graine, les valeurs obtenues sont différentes, et le calcul est plus lent. Le normal inverse gaussien et le Poisson composé n'ont pas de pont : pour eux, seule la méthode RWA est disponible.

Cas 1 : une trajectoire

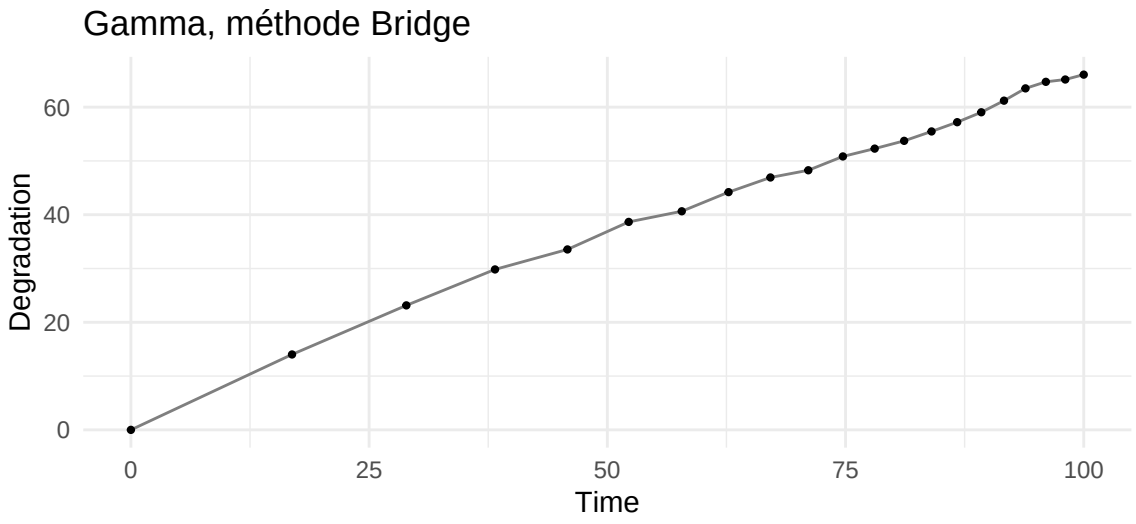
Les inspections sont celles du cas 2 de la partie I (pas décroissants). L'affichage de `x` indique la méthode utilisée.

Modèle 1 : gamma

```
set.seed(401)
x <- sim_gamma(t2, a = 2, b = 3, method = "Bridge")
x
```

```
## 1 simulated trajectory(ies) of the gamma process, same inspection times (matrix)
## parameters: a = 2, b = 3
## simulation method: Bridge
##
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 14.01904
## [3,] 28.90648 23.14478
## [4,] 38.21229 29.82767
## [5,] 45.81569 33.54447
## [6,] 52.24427 38.66434
```

```
plot_traj(x, "Gamma, méthode Bridge")
```

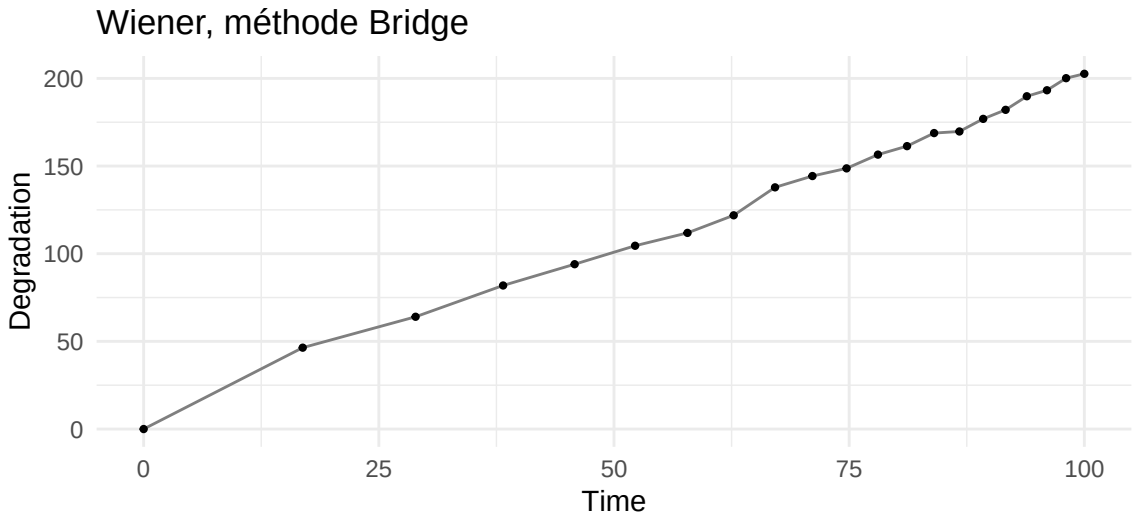


Modèle 2 : Wiener

```
set.seed(402)
x <- sim_wiener(t2, mu = 1.5, sigma = 2, method = "Bridge")
head(x)
```

```
##             t        X1
## [1,]  0.00000   0.00000
## [2,] 16.90921  46.41101
## [3,] 28.90648  64.03561
## [4,] 38.21229  81.87913
## [5,] 45.81569  94.01316
## [6,] 52.24427 104.53435
```

```
plot_traj(x, "Wiener, méthode Bridge")
```

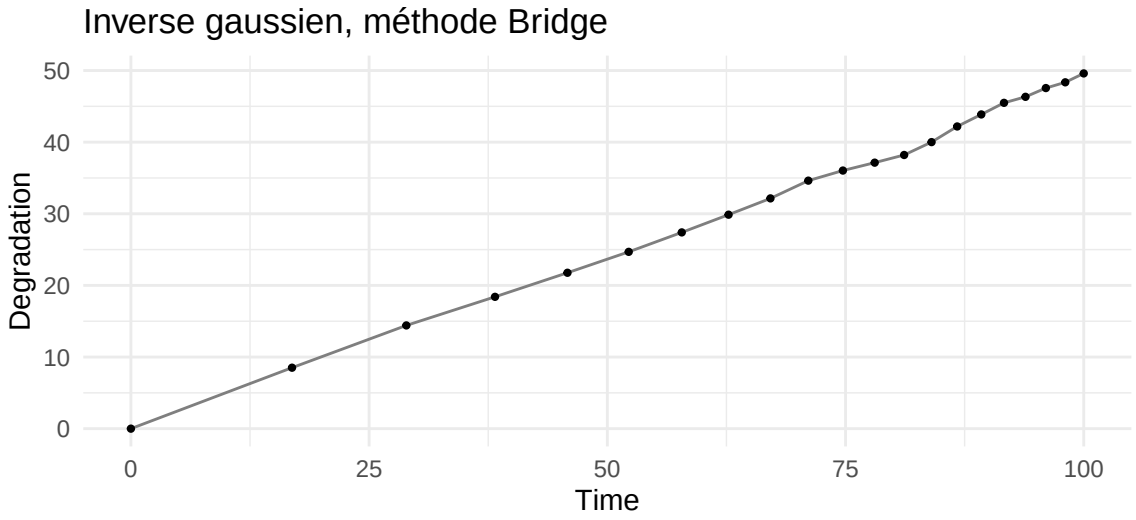


Modèle 3 : inverse gaussien

```
set.seed(403)
x <- sim_ig(t2, mu = 0.5, eta = 2, method = "Bridge")
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  8.510278
## [3,] 28.90648 14.411761
## [4,] 38.21229 18.405516
## [5,] 45.81569 21.764741
## [6,] 52.24427 24.685201
```

```r
plot_traj(x, "Inverse gaussien, méthode Bridge")
```

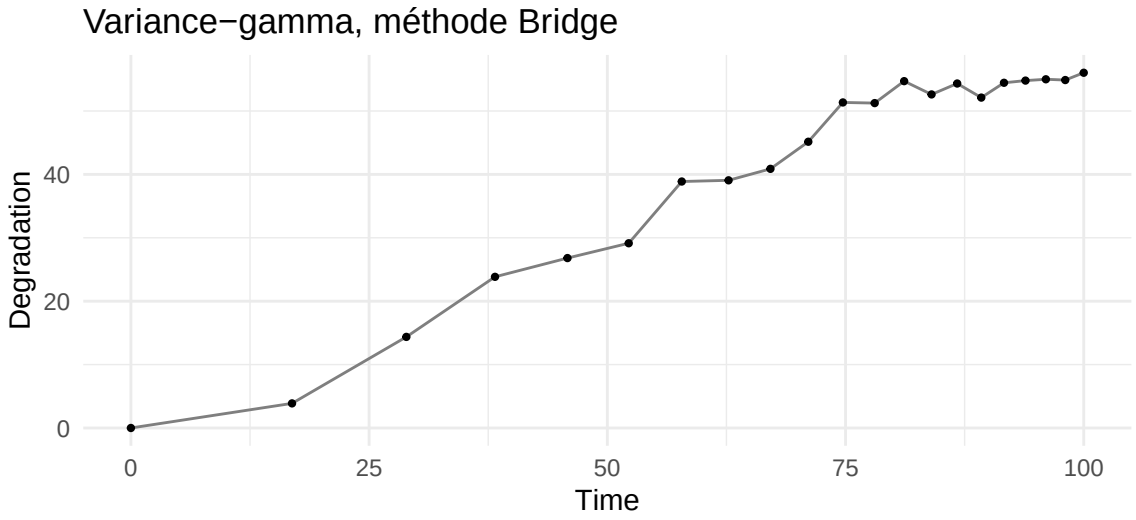


Modèle 4 : variance-gamma

```r
set.seed(404)
x <- sim_vg(t2, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  3.880716
## [3,] 28.90648 14.367985
## [4,] 38.21229 23.842293
## [5,] 45.81569 26.800224
## [6,] 52.24427 29.138386
```

```r
plot_traj(x, "Variance-gamma, méthode Bridge")
```

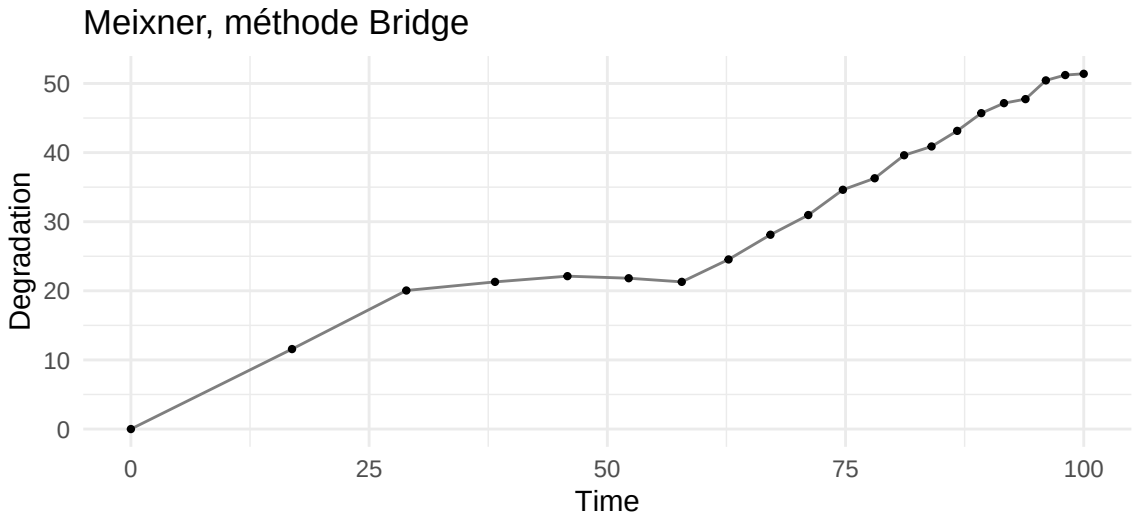


Modèle 5 : Meixner

```
set.seed(405)
x <- sim_meixner(t2, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 11.57110
## [3,] 28.90648 20.04560
## [4,] 38.21229 21.29191
## [5,] 45.81569 22.12137
## [6,] 52.24427 21.82117
```

```
plot_traj(x, "Meixner, méthode Bridge")
```



Cas 2 : plusieurs trajectoires

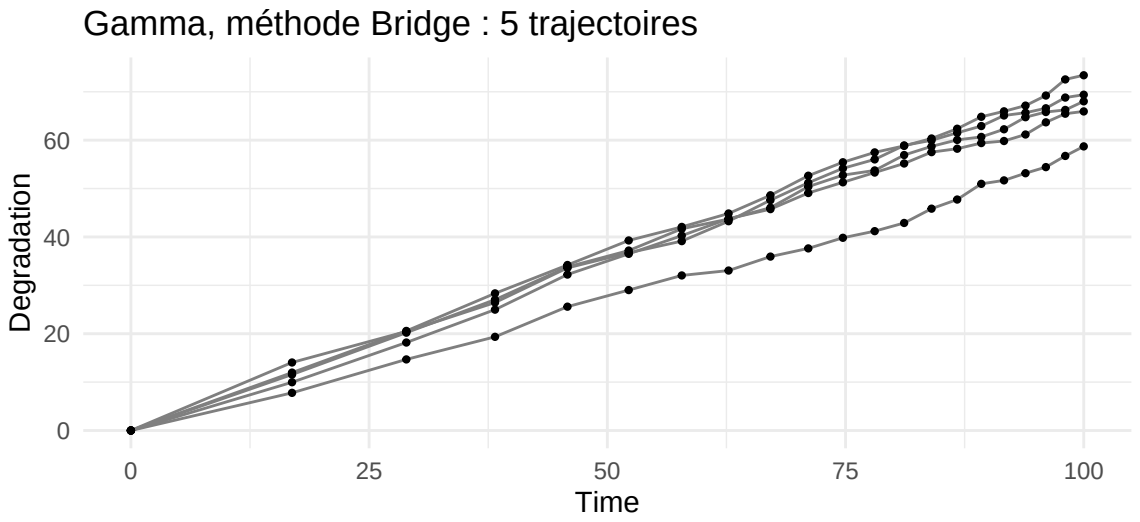
Les inspections sont celles du cas 2 de la partie II : 5 trajectoires, pas décroissants.

Modèle 1 : gamma

```
set.seed(411)
x <- sim_gamma(p2, a = 2, b = 3, method = "Bridge")
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 16.90921 14.05491 11.56010  7.78051
## [3,] 28.90648 20.46629 20.21810 14.68894
## [4,] 38.21229 26.43952 27.03148 19.35558
## [5,] 45.81569 33.60512 33.79778 25.58765
## [6,] 52.24427 36.72170 37.18542 29.02002
```

```
plot_traj(x, "Gamma, méthode Bridge : 5 trajectoires")
```



Modèle 2 : Wiener

```
set.seed(412)
x <- sim_wiener(p2, mu = 1.5, sigma = 2, method = "Bridge")
head(x[, 1:4])
```

```
##             t       X1        X2       X3
## [1,]  0.00000  0.00000  0.000000  0.00000
## [2,] 16.90921 26.48238  5.394106 29.62556
## [3,] 28.90648 24.86890 20.893442 54.72752
## [4,] 38.21229 50.43032 26.063968 62.25786
## [5,] 45.81569 54.01128 32.219544 76.40714
## [6,] 52.24427 69.95381 31.968367 86.62707
```

```
plot_traj(x, "Wiener, méthode Bridge : 5 trajectoires")
```

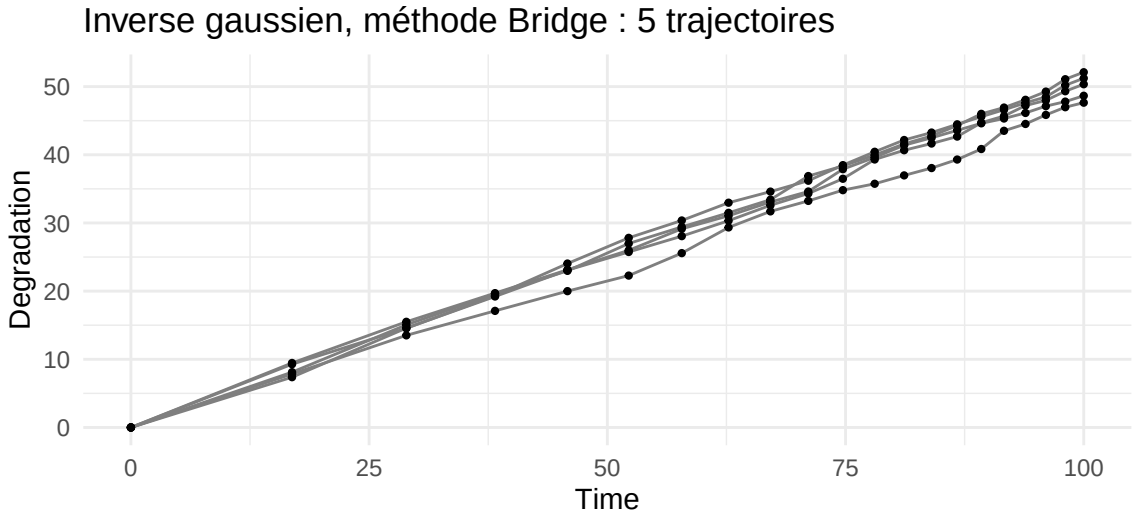


Modèle 3 : inverse gaussien

```
set.seed(413)
x <- sim_ig(p2, mu = 0.5, eta = 2, method = "Bridge")
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 16.90921  7.374389  9.458903  7.845739
## [3,] 28.90648 14.578842 15.499991 13.511093
## [4,] 38.21229 19.211310 19.701243 17.101138
## [5,] 45.81569 23.073620 23.128937 19.995119
## [6,] 52.24427 25.742532 25.984227 22.278263
```

```
plot_traj(x, "Inverse gaussien, méthode Bridge : 5 trajectoires")
```

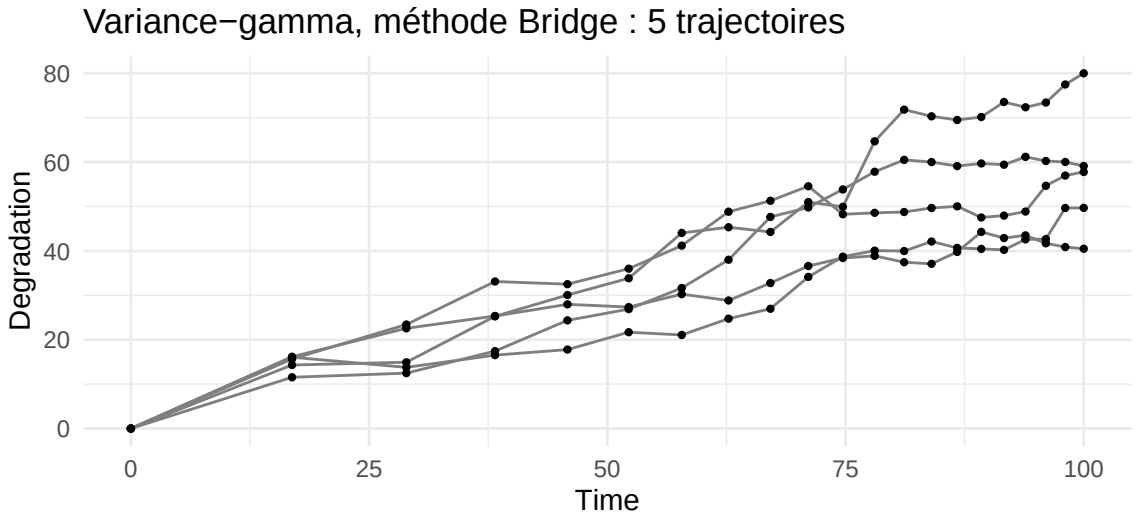


Modèle 4 : variance-gamma

```
set.seed(414)
x <- sim_vg(p2, theta = 0.5, sigma = 1.5, nu = 2, method = "Bridge")
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 16.90921 14.30362 16.16079 15.64419
## [3,] 28.90648 14.92431 22.56686 23.41661
## [4,] 38.21229 25.26229 25.33245 33.10603
## [5,] 45.81569 30.05651 27.96545 32.50451
## [6,] 52.24427 33.82919 27.35094 35.99325
```

```
plot_traj(x, "Variance-gamma, méthode Bridge : 5 trajectoires")
```

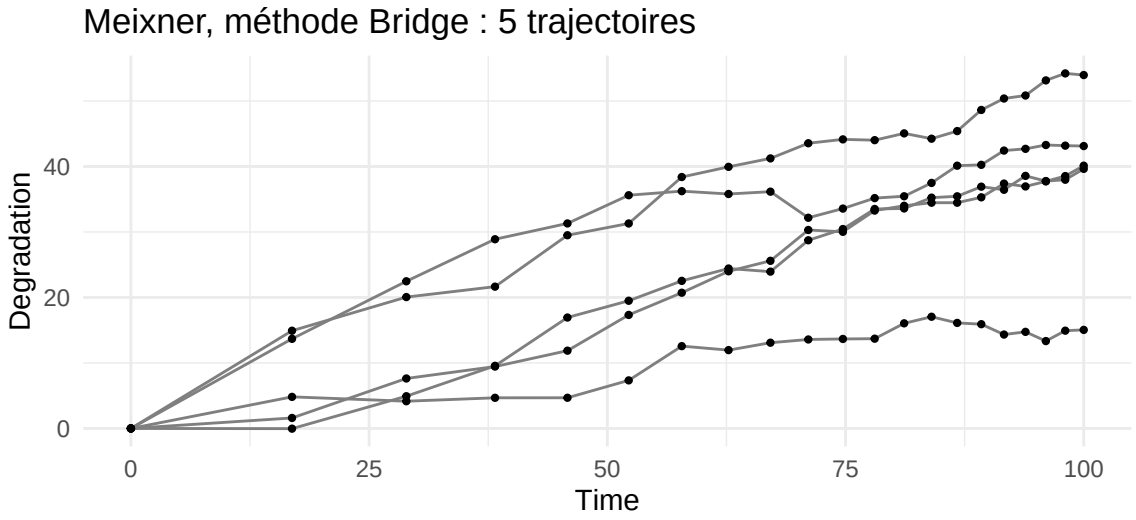


Modèle 5 : Meixner

```
set.seed(415)
x <- sim_meixner(p2, a = 1, b = 0.3, d = 1.5, m = 0.2, method = "Bridge")
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000 0.000000
## [2,] 16.90921 13.68310 14.91878 4.817240
## [3,] 28.90648 22.46950 20.05705 4.154410
## [4,] 38.21229 28.88748 21.64699 4.676184
## [5,] 45.81569 31.29980 29.49955 4.682324
## [6,] 52.24427 35.60779 31.29381 7.337968
```

```
plot_traj(x, "Meixner, méthode Bridge : 5 trajectoires")
```



Cas 3 : processus à échelle de temps

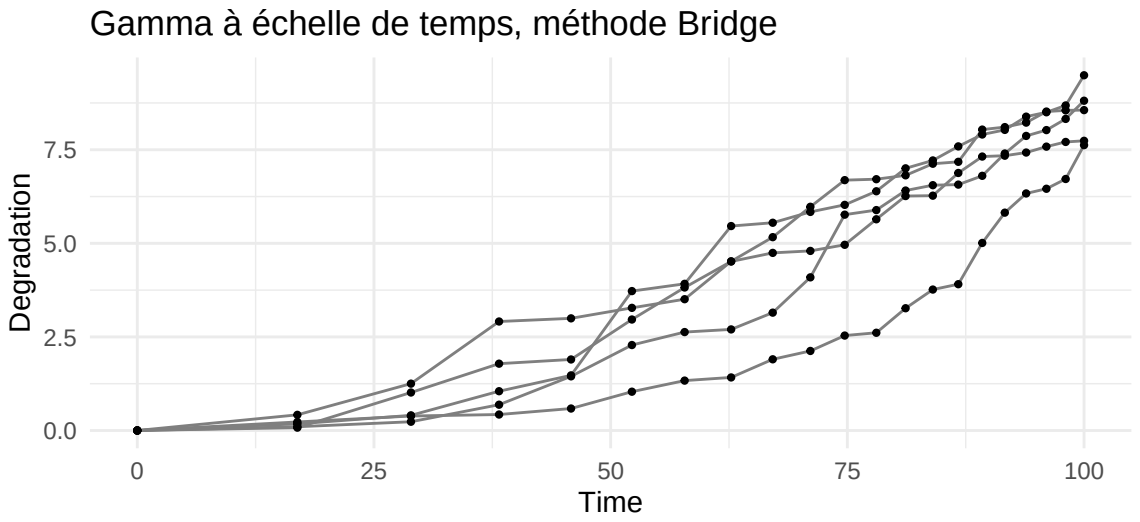
Les processus et la fonction `A` sont ceux de la partie III. Le pont est alors tiré sur l'échelle de temps `A(t)`.

Modèle 1 : gamma à échelle de temps

```
set.seed(421)
x <- sim_gamma(p2, b = 3, A = A, method = "Bridge")
head(x[, 1:4])
```

```
##             t         X1        X2        X3
## [1,]  0.00000 0.00000000 0.0000000 0.0000000
## [2,] 16.90921 0.07629904 0.2281078 0.1773605
## [3,] 28.90648 1.01502250 0.3866171 0.4044226
## [4,] 38.21229 1.78504539 0.4265079 1.0495026
## [5,] 45.81569 1.89745918 0.5869177 1.4728988
## [6,] 52.24427 2.96484466 1.0365544 3.7242360
```

```
plot_traj(x, "Gamma à échelle de temps, méthode Bridge")
```

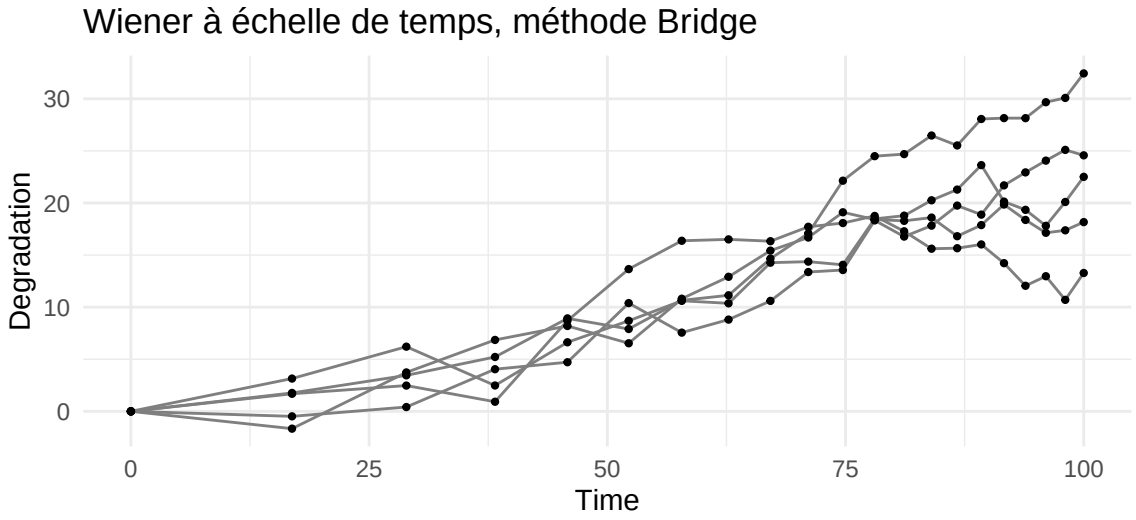


Modèle 2 : Wiener à échelle de temps

```
set.seed(422)
x <- sim_wiener(p2, sigma = 2, A = A, method = "Bridge")
head(x[, 1:4])
```

```
##             t        X1       X2         X3
## [1,]  0.00000  0.000000 0.000000  0.0000000
## [2,] 16.90921 -1.659401 1.767597 -0.4774920
## [3,] 28.90648  3.736923 3.456782  0.4152826
## [4,] 38.21229  6.855555 5.220100  4.0458507
## [5,] 45.81569  8.195915 8.923404  4.7171380
## [6,] 52.24427  6.534788 7.907112 10.3908497
```

```
plot_traj(x, "Wiener à échelle de temps, méthode Bridge")
```

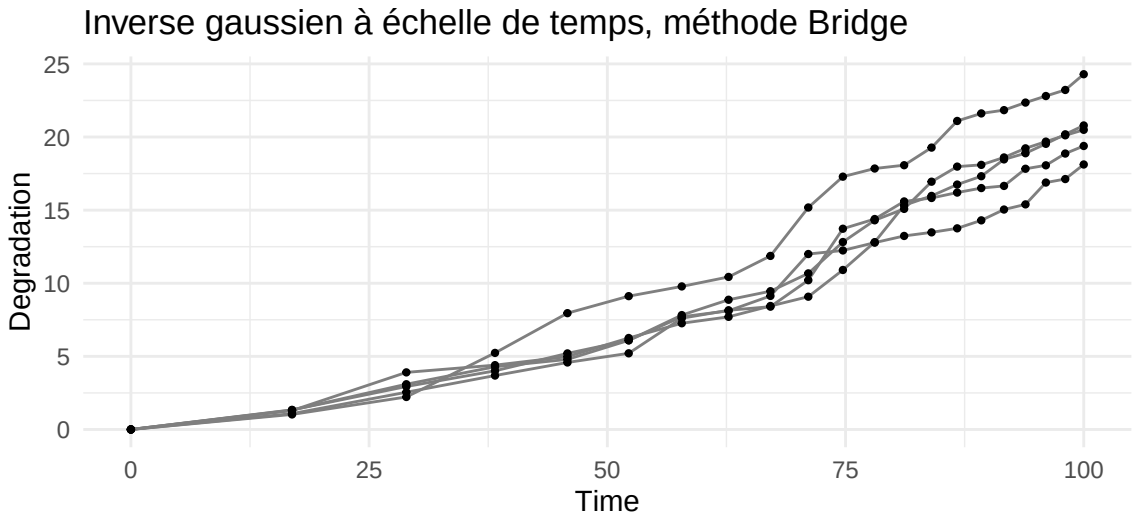


Modèle 3 : inverse gaussien à échelle de temps

```
set.seed(423)
x <- sim_ig(p2, eta = 2, A = A, method = "Bridge")
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000 0.000000 0.000000 0.000000
## [2,] 16.90921 1.028009 1.324015 1.340367
## [3,] 28.90648 2.221875 3.093814 2.937737
## [4,] 38.21229 5.235950 4.288924 4.010464
## [5,] 45.81569 7.953884 4.791547 5.201757
## [6,] 52.24427 9.116765 6.095420 6.095561
```

```
plot_traj(x, "Inverse gaussien à échelle de temps, méthode Bridge")
```



Vérification : la densité des accroissements

Pour chaque processus, on simule 100 trajectoires inspectées tous les 5. L'histogramme montre les accroissements simulés avec `method = "Bridge"`, et la courbe rouge la densité exacte de l'accroissement sur cette durée, calculée sans simulation. L'histogramme doit suivre la courbe.

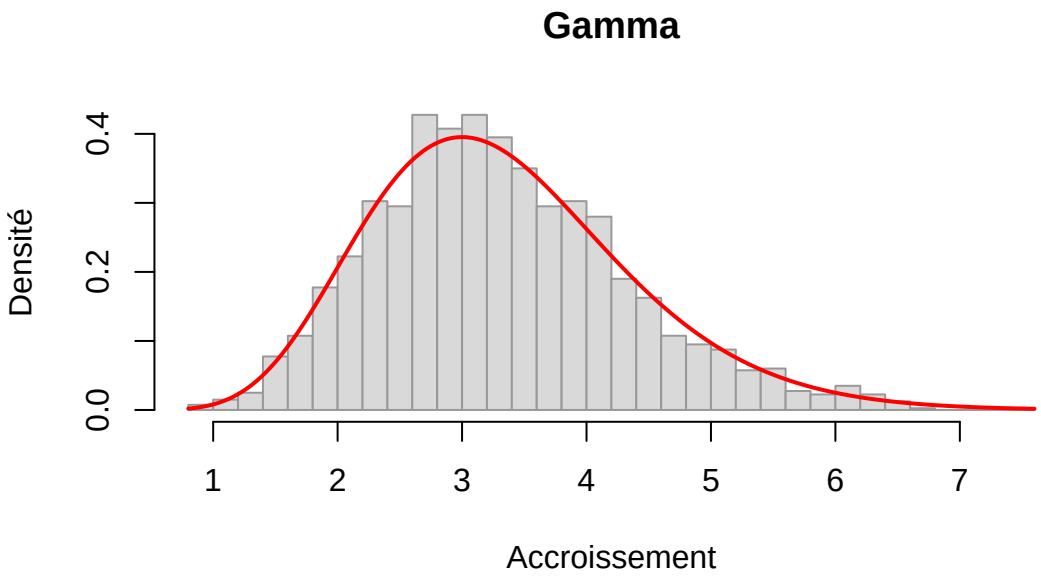
La forme de la densité dépend de la durée du pas : un accroissement gamma, par exemple, suit une loi gamma de forme `a * pas`, décroissante sur un pas court et en cloche sur un pas long.

```
pas <- 5
t6 <- seq(0, 100, by = pas)

trace_densite <- function(x, densite, titre) {
  dx <- traj_df(x, increments = TRUE)$dX
  dx <- dx[!is.na(dx)]
  hist(dx, breaks = 40, freq = FALSE, col = "grey85", border = "grey60",
       main = titre, xlab = "Accroissement", ylab = "Densité")
  curve(densite(x), add = TRUE, col = "red", lwd = 2, n = 500)
}
```

Modèle 1 : gamma

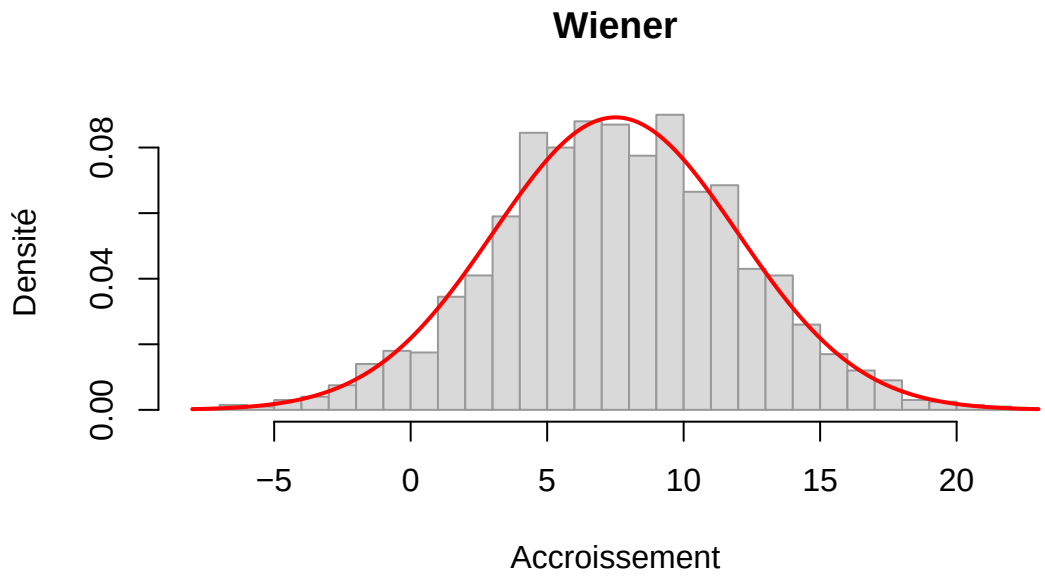
```
set.seed(431)
x <- sim_gamma(t6, a = 2, b = 3, n_traj = 100, method = "Bridge")
trace_densite(x, function(y) dgamma(y, shape = 2 * pas, rate = 3), "Gamma")
```



Modèle 2 : Wiener

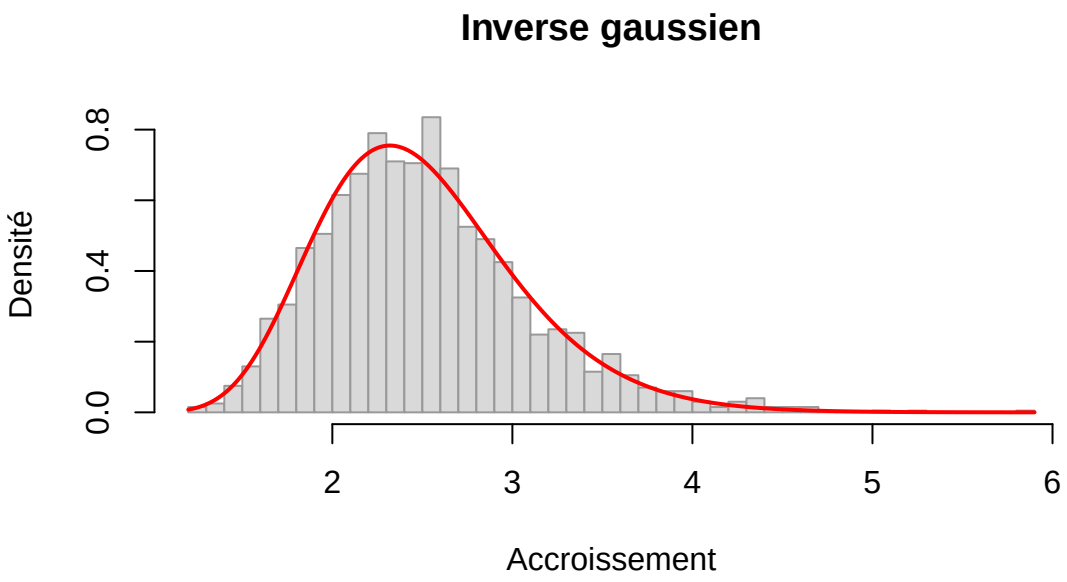
```
set.seed(432)
x <- sim_wiener(t6, mu = 1.5, sigma = 2, n_traj = 100, method = "Bridge")
```

```r
trace_densite(x, function(y) dnorm(y, mean = 1.5 * pas, sd = 2 * sqrt(pas)), "Wiener")
```



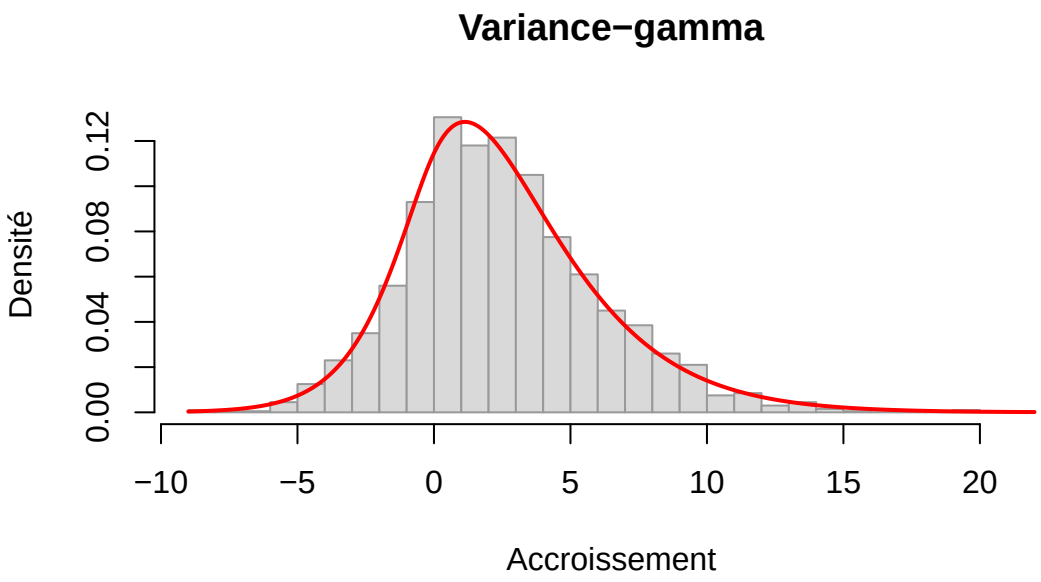
Modèle 3 : inverse gaussien

```r
set.seed(433)
x <- sim_ig(t6, mu = 0.5, eta = 2, n_traj = 100, method = "Bridge")
dig <- function(y) {
  m <- 0.5 * pas
  l <- 2 * pas^2
  sqrt(l / (2 * pi * y^3)) * exp(-l * (y - m)^2 / (2 * m^2 * y))
}
trace_densite(x, dig, "Inverse gaussien")
```



Modèle 4 : variance-gamma

```
set.seed(434)
x <- sim_vg(t6, theta = 0.5, sigma = 1.5, nu = 2, n_traj = 100, method = "Bridge")
dvg <- function(y, theta = 0.5, sigma = 1.5, nu = 2) {
  r <- sqrt(theta^2 + 2 * sigma^2 / nu)
  z <- abs(y) * r / sigma^2
  k <- pas / nu
  2 / (nu^k * sqrt(2 * pi) * sigma * gamma(k)) * (abs(y) / r)^(k - 0.5) *
    besselK(z, k - 0.5, expon.scaled = TRUE) * exp(theta * y / sigma^2 - z)
}
trace_densite(x, dvg, "Variance-gamma")
```



Modèle 5 : Meixner

```
set.seed(435)
x <- sim_meixner(t6, a = 1, b = 0.3, d = 1.5, m = 0.2, n_traj = 100, method = "Bridge")
loi <- Runuran::udmeixner(alpha = 1, beta = 0.3, delta = 1.5 * pas, mu = 0.2 * pas)
trace_densite(x, function(y) Runuran::ud(loi, y), "Meixner")
```

